THE POLITICAL ORIGINS AND ECONOMIC
CONSEQUENCES OF EARLY U.S. ANTITRUST LAW

By

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Competitive Effects of State Antitrust Laws: Evidence from the Progressive Era

1.1 Introduction

Over the last four decades, the U.S. economy has seen rising concentration. Several studies, drawing on a variety of data sources and methodologies, document this pattern across a broad range of industries (De Loecker et al., 2020; Grullon et al., 2019; Smith and Ocampo, 2025; Bajgar et al., 2019; Zeballos et al., 2023; Philippon, 2019). Research also links higher concentration to a declining share of income going to labor (Barkai, 2020; Autor et al., 2020; Aghion et al., 2023). Together, these observations have spurred concerns about the possibility of reduced competition and increased market power in the United States.[1]

In light of these trends, and amid broad public support for stronger oversight, some have called for reforms to the nation's antitrust policies.[2] Antitrust laws restrict firms' ability to monopolize an industry or collude with each other and are designed to protect competition among firms. In turn, competition among firms can create benefits for consumers, such as downward pressure on prices and incentives for firms to maintain product quality. However, the empirical literature offers mixed evidence on the effectiveness of antitrust interventions in promoting competition. Consequently, whether more aggressive enforcement would reverse the increase in concentration observed in recent decades is unclear. To shed light on this question, I analyze a natural experiment in which 39 U.S. states enacted antitrust statutes over the course of the late nineteenth and early twentieth centuries. In particular, I examine the effects of these laws on competition and innovation, which allows me to offer evidence that is relevant to modern debates about the role of antitrust policy.

The origins of U.S. antitrust law lie at the state level of government. Thirteen states had already adopted an antitrust statute of their own by the time the Sherman Act—the first federal antitrust law—was enacted in 1890. Although federal antitrust law is often regarded as paramount, state antitrust statutes remained important well after the Sherman Act's passage. In fact, most states enacted their antitrust statutes after 1890, underscoring that state law was seen as a necessary complement to federal legislation, not a redundancy. The enactment of these statutes and the enforcement actions that followed often generated substantial media cov-

[1]There is active discussion over the extent to which these trends truly indicate greater market power. See, for example, Shapiro and Yurukoglu (2024); Miller (2025); Werden and Froeb (2018).

[2]Support for reform is reflected in proposals from prominent scholars and politicians as well as in survey data on economists' and the public's views. For examples of calls to strengthen antitrust laws and enforcement, see Khan (2017); Wu (2018); Baker et al. (2020); Warren (2024); Hawley (2025). Likewise, in a recent poll of leading economists, a majority of those surveyed agreed that the dominance of large technology firms merits either new regulatory measures or substantial reforms to antitrust policy (Kent A. Clark Center for Global Markets, 2020). There is also strong support for regulation among the general public. In one recent poll, 68 percent of respondents said they support antitrust laws, and pluralities said stricter enforcement would benefit consumers, workers, small businesses, and the U.S. economy as a whole (YouGov, 2023).

erage, reinforcing the perception that state legislatures were leading the fight against trusts.[3] That perception was also grounded in a steady stream of legal activity at the state level. State attorneys general brought numerous cases, including actions against large interstate firms; courts often interpreted state statutes expansively; and penalties in some cases were severe (May, 1987, 1990; Lamb, 2001; Bringhurst, 1979). Moreover, state courts were frequently the primary venues for addressing anticompetitive conduct in manufacturing and other key sectors until the New Deal expansion of federal commerce power (Columbia Law Review, 1961). This evidence indicates that state antitrust laws had the potential to influence economic outcomes, prompting the question of whether they actually did.

I answer this question using data from four main sources. The first source pertains to state antitrust laws themselves. To identify state antitrust statutes in force in the late nineteenth and early twentieth centuries, I reviewed state session laws and legal codes from 1860 through 1940 and identified provisions restricting anticompetitive conduct, such as price fixing. After identifying all such provisions, I coded their contents to capture their main features. The second data source relates to manufacturing. To assess the impact of state antitrust laws on the manufacturing sector, I use newly digitized state-by-industry tabulations from the census of manufactures for 1850 through 1940 (Barkai et al., 2025). The main outcomes I examine are the number of manufacturing establishments, the profits and labor demand of these establishments, and the labor share. The manufacturing sector comprised a relatively large share of the U.S. economy during the study period, which motivates my focus on manufacturing in this paper.[4] Third, I use data from the Comprehensive Universe of U.S. Patents (CUSP) to assess the impact of state antitrust laws on innovation, as proxied by patenting behavior (Berkes, 2018). Finally, to identify newspaper articles mentioning one or more state antitrust laws, I searched two popular digital archives for terms relating to state antitrust laws. These articles allow me to understand how state antitrust laws were perceived by the general public and enforced by state governments during the study period.

I use "stacked" difference-in-differences models, which account for the staggered nature of treatment timing, to study the effects of state antitrust laws. This method involves stacking cohorts of states by adoption date and estimating treatment effects relative to untreated states, thereby avoiding biases common in conventional two-way fixed effects specifications. As is typical in difference-in-differences models, this approach exploits variation in the timing of states' adoption of antitrust laws and compares outcomes in states with an-

[3]For example, California newspapers described that state's law as "perhaps the most far reaching in effect of any of the laws of the legislature" (*San Francisco Call*, May 24, 1907) and reported that "trusts [went] 'bust' in just one week" under the new antitrust law (*Los Angeles Times*, May 16, 1907). Other coverage emphasized the expansive nature of state statutes, with the *Owosso Times* noting that Michigan's antitrust law was "sweeping in its nature" (July 19, 1889), the *New York Times* reporting Kentucky indictments against more than forty companies for "banding together and fixing" prices (March 23, 1899), and the *Iowa Plain Dealer* announcing that Missouri had revoked the charters of "about seven hundred trusts" for violating the state's antitrust law (November 23, 1889). These kinds of reports conveyed to readers that state antitrust laws represented a dramatic and far-reaching assault on big business.

[4]Figure 1.1 shows how the manufacturing sector's share of GDP evolved between 1850 and 2023. In 1890, shortly after the first state antitrust laws took effect, manufacturing comprised about 27 percent of nominal GDP.

titrust laws to outcomes in states without them, before and after adoption. Because the timing of effects is of interest, I also employ event study models in my analysis. A key identification advantage of studying antitrust in a historical context is the presence of both a period without any antitrust laws and a clear control group of states that never adopted antitrust statutes, which papers studying antitrust in modern settings typically lack.

My results suggest that, for all the attention surrounding their enactment, state antitrust laws fell short of meaningfully constraining powerful incumbents. I find that these statutes increased the number of manufacturing establishments by roughly 10 percent, but this growth was likely driven by the expansion of incumbent firms rather than new entry. At the same time, the labor share declined by 4 percent overall and 7 percent in trust-affiliated industries, while profits and employment remained unchanged, indicating that dominant firms fortified their positions despite new legal constraints on anticompetitive conduct. I fail to find systematic evidence that greater enforcement intensity or statutory stringency mitigated these effects. I also document an increase in patenting, pointing to positive spillovers on innovation and supporting the idea that concentration and innovation often coincide. These findings suggest that while state antitrust laws encouraged innovation, they did not meaningfully curtail the market power of dominant firms.

This paper is the first quasi-experimental, quantitative study to examine the effects of state antitrust laws around the time of their enactment. In doing so, I engage with several strands of literature. Among these, I speak to longstanding debates on the effectiveness of antitrust policy, where existing studies report mixed effects on competitive outcomes in both historical and modern contexts (e.g., Stigler, 1966; Burns, 1977; Bosch and Eckard, 1991; Sproul, 1993; Babina et al., 2023). I also inform discourse on antitrust and innovation (e.g., Watzinger et al., 2020; Kang, 2025) and on collusion and firm behavior in historical U.S. settings (e.g., Porter, 1983; Alexander, 1994; Vickers and Ziebarth, 2014; Gross, 2020). Finally, I extend a body of work by economic and legal historians on the role of states in shaping competition during the late nineteenth and early twentieth centuries (e.g., May, 1987; Lamoreaux and Phillips-Sawyer, 2021; Troesken, 2000; Lamoreaux, 2023).

My central contributions to these literatures are fourfold. First, unlike studies that focus on incremental policy changes or individual lawsuits, my setting allows me to estimate the overall impact of a new antitrust regime, including potential deterrence effects. Antitrust law can affect firms' outcomes through both direct channels, which fall on firms subject to enforcement actions, and indirect channels, which operate through deterrence: even firms not targeted may alter behavior in anticipation of scrutiny.[5] Second, I broaden the scope of prior scholarship by studying the long-term effects of antitrust, employing data covering nearly a

[5] The ability to capture deterrence is a unique advantage of my setting. By discouraging firms from engaging in anticompetitive conduct in the first place, antitrust laws can influence not only the firms in the courtroom but also the broader economy. However, because most antitrust research focuses on individual lawsuits, estimated effects typically concern the parties to a case or their industries, while the broader consequences are harder to observe.

century, and examining effects on the entire manufacturing sector rather than focusing on individual industries. Third, I highlight the formative role of state competition policy, which has received less attention in the literature than the federal antitrust regime despite the fact that states pioneered antitrust regulation in the United States. Fourth, this paper advances our understanding of the Progressive Era, a period in American history characterized by sweeping reform. As society undergoes what some are calling a second Gilded Age,[6] lessons from this period provide valuable insight into the challenges of today's world.

1.2 Background

1.2.1 Historical Context

The antitrust movement emerged during a period of rising concentration, as large firms increasingly displaced smaller competitors in the U.S. economy. As shown in Figure 1.2, references to "trusts" rose sharply in the mid- to late 1880s, underscoring the growing public attention directed toward big business.[7] This public concern emerged alongside major changes in transportation, communications, and other technologies that encouraged consolidation and eroded the position of small firms (Higgs, 1971; James, 1983; Atack, 1985). Railroads, which made long-distance shipping easier and cheaper, allowed firms to serve increasingly distant markets (Donaldson and Hornbeck, 2016; Hornbeck and Rotemberg, 2019, 2024). New communications technologies, such as the telegraph (and later, the telephone), allowed national markets to emerge as firms could more easily do business with partners in far-flung places (Yates (1986)). New production technologies, such as the Bessemer method of steelmaking, favored mass production (Rogers, 2009). The efficiencies associated with these technological advances enabled firms to dramatically increase output, often surpassing demand and resulting in widespread industrial overcapacity that pushed firms toward consolidation as a strategy for stabilizing prices and surviving competition (McCraw, 1981). As a result, large firms began to capture increasing market shares; as Figure 1.3 illustrates, concentration in the manufacturing sector rose steadily in the decades preceding the antitrust movement. As large firms became more numerous, concerns about their growing power set the stage for the rise of the antitrust movement.

The relative decline of agriculture also played a role in the rise of antitrust. According to the standard "agrarian discontent" hypothesis, farmers found themselves increasingly vulnerable to the power of large corporations as the United States industrialized, leading them to lend strong support for antitrust legislation

[6] See, for example, Wu (2018).

[7] "Trust" became a catch-all term for industrial combinations perceived as threatening competition during the late nineteenth century (Werden, 2020, 11-17). Originally, however, a trust referred to a legal tool used to coordinate cooperative actions among parties that would otherwise compete. Under a trust, member firms placed stock under the control of appointed trustees, who acted collectively to manage operations and limit competition among participants.

(Thorelli, 1955, 58).[8] The geographic pattern of early antitrust legislation reflects the agricultural roots of the movement. As Figure 1.4 shows, the first states to adopt antitrust laws were concentrated in the largely agricultural Midwestern United States. Monopolization among railroads was an acute concern for farmers, who relied on railroads and railroad-owned grain elevators for the transportation and storage of their crops (Thorelli, 1955, 143); (Phillips-Sawyer, 2019, 4)). In particular, farmers alleged that monopolistic railroads charged unfair prices for their services.[9] Libecap (1992) also highlights the role of the Chicago meat packers in depressing cattle prices, which led Midwestern farmers to lobby for antitrust legislation as a way to counteract what they saw as monopsonistic abuses in livestock markets.

States were the first to respond to popular cries for antitrust legislation, with 13 enacting statutes before the federal government.[10] Although state legislatures do not consistently record vote counts,[11] the historical record suggests that state antitrust laws often enjoyed broad, bipartisan support. For example, Texas's 1889 antitrust statute passed unanimously.[12] When that statute was later overturned, the legislature responded by passing a new antitrust law in 1903, which again received overwhelming support—unanimous approval in the Senate and a 103–2 vote in the House.[13]

This state-level action laid the groundwork for antitrust reform at the federal level, beginning with the Sherman Antitrust Act of 1890. There is some evidence that trusts welcomed the passage of the Sherman Act because they believed its passage would thwart state antitrust policy, which they perceived to be a greater threat. For example, Troesken (2000) finds that trust stocks fell significantly in response to state antitrust

[8] Despite extensive scholarly attention, no consensus exists on what—or who—actually drove the antitrust movement. Some scholars have expressed doubt that farmers' cries for help were the result of genuine anticompetitive abuses and have alleged that the real driving force behind antitrust legislation was interest-group lobbying by agricultural groups (Dilorenzo, 1985; Boudreaux and DiLorenzo, 1993; Boudreaux et al., 1995; Gilligan et al., 1989; Dilorenzo and High, 1988). Other scholars, however, reject the idea that agricultural groups were the primary political force behind early antitrust legislation, instead arguing that urban small business interests, especially those facing displacement by more efficient firms, played a more decisive and sustained role (Baxter, 1980; Stigler, 1985; John, 2017). A third group emphasizes that political pressure to respond to perceived monopoly abuses, especially in the wake of widespread public outrage over trusts like Standard Oil and the Whiskey Trust, played a key role in prompting both state and federal antitrust legislation–even in the absence of clear evidence that market concentration had materially increased (Hofstadter, 1979; Lande, 1982; Millon, 1988; Werden, 2020; Letwin, 1965; Flynn, 1988; Lande, 1988). A fourth view, associated with Bork (1966) and defended more recently by Crane (2014), holds that the Sherman Act was a "consumer welfare prescription" intended to prevent only those restraints that harmed allocative efficiency, regardless of the political coalitions that gave rise to its passage. A fifth view emphasizes that the Sherman Act was a political compromise among factions with conflicting goals and that its vague language was designed to paper over disagreement, making any unified theory of original intent—such as Bork's consumer welfare thesis—fundamentally untenable (Grandy, 1993; Hazlett, 1992).

[9] Economic historians have provided empirical support for farmers' allegations of high railroad freight rates. Higgs (1970) argues that, despite nominal declines in freight rates, the real cost of railroad transportation relative to crop prices remained essentially flat from the 1860s through the 1890s. Responding to Higgs's paper, Aldrich (1980) uses transaction data from the Chicago Board of Trade and published freight rates to argue that real freight rates fell until the early 1880s, then stabilized or rose during the late 1880s and 1890s—a pattern consistent with the political science theory of "relative deprivation," which posits that protest arises not from steady hardship but from the reversal of improving conditions.

[10] Antitrust legislation was neither the first nor the sole form of government regulation that state legislatures pursued in response to political pressure by aggrieved farmers in the mid-nineteenth century. In the 1870s, over a decade before Iowa became the first state to adopt an antitrust statute in 1888, several state legislatures enacted "Granger" laws to regulate the prices and practices of railroads and grain elevators (Noll and Kanazawa, 1994). In 1889, state legislatures in several Midwestern states also passed or considered livestock inspection laws to respond to farmer concerns about falling cattle prices and the market power of Chicago meat packers (Libecap, 1992).

[11] Texas is an exception; its published session laws include vote counts for each legislative chamber.

[12] Act of March 30, 1889, Chapter 117, 1889 Texas Acts 141.

[13] Act of March 31, 1903, Chapter 94, 1903 Texas Acts 119.

enforcement, while trust stocks increased in value after two different federal antitrust laws were proposed, suggesting investors favored at least some forms of federal antitrust legislation. The apparent preference of trusts for federal antitrust legislation over state enforcement underscores the initial ineffectiveness of the Sherman Act. In its early years, federal enforcement was notoriously weak, hampered by a lack of resources and political will. It was not until 1903 that Congress appropriated funds to support federal antitrust enforcement ((Thorelli, 1955, 3)). Even then, staffing remained minimal; just two professional staff members responsible for antitrust enforcement are identified in the 1904 *Department of Justice Register* ((Werden, 2018, 425)). Moreover, the Antitrust Division within the U.S. Department of Justice was not created until 1919 (Werden (2018)). Thus, despite its landmark status, the Sherman Act's immediate impact on trust-busting was limited by inadequate implementation.

Relative to the federal government's meager enforcement efforts, state antitrust enforcement was robust in the late nineteenth and early twentieth centuries. State enforcement actions were numerous and sometimes targeted major interstate combinations (May, 1987, 1990). For example, Standard Oil faced 24 separate state actions (Bringhurst, 1979, 204).[14] Weak federal enforcement also meant that state action was often the only remedy available to address anticompetive behavior. For example, in *United States v. E. C. Knight Co.*, the U.S. Supreme Court ruled that the Sherman Act did not apply to manufacturing.[15] Through this ruling, the Court effectively made state courts the primary venue for addressing anticompetitive conduct in this crucial sector of the economy. In general, antitrust enforcement lied largely in state hands until the expansion of federal commerce power in the New Deal era (Columbia Law Review, 1961).

Federal antitrust law was intended to supplement existing state laws. Senator John Sherman advocated for the passage of the Sherman Act, which was named after him, by arguing that the law would apply at the national level law that already existed within several states. As Hovenkamp (1983, 375) writes, "the legislative history of the federal antitrust law indicates that Congress intended to leave state antitrust enforcement more or less intact but to provide an additional federal forum for dealing with restraints of trade which exceeded the jurisdiction of the courts of any particular state." The enduring importance of state antitrust laws, even after the enactment of federal legislation, is reflected in early court decisions that often interpreted state antitrust

[14] May (1987, 537) notes that Texas collected $1,623,500 in penalties in a single case against the Texas affiliate of the Standard Oil Company in *Waters-Pierce Oil Co. v. Texas*. The state fined the defendant $1,500 per day for a total of 1,033 days in violation of a 1899 law and $50 per day for a total of 1,480 days in violation of a 1903 law. The fine amounts to about $57 million in 2024 dollars (Federal Reserve Bank of Minneapolis, Federal Reserve Bank of Minneapolis). The defendant paid 32 percent of the fine and Standard Oil of New Jersey paid the remaining balance. 212 U.S. 86 (1909). Act of May 25, 1899, Chapter 146, 1899 Texas Acts 246. Act of March 31, 1903, Chapter 94, 1903 Texas Acts 119.

[15] 156 U.S. 1 (1895).

statutes expansively (May, 1990; Lamb, 2001).[16] As Brinkerhoff (2017) shows, courts typically focused on whether restraints themselves crossed state lines—not whether their effects did—allowing for meaningful state enforcement even when alleged anticompetitive conduct affected interstate commerce.

1.2.2 Existing Literature and Contributions

The effects of antitrust policy on competition have been extensively studied, yet the literature remains divided on the upshot.[17] Several papers studying antitrust policy in historical contexts suggest that the Sherman Act and its initial implementation had limited or even counterproductive effects. These papers argue that landmark antitrust interventions—including early enforcement of the Sherman Act (Stigler, 1966); the 1911 dissolutions of Standard Oil, American Tobacco, and American Snuff (Burns, 1977); Supreme Court decisions against railroad combinations (Binder, 1988); and the 1948 Paramount decision mandating vertical divestiture in the film industry (Gil, 2015)—failed to meaningfully alter market structures, suppress anticompetitive behavior, or lower prices. Bittlingmayer (1993, 1992) echoes this skepticism by arguing that uncertainty surrounding antitrust enforcement disrupted financial markets and real investment, potentially causing unintended contractionary effects on the economy. In contrast, subsequent studies argue that historical episodes of antitrust enforcement constrained monopoly power and altered firm behavior. For example, Mullin et al. (1995) find that the initiation of the 1911 antitrust suit against U.S. Steel led to stock gains among the conglomerate's customers, indicating expectations that a dissolution would lower steel prices and increase output, and Mullin and Snyder (2021) find that competition increased following the 1952 DuPont patent-licensing remedy and the 1911 breakup of American Tobacco.[18] Moreover, both Prager (1992) and Baker et al. (2023) show that financial markets responded strongly to signals of heightened antitrust enforcement, suggesting that investors viewed stricter antitrust policy as a credible threat to firm profitability. Together, these studies

[16] For example, in *Hammond Packing Co. v. Arkansas*, the U.S. Supreme Court upheld Arkansas's power to condition an Illinois corporation's right to do business in Arkansas on not belonging to a price-fixing conspiracy *anywhere* (the conspiracy was alleged to have occurred in Illinois). 212 U.S. 322 (1909). Likewise, in *International Harvester Co. v. Missouri*, the U.S. Supreme Court upheld Missouri's antitrust action against an out-of-state trust that had formed a monopoly affecting local markets, concluding that the corporation's presence and conduct within the state subjected it to Missouri's jurisdiction. 234 U.S. 199 (1914).

[17] Though I focus on studies of U.S. antitrust policy in this literature review, studies from international contexts also provide insights. Across multiple papers, George Symeonidis finds that cartelization generally harmed economic performance in the United Kingdom, while cartel policy reforms led to more intense competition, which in turn increased labor productivity growth, had no adverse effect on wages, shifted firms toward price rather than non-price competition, encouraged innovation, and sometimes even forced unprofitable cartels to disband (Symeonidis, 2000, 2008, 2019, 2024). Geloso (2020) and Crane (2023) also study historical decartelization policies in Canada and Germany, respectively, while Reed et al. (2022) examine modern Mexico and Aguzzoni et al. (2013) study the European Union. Several studies examine how antitrust laws influence economic outcomes across countries, analogous to this paper's focus on variation across U.S. states. They find that antitrust policies reduce firms' complaints about anticompetitive obstacles (Mayo and Schiffer, 2006); increase productivity growth (Buccirossi et al., 2013); induce firms to raise investment and external financing in response to greater competition (Dasgupta and Žaldokas, 2019); reduce profit margins but encourage mergers as a strategic response to enforcement (Dong et al., 2019); and are effective at constraining market power in nontradable sectors, where competitive pressures from international trade are absent (Besley et al., 2021). Another set of papers shows that effective enforcement—not just the adoption of antitrust laws—ultimately shapes outcomes (Hylton and Deng, 2007; Bradford et al., 2025).

[18] The work of Mullin and Snyder (2021) contrasts with the earlier assessment by Burns (1977) that the dissolution of American Tobacco had little effect on the industry's structure or anticompetitive conduct. Mullin and Snyder (2021) do not cite Burns (1977) or discuss why their conclusions differ.

highlight ongoing disagreement about whether early antitrust enforcement meaningfully affected markets.

This pattern of disagreement persists in the literature studying antitrust policy in relatively more modern settings.[19] Several studies question the effectiveness of antitrust enforcement, finding that prices often rise following major interventions (Sproul, 1993); that enforcement functions as a negative productivity shock (Young and Shughart, 2010); and that lax merger policy was not to blame for rising industrial concentration in the late twentieth century (Peltzman, 2014). However, other scholars have challenged these conclusions. Bosch and Eckard (1991) show that price-fixing indictments led to substantial equity losses, consistent with the expectation that enforcement would curb monopoly profits, and Shepherd (1982) attributes declining industrial concentration in the mid-twentieth century partly to efficacious antitrust policy. Similarly, Babina et al. (2023) find that federal antitrust cases between 1971 and 2018 increased employment, wages, and business formation, suggesting broad economic benefits. Thus, despite extensive study, no consensus exists on the effectiveness of antitrust enforcement in promoting competition.

Beyond effects on competition, the effect of antitrust policy on innovation has also been a subject of scholarly inquiry, with studies yielding mixed results. A series of studies find that antitrust actions against Bell Labs (Watzinger et al., 2020), the Bell System (Watzinger and Schnitzer, 2022), and Microsoft (Thatchenkery and Katila, 2023) led to increased innovation—as proxied by patenting behavior—in affected sectors. Likewise, Kwon and Marco (2021) find that antitrust regulation of patent transfers curbs the negative impact of patent consolidation on rivals' innovation. However, other studies complicate this picture. Kang (2025) finds that collusion, which antitrust aims to prevent, raises both the quantity and quality of patents among cartel members. Similarly, Hashmi (2013) reports a mildly negative association between competition and innovation. This unresolved debate highlights the need for further empirical work on when and how antitrust policy affects economic outcomes.

Responding to this need, this paper advances both the literature on historical antitrust enforcement and the broader literature on how antitrust policy affects competition and innovation in several ways. First, unlike previous studies focusing on incremental changes or specific antitrust cases, I examine the overall impact of implementing an antitrust regime where none existed before. This approach allows for a more comprehensive assessment of effects, including potential deterrence, by analyzing broader market outcomes rather than focusing solely on parties involved in particular cases.[20] Second, while many existing studies concentrate on individual industries or a small number of industries, I provide a wider perspective by estimating effects

[19] For literature reviews representing opposing perspectives on the effectiveness of antitrust enforcement, see Crandall and Winston (2003) for a skeptical view and Baker (2003) for a pro-enforcement one.

[20] Antitrust laws influence firm behavior by deterring anticompetitive conduct, even among firms never prosecuted for violations. Just as most taxpayers never face an audit but still comply with tax laws because of the threat of scrutiny, antitrust laws may deter firms from engaging in anticompetitive conduct even if enforcement actions are uncommon. While the direct effects of antitrust enforcement are visible in prosecuted cases, deterrence is harder to observe, which underscores the significance of this paper's ability to estimate these broader, indirect effects.

across the entire manufacturing sector. Third, I analyze long-term effects over nearly a century, in contrast to the narrower time windows examined in most of the existing literature. This extended time frame provides a better understanding of the enduring impact of antitrust policy. Fourth, while most existing papers focus on the federal antitrust regime; I instead examine the role of state antitrust policy, allowing me to bring new evidence to bear on this ongoing and important economic debate.

This paper also contributes to the broader literature on collusion and firm behavior in the historical United States. A central theme in this literature is that collusion can be both unstable and highly adaptive to institutional and market conditions. Porter (1983) highlights the fragility of cartel arrangements, showing that the nineteenth-century railroad cartel known as the Joint Executive Committee repeatedly broke down in response to unexpected shocks. In other cases, collusion has proven more durable. Alexander (1994), Chicu et al. (2013), and Vickers and Ziebarth (2014) show that the National Industrial Recovery Act enabled widespread and sustained collusive behavior across multiple industries, with firms reducing price responsiveness to cost changes and ceasing competitive price adjustments altogether. Another strand of the literature examines how collusion interacts with investment and entry. Gross (2020) shows that coordination among Southern railroads facilitated rapid technological standardization, but cartel discipline prevented efficiency gains from reaching consumers. Similarly, Syverson (2024) finds that railroads used strategic duplication to deter entry and temporarily raise prices for grain sellers, but this came at the cost of wasteful overinvestment. Finally, Huang (2025) shows that the Beef Trust manipulated price signals by temporarily raising cattle prices to attract supply, then lowering them, causing more harm to sellers than standard monopsony would predict. Together, these studies underscore that collusive behavior is not only shaped by its legal environment, but can also distort investment, entry, and the information available to market participants.

Finally, this paper builds on work by economic and legal historians, who have written about how states shaped the competitive landscape during the late nineteenth and early twentieth centuries. Most papers in this literature show that state enforcement actions, including lawsuits by attorneys general and shareholders, effectively curtailed anticompetitive conduct during this period (Pratt, 1980; May, 1987; Clay and Troesken, 2002; Nolette, 2012; Phillips-Sawyer, 2018; Lamoreaux and Phillips-Sawyer, 2021; Lamoreaux, 2023).[21] I build upon and extend this literature in several important ways. First, I bring quasi-experimental evidence to bear on the effects of state antitrust laws, focusing on the identification of causal effects. Furthermore, unlike much of the existing scholarship that concentrates on legal outcomes such as individual enforcement actions and cases won against trusts in court, I measure "success" by analyzing the extent to which antitrust laws had

[21] Two papers by Werner Troesken offer a perspective that differs from this prevailing view. In a study of the 1889 dissolution of the Chicago Gas Trust, Troesken (1995) shows that the breakup, carried out through a *quo warranto* proceeding under corporate law rather than antitrust law, had little effect on market structure or performance. Similarly, Troesken (1998) argues that competition—not state antitrust enforcement—disciplined the Whiskey Trust.

a discernible effect on important economic indicators. Finally, by comprehensively tracing the evolution of state antitrust laws over time, I offer a more detailed description of state antitrust law in its emergent stage than previously available in the literature. These contributions allow me to present new evidence on how state antitrust laws shaped economic outcomes during the formative era of U.S. competition policy.

1.2.3 Theoretical Predictions

Before turning to the empirical analysis, it is helpful to consider what standard economic theories suggest about the expected effects of state antitrust laws on the outcomes examined in this paper. Because early antitrust policy primarily targeted cartels and other horizontal restraints, state antitrust laws can be understood as interventions designed to shift markets from monopolistic or oligopolistic structures toward more competitive ones. Accordingly, they are expected *ex-ante* to reduce—or at least not increase—firms' profits.[22] If a new state antitrust law meaningfully curtails ongoing anticompetitive behavior or deters new conduct from emerging, markets may shift away from monopoly or oligopoly—where profits are positive—towards perfect competition—where economic profits are zero under standard assumptions. From a welfare perspective, while profits may decline, surplus may be reallocated from producers to consumers, and aggregate surplus may increase. *Ceteris paribus*, increased competition should reduce firms' profits. However, if antitrust laws also enhance efficiency or spur innovation, profits could rise even as market power declines. Further, if enforcement is weak or firms perceive little risk of detection and punishment—or if little anticompetitive conduct existed in the first place—profits may remain unchanged.

Greater competition may also lead to more firms operating in a market. A sizable literature in industrial organization documents a positive relationship between the number of firms and competition (Bresnahan and Reiss, 1991; Tirole, 1988; Syverson, 2004; Melitz and Ottaviano, 2008). Standard Cournot models of oligopolistic competition, which show that price converges to marginal cost as the number of firms tends towards infinity, reinforce this view. Additionally, antitrust jurisprudence tends to view a larger number of firms in a market as a sign of increased competition (Hovenkamp and Areeda, 2022). Thus, to the extent that state antitrust laws increased competition by weakening the market power of large firms, one would expect them to also increase the number of firms by encouraging new entrants. One important caveat to this theory is that most state antitrust laws explicitly prohibited horizontal price-fixing arrangements but did not affect firms' ability to merge with each other. Thus, if firms responded to a new antitrust law in their state by merging with competitors to avoid prosecution for cartel behavior, state antitrust laws may have

[22] Similarly, they are expected *ex-ante* to reduce or leave unchanged the retail prices consumers pay. Though not the focus of this paper, I examine effects on food retail prices in Appendix 1.10.

reduced the number of firms.[23] My ability to test this theory is partially limited by the fact that the census of manufactures reports establishment counts rather than firm counts. While a firm refers to a distinct business entity, an establishment refers to a single physical location where manufacturing takes place; a single firm can operate multiple establishments. Thus, any observed increase in the number of establishments could stem from new firm entry or growth by existing firms, a distinction that cannot be directly observed in the data.

Beyond market structure, economic theory also predicts that increased competition from antitrust policy should affect production and labor demand. Specifically, antitrust laws should lead to higher employment by lifting artificial constraints on output imposed by monopolistic or collusive firms. This prediction reflects a well-established result in industrial organization, which holds that collusion and anticompetitive mergers suppress output (Asker and Nocke, 2021; Whinston, 2006).[24] In markets where firms had previously restricted output to maintain high prices, antitrust can push firms to expand production. As output rises, so too does the demand for inputs, including labor, resulting in higher employment.

Standard models leave the net effect on wages indeterminate but suggest that labor's share of income should increase if antitrust policy increases competition. For wages, an expansion of output induced by stronger competition should increase labor demand, as discussed above, placing upward pressure on wages. However, heightened competition also reduces firms' profit margins, which may constrain their ability or willingness to raise wages. For the labor share, the total wage bill should grow as output expands and employment rises.[25] At the same time, thinner profit margins imply that the share of total income accruing to capital falls. Taken together, these forces suggest that labor's share of income is likely to increase, even if average wages do not rise appreciably.

A further dimension of firm behavior potentially affected by antitrust policy is innovation. The adoption of an antitrust statute may increase or decrease a firm's incentive to innovate. On one hand, when firms can no longer rely on monopoly power or collusive arrangements to sustain profits, they face stronger pressure to improve their products and processes to stay ahead of rivals. Antitrust policy can also reduce barriers to entry, enabling new firms to challenge incumbents and spurring industry-wide innovation. In such an environment, innovation becomes a key strategy for differentiation and survival.[26] On the other hand, monopolies may foster innovation by investing their substantial resources in research and development (Schumpeter, 1942;

[23] Federal lawmakers similarly declined to limit firms' ability to consolidate with competitors when they passed the Sherman Act in 1890. Lamoreaux (1985) and Bittlingmayer (1985) provide evidence that the wave of mergers that occurred around the turn of the twentieth century was partially spurred by the Sherman Act.

[24] Similarly, Jacquemin and Slade (1989) present a model implying collusin reduces employment and wages.

[25] Contrary to this view, Berger et al. (2022) argue concentration can raise the labor share.

[26] A twenty-first-century observer might expect the adoption of a state antitrust law to make firms more cautious in their patenting behavior, encouraging a focus on genuine innovation rather than the use of patents for market control. This expectation reflects a modern legal landscape in which tensions between patent and antitrust law are well recognized, particularly when patents are used to exclude rivals or entrench market power. However, early twentieth-century courts generally treated patents as government-sanctioned monopolies that were immune from antitrust scrutiny (Jacobson, 2018), making a reduction in patenting motivated by legal risk unlikely at this time.

Hashmi, 2013). A possible reconciliation of these two theories posits that the relationship between product market competition and innovation follows an inverted-U shape—that is, innovation increases with competition at low levels of competition, but decreases with competition at high levels (Aghion et al., 2005). According to this view, innovation rises with competition at low levels because closely matched firms are motivated to innovate to gain market advantage, but it falls at high levels of competition because profit margins become too thin to justify the cost of innovation. Overall, these predictions show that antitrust laws' implications for several of the outcomes I study in this paper are theoretically uncertain, underscoring the need for empirical analysis.

1.3 Data Sources and Descriptive Statistics

1.3.1 State Antitrust Laws

I created a historical compendium of state antitrust statutes to trace the development of state antitrust law from 1888 (when the first state antitrust law was enacted) to 1940. My database includes statutes from every state except Alaska and Hawaii, which were not yet states during the study period, allowing for a comprehensive view of subnational competition law across the continental United States. To create this database, I reviewed state session laws and legal codes for each year from 1860 through 1940 and identified statutes outlawing anticompetitive conduct.[27] In total, I identified 168 statutes enacted between 1888 and 1940. This set of statutes includes amendments, which allows me to characterize the evolution of state antitrust regimes over the study period. To my knowledge, no other database documents both the initial enactment and subsequent evolution of state antitrust laws over the late nineteenth and early twentieth centuries. I do not regard statutes applying to a single industry, rather than to commerce more generally, as antitrust statutes in this paper.[28] I find that Iowa became the first state to adopt an antitrust statute in 1888.[29] I also compared my database to

[27]I considered any state statute outlawing restraints against trade, monopolization, horizontal price fixing, horizontal output restrictions, trustee control of corporations, anticompetitve stock purchases, refusals to deal, or some combination thereof to be a state antitrust statute. See Table 1.11 for definitions of these acts. I began my review with materials from 1860 onward to verify that no relevant statutes had been enacted prior to 1888.

[28]I exclude industry-specific antitrust statutes from my analysis because these laws were limited in scope, and the focus of this paper is to study the broad economic impacts of state antitrust laws. Moreover, applying a uniform identification strategy to statutes targeting very different industries would likely be challenging. For example, an 1897 Florida statute applied antitrust principles to the sale of beef or "other edible animal[s]," while several states, such as Georgia in 1891, passed laws restricting anticompetitive behavior in the business of insurance only. I do not include these statutes in my database of state antitrust laws. Act of June 11, 1897, Chapter 4534, 1897 Florida Acts 60. Act of October 21, 1891, Chapter 745, 1891 Georgia Acts 206.

[29]Scholars have reached different conclusions about which state laws should be considered antitrust statutes. For example, Millon (1990) considers an 1889 Kansas statute to be the United States' first antitrust statute. I consider the 1889 Kansas law to be Kansas' first antitrust statute but recognize Iowa's 1888 law directed at "the punishment of pools, trusts and conspiracies" as being first in the nation. Other scholars also conclude that Iowa was the first state to adopt an antitrust statute (Collins, 2013; Dameron, 2016). Any time I encountered an accounting of state antitrust statutes that disagreed with my own, I reviewed the differences to confirm the accuracy of my analysis. Act of March 9, 1889, Chapter 257, 1889 Kansas Acts 389. Act of April 26, 1888, Chapter 84, 1888 Iowa Acts 124.

contemporaneous compilations of state antitrust laws to ensure accuracy.[30] After identifying state antitrust statutes in effect between 1860 and 1940, I analyzed the content of states' laws. Specifically, I coded the anticompetitive acts outlawed; whether the law made violations a crime, a tort, or both; the minimum and maximum fines and prison sentences authorized; whether the law permitted both a fine and imprisonment to be imposed; any civil damages authorized; the official(s) tasked with the duty to enforce the antitrust law; any exemptions for labor unions and/or the agricultural sector; whether the statute stipulated that every day of violation constituted a separate offense; and any other penalties the statute authorized. Appendix 1.9 provides more information on how I identified and coded these laws.

To summarize the features of state antitrust laws with a single, continuous measure of stringency, I constructed an antitrust law index using Principal Component Analysis (PCA). To do this, I coded numeric measures of statutory severity for each state-by-year observation in which an antitrust statute was in force.[31] I then standardized these variables to have mean zero and unit variance to ensure comparability across measures expressed in different units. Next, I used PCA to extract the first principal component, which is the linear combination of the standardized variables that maximizes the proportion of the total variance explained. The resulting first component has positive loadings on all measures of statutory strength, indicating that higher scores reflect more comprehensive prohibitions and stronger penalties.[32] I then rescaled the first principal component to range from 0 to 1, with higher values indicating greater statutory stringency. In the final index, all state-by-year observations without an antitrust law receive an index value of 0, while state-by-year observations with antitrust laws have positive index values ranging from about 0.22 to 1 that reflect their varying levels of statutory severity.

A descriptive analysis of my state antitrust law database reveals that statutes evolved along diverse paths but frequently converged around a few key regulatory priorities. Figure 1.5 traces the enactment and subsequent trajectories of state antitrust statutes during the study period, illustrating that nearly every state amended its statute at least once, and many modified their statutes repeatedly. Although the content of states' laws varied significantly, some trends emerge. For example, as Figure 1.6 shows, state antitrust laws most commonly prohibited horizontal arrangements to fix prices or limit quantity. Such provisions were aimed squarely at dismantling cartels, which were the main target of early antitrust law. Several states also included provisions mirroring section 1 of the Sherman Act, which broadly prohibits restraints of trade, in their antitrust statutes.

[30] Sources I used to check my work include Forrest (1896), von Halle (1899), Bureau of Corporations (1915), and Works Progress Administration (1940). I also compared my compilation of state antitrust statutes to internal memoranda from the now-defunct Bureau of Corporations. The Bureau of Corporations was a government agency that existed from 1903 to 1915, had broad authority to conduct investigations and gather data on the operations of interstate corporations, and predated the Federal Trade Commission. Given the Bureau's focus on investigating large corporations and trusts, I identified a number of internal agency documents relating to state and federal antitrust laws during a visit to the National Archives. These documents helped me ensure the accuracy of my state antitrust database.

[31] Table 1.1 lists and describes the 19 variables I used to construct my antitrust law index.

[32] Table 1.1 provides the weights associated with the first principal component, which I used to construct my index.

Provisions mirroring section 2 of the Sherman Act, which condemns monopolization, were somewhat less common at the state level, particularly in the first several decades states began adopting antitrust statutes. Several states also prohibited the sale of trust certificates, which were commonly used to coordinate horizontal restraints starting in the late nineteenth century. Fewer states adopted provisions barring firms from buying their competitors' stock or refusing to do business with buyers or sellers that also transacted with a competitor. Moreover, because unions could be construed as combinations that restrained trade through the collective organization of workers, at least 13 states explicitly exempted labor unions from their antitrust statutes to prevent the laws from being used to weaken organized labor. Similarly, at least 11 states explicitly exempted the agricultural sector from their antitrust statutes to protect collective price-setting by farmers.

States' antitrust statutes also varied not only in the types of remedies they provided but also in who was empowered to enforce them. As Figure 1.7 shows, the first states that adopted antitrust laws classified violations as criminal offenses, punishable by fines or imprisonment.[33] In contrast, when antitrust violations were treated as torts, injured parties could seek compensation through private lawsuits. Civil remedies shifted the burden of enforcement away from the state and toward private actors, potentially expanding the reach of antitrust law. Perhaps for this reason, later adopters tended to include means of both civil and criminal recourse in their statutes. Further, as civil enforcement gained traction, states began to replace remedies that enabled compensation for actual harm with more punitive measures aimed at deterrence. As seen in Figure 1.8, the first crop of states that allowed parties injured by antitrust violations to sue for damages tended to limit damages to the amount of actual losses incurred. Enhanced remedies—such as double or treble damages—became more popular in the early 1900s. States also differed in who they tasked with enforcement. As shown in Figure 1.9, a small share of states did not designate enforcement to anyone. Most states designated enforcement authority jointly to the Attorney General and District Attorneys, while others relied on only one of these offices. Over time, the share of states assigning joint authority increased, while the share assigning enforcement solely to District Attorneys declined. Variation in enforcement responsibility may have shaped the vigor with which laws were applied, with states assigning responsibility to multiple offices potentially

[33]For states in which antitrust violations were criminal offenses, Figure 1.10 illustrates the fines and prison sentences authorized under state antitrust statutes between 1888 and 1940. An important caveat is that Figure 1.10 illustrates the *statutory* minima and maxima for fines and prison sentences. Systematic data on the fines and prison sentences actually imposed under state antitrust laws are not available. However, contemporaneous press coverage suggests that sizable fines were not uncommon, and that at least some individuals were sentenced to jail for antitrust violations. As Figure 1.10 shows, the median maximum fine was $5,000 among states with antitrust statutes during the study period. (According to historical inflation data, $5,000 in 1890 dollars amounts to approximately $174,981 in 2024 dollars (Federal Reserve Bank of Minneapolis, Federal Reserve Bank of Minneapolis).) The median minimum fine hovered around $0 during the study period; that is, the average state did not set a minimum fine for antitrust violations in its antitrust statute. Moreover, as Figure 1.10 shows, the median maximum prison sentence was 12 months during most years between 1888 and 1840. The year 1897 provides a notable exception; the median maximum prison sentence jumped to 24 months that year when three states (Arkansas, Indiana, and South Carolina) enacted antitrust statutes imposing 10-year maximum prison sentences. As with fines, the median minimum prison sentence also hovered around $0 during the study period, implying that the average state did not set a minimum prison sentence for antitrust violations. As shown in Figure 1.11, many statutes considered each day a party was in violation of statute to be a separate offense.

able to mount broader enforcement efforts.

Figure 1.12 plots my antitrust law index between 1888 and 1940, further underscoring the trend of increasing statutory severity over time. The solid black line traces the mean index value over the study period among states with an antitrust law in force. The steady upward movement of the mean from the late 1880s through the early 1900s reflects the initial wave of enactments and subsequent amendments that expanded prohibitions and strengthened penalties. After about 1910, the mean stabilizes, suggesting that most states had by then settled on a regulatory framework. Moreover, the wide dispersion of index values over time indicates that substantial differences in the comprehensiveness and severity of antitrust laws persisted throughout the study period. These trends are also shown by state in Figure 1.13.

1.3.2 State-by-Industry Manufacturing Outcomes

To assess the impact of state antitrust laws on competition within the manufacturing sector, I draw on state-by-industry tabulations from the census of manufactures, which I digitized in related co-authored work (Barkai et al., 2025).[34] These data cover the full scope of manufacturing activity in the United States and represent the most comprehensive data source on manufacturing for this period that is disaggregated by both geography and industry. The data span each of the 20 manufacturing censuses that took place between 1850 and 1940, allowing for a systematic examination of long-run trends in manufacturing activity across states and industries.[35] The main outcomes I examine are the number of manufacturing establishments, as well as the profits and labor utilization of these establishments. To obtain profits per state-by-industry cell, I compute

$$\pi_{ist} = Y_{ist} - M_{ist} - W_{ist} \tag{1.1}$$

where i indexes an industry, s indexes a state, and t indexes the year. Revenue, the value of materials (i.e., inputs), and the total wage bill are denoted by Y_{ict}, M_{ict}, and W_{ict}, respectively. Of course, this approach yields a measure of *accounting* profits rather than *economic* profits. Accounting profits imperfectly proxy real economic returns (Fisher and McGowan, 1983). However, the aggregated census of manufactures data I employ in this paper leave me with few feasible alternatives.

Though the census of manufactures provides valuable state-by-industry data, it is not without its limitations. One challenge associated with these data is that the census of manufactures did not include industry

[34] Establishment-level schedules from the census of manufactures are presumed lost for all years after 1880 and prior to 1929. As Vickers and Ziebarth (2019, 1705) describe, "the underlying schedules [from censuses of manufactures taken between 1880 and 1929] have been lost due to accidents such as the fire that destroyed much of the 1890 Population Census, bureaucratic neglect, or active destruction to conserve space at the National Archives." As a result, I use aggregated tabulations that appeared in published census volumes for this research. Without establishment- or firm-level data, some outcomes of interest—such as the Herfindahl-Hirschman Index of industry concentration—are not possible to compute.

[35] Figure 1.14 shows these years and the number of states included in the state-by-industry data each year.

codes until 1947, when it adopted the Standard Industrial Classification (SIC) system. Accordingly, I use the industry descriptions included in the census of manufactures tabulations to map industries to their corresponding four-digit SIC codes.[36] To do this, I use the detailed narratives explaining each four-digit SIC code in a contemporaneous version of the *Standard Industrial Classification Manual* (Technical Committee on Industrial Classification, 1945). In addition, some censuses report only the number of workers—that is, production laborers such as operatives, "hands," and wage earners—rather than the total number of employees, which would also include salaried staff like managers, superintendents, and clerks. Because comprehensive employee counts do not appear consistently across censuses, I instead rely on the number of workers throughout the analysis.

1.3.3 Trust Status

To identify and characterize major industrial combinations active around the turn of the twentieth century, I rely on *The Truth About the Trusts*, which documents the structure, capitalization, and constituent firms of trusts in the United States at the height of the merger wave (Moody, 1904).[37] This source provides detailed information on 445 trusts, which I use to infer each trust's primary industry and assign a four-digit SIC code to enable linkage with other historical economic data. In the absence of comprehensive firm-level data from the study period, I use this source to approximate industry-level concentration, the dimension most likely to determine whether state antitrust laws constituted a binding constraint.[38] *The Truth About the Trusts* has been used in major works such as those by Nutter and Einhorn (1969) and Lamoreaux (1985) and primarily draws on *Moody's Manual of Corporation Securities*, supplemented by the *Commercial & Financial Chronicle*, the *Boston News Bureau*, the *Wall Street Journal*, and other sources (Moody, 1904, xxii). Subsequent research shows that *The Truth About the Trusts* also relies on the *Manual of Statistics: Stock Exchange Handbook* (Bunting, 1971). In keeping with contemporaneous usage of the word "trust," Moody (1904, xiii) defines a

[36]As Pittman and Werden (1990) show, four-digit SIC codes often diverge from the product market definitions used in antitrust enforcement. However, detailed case-specific market definitions are rarely available across a wide range of industries, making four-digit SIC codes the most practical and consistent alternative.

[37]I digitized Table 1 in section VI, *Classified Statistics of All Trusts*, to create a firm-level dataset on major industrial trusts (Moody, 1904, 453-477). Trusts are designated as "greater industrial trusts," "lesser industrial trusts," "important industrial trusts in process of reorganiztion or readjustment," "leading franchise trusts" (which are comprised of "telephone and telegraph consolidations" and "electric light, railway, and gas companies"), "the great steam railroad groups," or "'allied independent' steam railroad systems."

[38]I also considered using the surviving schedules from the 1850–1880 censuses of manufactures (Hornbeck et al., 2024) to approximate industry-level concentration in the manufacturing sector prior to the rise of the antitrust movement, but I ultimately chose not to do so, for reasons discussed below. The surviving schedules from 1850 to 1880 provide the last nationwide, establishment-level data on U.S. manufacturing prior to 1929 because establishment-level schedules from the census of manufactures are presumed lost for all years after 1880 and before 1929. However, a key limitation of these data is the absence of firm identifiers, which prevents aggregation of establishments under common ownership. As a result, attempts to compute a Herfindahl-Hirschman Index (HHI) will likely understate true market concentration, especially in cases where many establishments are in fact owned by a single firm. While an industry with a small number of establishments correctly indicates high concentration, a diffuse structure of small establishments need not imply a competitive market if ownership is consolidated. The limitations of establishment-based HHI are reinforced by Figure 1.15, which shows that trust-affiliated industries had consistently *lower* HHI than non–trust-affiliated industries from 1850 to 1880—the opposite of what would be expected if concentration were measured accurately. This pattern suggests that the absence of firm identifiers is a material concern, motivating my decision not to measure pre-enactment concentration using these data.

trust not by its legal form but by its perceived market power; the definition encompasses any combination believed to have the power, intent, or tendency to monopolize, restrict trade, or influence prices.

Figure 1.16 presents the distribution of capital and establishments by sector for the trusts cataloged in Moody (1904). Panel A, which covers all 445 trusts listed in Moody (1904), highlights the dominance of transportation and utilities in terms of aggregate capitalization. These sectors absorbed a disproportionate share of total investment during the trust era, reflecting both their high fixed costs and the tendency for consolidation to occur where natural monopoly conditions prevailed. Panel B hones in on the 305 entities designated by Moody as "industrial" trusts. Industrial trusts were heavily concentrated in manufacturing, which motivates my decision to focus on that sector throughout the analysis. The magnitude of these trusts' economic footprint is remarkable. In 1904, four-digit SIC industries that included at least one trust accounted for approximately 59 percent of total manufacturing revenue. This fact highlights the outsized role that industrial trusts played in the early twentieth-century economy.

1.3.4 Stock Returns

To understand how the stock market reacted to the passage of state antitrust laws, I use data on daily stock returns to the Dow Jones composite portfolio (Schwert, 1990). The dataset covers 1885 to 1962 and reports a daily time series over this period, capturing average returns, capital gains, and dividend yields for the composite portfolio. The index is composed of 12 to 50 industrial and transportation stocks. While the index's composition changed over time, it consistently aimed to include the most prominent stocks in terms of trading volume and market capitalization. The resulting data thus enable analysis of market-level reactions to changes in antitrust policy. Because stock prices reflect investors' expectations about firms' future earnings, stock market responses to the passage of state antitrust statutes can provide insight into whether these laws were anticipated to constrain or enhance profitability.[39] One limitation of using these data to detect changes in investors' expectations is that the index is reported only at the national level and does not identify which constituent firms had significant operations in states enacting antitrust laws. Nevertheless, the index allows for inference about how investors, in the aggregate, perceived the broader economic implications of state antitrust legislation.

1.3.5 Patents

To identify the effects of state antitrust laws on innovation, I draw on data from the Comprehensive Universe of U.S. Patents (CUSP) (Berkes, 2018). The CUSP database contains the full population of patents issued by the U.S. Patent and Trademark Office (USPTO) between 1836 and 2016, enabling consistent measurement

[39] As discussed in Section 1.2.2, a number of previous studies have documented negative stock market reactions to the implementation of antitrust policy(e.g., Prager, 1992; Baker et al., 2023).

of inventive activity over nearly two centuries. This long time horizon makes it possible to observe both the pre- and post-enactment periods of state antitrust laws and to analyze the evolution of innovation at geographically disaggregated levels. The dataset provides a rich set of variables for each patent, including issue and filing years, technology classifications, backward and forward citations, and detailed information on inventors and assignees such as names and geocoded addresses. Berkes (2018) constructed the database by merging records from multiple sources, including digitized USPTO files, state-level archives, and optical character recognition of historical patent documents. The resulting dataset has been extensively validated through comparisons with official USPTO statistics and other widely used historical sources, such as HistPat (Petralia et al., 2016).

1.3.6 Newspaper Articles on State Antitrust Topics

I assembled a collection of 118,560 historical newspaper articles that mention one or more state antitrust laws during the study period.[40] To identify relevant articles, I implemented a search strategy designed to cast a wide net while still targeting content explicitly related to state antitrust laws. Specifically, I searched for articles that (1) contain either `antitrust` or `anti-trust`, (2) contain a state name or the title of a named state antitrust law (e.g., `cartwright` in reference to California's Cartwright Act, `donnelly` in reference to New York's Donnelly Act, or `valentine` in reference to Ohio's Valentine Act), and (3) were published between 1850 and 1940. The resulting corpus spans a wide range of geographies, capturing both local reporting in small-town papers and coverage in major metropolitan dailies.[41] It also complements the statutory and economic data I use by revealing how legal changes were communicated to and interpreted by the public.

Figure 1.17 plots the number of articles in my dataset by year of publication and by the treatment status of the state mentioned in the article.[42] The figure corroborates that the collection of newspaper articles I assembled generally reflects discussion of state antitrust laws. Because never-treated states never passed a

[40]These articles come from two digital newspaper archives. The first is the *Chronicling America* project established by the Library of Congress, and the second is the ProQuest Historical Newspapers database. Together, these sources' rich holdings comprise over 73 million digitized pages. *Chronicling America* includes about 21 million pages from 4,051 newspapers, many of which are small-town publications (Library of Congress, 2024). ProQuest, meanwhile, includes about 52 million pages from 39 major newspapers (ProQuest, 2024). Just three newspapers appear in both datasets; I remove duplicate articles from these newspapers to ensure no publication is over-represented. Combining articles from these two databases provides comprehensive coverage of newspaper media during the study period. It is also worth noting that *Chronicling America* provides Optical Character Recognition (OCR) text at the page level rather than for individual articles, while ProQuest provides article-level data but only makes OCR text available for an "abstract" consisting of roughly the first paragraph of each article. Importantly, these differences in OCR structure do not affect my ability to produce accurate article counts across time and space.

[41]Geographic coverage is not perfectly even across states and years, reflecting the holdings and digitization priorities of the two archives. *Chronicling America* has more extensive coverage in some states, while ProQuest emphasizes large urban centers. Moreover, as with all keyword-based searches, some relevant articles may be missed if they use unexpected terminology, and some irrelevant articles may be included if keywords appear in unrelated contexts.

[42]Figure 1.18 provides several examples of articles that appear in my dataset. As the examples illustrate, newspaper coverage of state antitrust laws ranged from brief notices announcing newly passed statutes, to opinion pieces lamenting weak enforcement, to detailed accounts of court cases, trials, and other enforcement actions.

statute, journalists in those states had fewer reasons to cover antitrust-related topics than their counterparts in states that eventually adopted such laws.[43] Accordingly, these states have very few mentions in the collection of newspaper articles I assembled, particularly relative to ever-treated states. These articles allow me to validate that the passage of a statute was publicly visible—that is, people in treated states were likely aware of the law's existence. Beyond documenting visibility, these articles also provide a window into contemporary enforcement activity, political debate, and public attitudes toward state antitrust laws.

1.3.7 Controls

Throughout the analysis, I control for the population, literacy rate, median occupational score, and personal income in state s and year t to account for potential confounding demographic and economic conditions. I constructed the first three variables from full-count U.S. population census microdata from 1850 to 1940 (Ruggles et al., 2021), and I created a composite series on state personal income drawn from four sources (Easterlin, 1957, 1960; Bureau of Economic Analysis, 2025; Klein, 2009).[44] Because the control variables are observed in years that do not always align with the census of manufactures, I linearly interpolate values between data points to generate consistent annual controls. Population captures local market size, literacy proxies human capital, and occupational score—a widely used metric in historical research—serves as a proxy for socioeconomic status in the absence of individual income data. However, occupational score may reflect social prestige or long-run status more than current economic conditions. Thus, to more directly account for contemporaneous state-level economic performance, I also control for personal income.

In robustness checks, I control for a number of additional variables to rule out alternative explanations for the results I document. The first of these is the share of employment in agriculture, which I compute from full-count U.S. population census microdata (Ruggles et al., 2021). Given the agrarian roots of the antitrust movement, states with larger agricultural sectors likely faced stronger pressure to enact and enforce antitrust laws; controlling for agricultural employment share helps account for these underlying economic and political dynamics. Second, information I collected on whether the state's constitution included an anti-monopoly provision helps to fully account for the legal environment vis-à-vis competition policy during the study period.[45] These provisions were short (63 words on average) and typically quite general in nature. Most frequently, these provisions would simply direct the state legislature to pass laws to regulate monopolies. They would

[43] Articles mentioning antitrust law in never-treated states may include, for example, editorials calling on the state's legislature to pass an antitrust statute. Some false positives are likely also included in the dataset and may reflect mentions of *federal* antitrust enforcement within never-treated states (and ever-treated states). To address this issue, I attempt to identify and remove articles containing words and names (such as the names of U.S. attorneys general) that likely reflected federal actions.

[44] Figure 1.19 shows how I pieced these sources together over time.

[45] I searched a database of historical state constitutions for keywords related to competition policy to identify all anti-monopoly provisions in state constitutions from 1776 (when the first such provision was ratified) through 1940. Figure 1.20 illustrates states' adoption of anti-monopoly constitutional provisions. Figure 1.21 illustrates the contents of states' anti-monopoly constitutional provisions, where each type of provision is defined as in Table 1.2.

also commonly decry the existence of monopolies or declare combinations, trusts, and/or other restraints of trade to be illegal. These provisions made restraints of trade neither criminal nor tortious, and designating a state officer, such as the Attorney General, to enforce a constitutional provision against monopolies, was rare. Third, using data from Barkai et al. (2025), I control for antitrust cases initiated by the U.S. Department of Justice during the study period. These data capture contemporaneous federal enforcement activity, which could influence manufacturing outcomes either directly—by targeting manufacturing firms—or indirectly. Including these controls increases confidence that my estimates isolate the effects of state antitrust laws from other factors that could confound the results.

1.4 Identification Strategy

The following two-way fixed effects model provides a framework for identifying the effects of state antitrust laws on manufacturing outcomes. Specifically, I base my analysis on:

$$y_{ist} = \beta_0 + \beta_1(statelaw_s \times post_{st}) + \Gamma X_{st} + \delta_{is} + \delta_{it} + \varepsilon_{ist} \tag{1.2}$$

where i denotes an industry, s denotes a state, and t denotes the year. The model includes state-by-industry fixed effects δ_{is}, industry-by-year fixed effects δ_{it}, and a vector of time-varying state-level controls X_{st} in most specifications. The variable y_{ist} denotes the outcome, $statelaw_s$ indicates whether state s adopted an antitrust law, and $post_{st}$ indicates the time period after which a state antitrust law was adopted.[46] Because no antitrust statutes existed in the United States prior to 1888, this approach allows for comparison to a period without any antitrust statutes, which papers studying antitrust in modern settings are unable to do.[47] Throughout the analysis, I cluster standard errors by state to correct for serial correlation and because treatment occurs at the state level. Because the timing of effects is of interest, I also employ event study models in this paper. The two-way fixed effects version of the event study model I estimate is given by:

$$y_{ist} = \sum_{j \neq 0} \beta_j 1[t = j] \times statelaw_s + \Gamma X_{st} + \delta_{is} + \delta_{it} + \varepsilon_{ist} \tag{1.3}$$

where variables are defined as above.

Recent research in econometrics has documented that two-way fixed effects specifications may produce biased estimates of average treatment effects in difference-in-differences and event study models when treat-

[46] As illustrated by Figure 1.5, several states' antitrust laws were overturned by a court ruling or repealed by a legislative act during the period I study. Several state legislatures also repealed their states' antitrust laws during the study period. To avoid treating these states in the same manner as states that never adopted any antitrust statute, I exclude states whose antitrust laws were overturned or repealed from my analysis.

[47] Though Iowa became the first state to adopt an antitrust statute in 1888, courts had long declined to enforce contracts in restraint of trade prior by 1888. Antitrust statutes supplemented these common law principles by making restraints against trade crimes (and often torts as well).

ment timing is staggered and treatment effects may be heterogeneous across groups or over time (de Chaisemartin and D'Haultfœuille, 2020; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Goodman-Bacon, 2021; Borusyak et al., 2024). Given that states adopted antitrust statutes at different times, this concern is relevant in the present context. As a result, I estimate average treatment effects using a stacked difference-in-differences design that accommodates variation in treatment timing across states (Gormley and Matsa, 2011; Deshpande and Li, 2019; Cengiz et al., 2019; Baker et al., 2022). This approach mitigates the bias from negative weighting and treatment effect heterogeneity, yielding more credible estimates of the impact of state antitrust laws. Two-way fixed effects estimates produce similar results.

To implement this methodology, I organize my data into stacks based on year-of-enactment cohort. That is, each stack consists of states that adopted their antitrust statute at the same time. For any given stack, both never-treated and not-yet-treated states serve as controls. I then append the stacks together and estimate the following equation:

$$y_{aist} = \beta_0 + \beta_1(statelaw_s \times post_{st}) + \Gamma X_{st} + \delta_{as} + \delta_{at} + \delta_{is} + \delta_{it} + \varepsilon_{aist} \tag{1.4}$$

where a denotes a stack and other variables are defined as above. Equation 1.4 includes several sets of fixed effects to account for time-invariant and time-varying unobserved heterogeneity. The stack-by-state fixed effects δ_{as} control for permanent differences across states within a given stack, such as differences in state-level economic structure, political preferences, legal traditions, or enforcement capacity that persist over time. The stack-by-year fixed effects δ_{at} absorb year-specific shocks that are common to all units within a stack, such as macroeconomic conditions. The state-by-industry fixed effects δ_{is} account for persistent differences in industry structure or other industry-specific economic characteristics across states, while the industry-by-year fixed effects δ_{it} control for time-varying shocks that affect specific industries nationally, such as technological change or national demand fluctuations. Together, these fixed effects allow me to isolate treatment effects by leveraging within-stack variation while controlling for a rich set of potential confounders. Throughout the paper, I weight observations by both the inverse frequency of the census of manufactures and revenue. Specifically, observations from decennial censuses receive a weight of 1, those from quinquennial censuses receive a weight of 0.5, and those from biennial censuses receive a weight of 0.2. I then weight observations by their share of manufacturing revenue in the observed census year.[48]

The key identification assumption underlying the difference-in-differences models I employ is that, in the absence of treatment, treated state-by-industry cells would have followed similar trends as control state-by-industry cells within the same industry and year. This assumption is made conditional on state-by-industry

[48] This period saw rapid economic growth, so weighting by raw or inflation-adjusted revenue disproportionately emphasizes later years. To mitigate this, I weight each observation by its share of total revenue in its census year.

and industry-by-year fixed effects. The assumption is most credible if unobserved shocks affecting treated and control units within the same industry evolve similarly over time. While this assumption is fundamentally untestable, I provide support for its plausibility in several ways. First, I present event study estimates to assess pre-trends. Second, I show that observable characteristics evolve similarly across treated and control states prior to treatment.[49] Finally, I test the sensitivity of my estimates to alternative specifications and control variables. Together, these strategies increase confidence in the plausibility of the parallel trends assumption.

My identification strategy has several limitations that merit discussion. First, manufacturing data are available only intermittently (every two, five, or ten years), which may obscure treatment effects that unfold rapidly following a law's adoption. Second, treatment effects may not be contained within state borders. If firms respond to antitrust laws by relocating or adjusting their behavior in untreated neighboring states, these spillovers could potentially bias estimates towards zero. Third, although my design controls for rich sets of fixed effects and observable covariates, I cannot rule out the possibility of selection into treatment. States that adopted antitrust laws may have differed systematically from those that did not in unobservable ways that also affect manufacturing outcomes. Finally, while this historical setting provides a rare opportunity to study the origins of antitrust policy, the economic principles, case law, and politics of antitrust policy have changed significantly since the late nineteenth century. As a result, my findings may not generalize to modern contexts in which antitrust law is embedded in a more complex regulatory environment.

1.5 Results

The initial enactment and ongoing enforcement of state antitrust laws generated noticeable coverage in contemporaneous newspapers, providing evidence that these statutes were publicly salient and known to economic actors. Figure 1.22 presents an event study estimating the effect of state antitrust law enactments on the number of newspaper articles discussing state antitrust laws. In the years leading up to enactment, the estimates remain close to zero until roughly five years prior, when mentions begin to rise. This pre-enactment uptick suggests that public discourse may have anticipated the introduction and passage of an antitrust statute. Following enactment, there is a sharp and sustained increase in antitrust mentions, suggesting that these laws received meaningful public attention. This pattern corroborates the idea that individuals and firms were aware of the new statutes and that their economic behavior could plausibly have been influenced in response.

The widespread visibility and public awareness of state antitrust laws raise the question of whether they produced measurable effects on markets. I find that they did, but that their impact was to entrench rather

[49]Table 1.3 compares pre-treatment characteristics in 1880—the last census before the first state antitrust law was adopted—between states that ever adopted an antitrust law and those that never did. Table 1.4 compares early- and late-adopting states. These comparisons reveal that treated and control states were broadly similar across a range of demographic, economic, and political characteristics prior to any state adopting an antitrust statute. Although a few differences are statistically significant, most variables show no significant differences. This balance provides suggestive evidence that treated and control states would have followed similar trends in the absence of treatment.

than erode industrial interests. In particular, I find that these laws induced growth in the manufacturing sector by increasing the number of manufacturing establishments, but, as described in further detail in the next paragraph, these new establishments most likely reflected expanded operations by incumbent firms rather than new entrants. As shown in column (4) of Table 1.5, the number of establishments rose by approximately 10 percent following the adoption of these laws. Notably, this expansion was concentrated in industries that did not have a trust. In these non-trust industries, the number of establishments increased by roughly 16 percent, while trust-affiliated industries saw no comparable growth. This pattern suggests that state antitrust laws did not curtail trusts' market power enough to allow new entrants to emerge in industries dominated by trusts, but they did foster growth in less concentrated industries. Event study estimates presented in Panel A of Figure 1.23 reveal that the establishment growth was relatively steady but intensified over time, indicating that the effects of these laws accumulated gradually rather than generating immediate shifts.

Although it is not fully clear whether the observed growth in manufacturing establishments reflects new firm entry or the expansion of existing firms, available evidence points toward the latter. To explore this, I examine changes in ownership patterns—specifically, the number of proprietors per county-by-industry cell. Unfortunately, this variable is only available in a limited set of census years.[50] If incumbent firms expanded by opening new establishments without taking on additional owners, we would expect a decline in the average number of proprietors per establishment. Consistent with this possibility, the estimated effect of antitrust law adoption on the average number of proprietors per establishment is negative across all specifications in Table 1.5, though statistically insignificant. However, more pronounced effects appear when the sample is restricted to industries without a trust. As shown in Table 1.6, the number of proprietors per establishment declined by roughly 17 percent in these industries—a magnitude similar to the 16 percent increase in the number of establishments. This pattern suggests that, in sectors not dominated by trusts, incumbent firms may have responded to antitrust statutes by expanding their operational footprints through internal growth, thereby reducing the proprietors-to-establishment ratio. Taken together, these findings offer suggestive evidence that the expansion in non-trust industries was driven not by *de novo* entry but by the extensive-margin expansion of existing firms. However, this interpretation should be viewed with caution given the limited temporal coverage of the ownership data.

Although the number of establishments increased, I find no significant effects on profits or employment, suggesting that incumbent firms were not forced to materially alter their pricing or production strategies in response to state antitrust laws. These null results hold across the full sample (see Panel B and Panel D in Table 1.5) as well as when disaggregated by trust status (see Panel B and Panel D in Table 1.6). Event

[50] The relatively small number of observations in panel C of Table 1.5 reflects the fact that the census of manufactures did not include questions about ownership in all years. Data on the number of proprietors are available in 1900, 1904, 1909, 1914, 1919, 1921, 1923, and 1939. By 1900, most states that would adopt an antitrust law had already done so, so this analysis is underpowered.

studies are also consistently flat (see Panel B in Figure 1.23 and Panel A in in Figure 1.24). The stability of profits and employment across specifications indicates that real output likely remained unchanged following the adoption of these laws. This, in turn, implies—at least at the aggregate level examined here—that markets did not transition from cartelized to more competitive structures as a result of new antitrust laws. Firms did not hire more workers to meet growing production needs, nor did they experience meaningful declines in profitability.[51] These patterns are difficult to reconcile with a strong competitive shock and instead suggest that firms with market power adapted to the new legal environment without meaningfully changing their behavior. Rather than eroding rents, the laws may have formalized certain boundaries on overtly collusive behavior while allowing existing structures of market power to persist.

Another notable and robust finding is the decline in labor share following the adoption of state antitrust laws.[52] As shown in Panel F of Table 1.5, the labor share decreased by an average of 4 percent post-enactment. This decline runs counter to the prediction that increased competition would raise labor's share of income. Figure 1.25 reveals that this decline is driven by increases in revenue outpacing increases in total compensation. Importantly, this pattern is driven by trust-affiliated industries, as demonstrated in Panel F of Table 1.6 The fact that revenues grew faster than compensation indicates that capital, rather than labor, captured the gains from expansion. This result calls the effectiveness of early state antitrust laws into question and suggests that the net effect of state antitrust laws may have been to entrench rather than disrupt market power. Rather than reducing the market power of dominant firms, the implementation of these laws appears to have left trust-affiliated firms largely unscathed—and may even have reinforced their position. This interpretation is consistent with the broader pattern of muted responses in profits and employment and highlights the limits of state antitrust policy in curbing entrenched economic power.

A reasonable next question to ask is whether differences in enforcement shaped these outcomes. To investigate this, I use contemporaneous newspaper discussions of state antitrust policy to distinguish between high- and low-enforcement states and estimate treatment effects separately for each group relative to untreated

[51] I also find null effects for productivity and trust incorporations. Across a range of TFP estimation methods, I find no evidence that the enactment of state antitrust laws affected productivity in Figure 1.26. I also show that the enactment of state antitrust laws likely played a minimal role in the Great Merger Movement. The Great Merger Movement was a period of rapid consolidation from about 1895 to 1904, when thousands of firms were combined into large trusts and holding companies (Lamoreaux, 1985). As Figure 1.27 shows, much of this consolidation occurred when firms reincorporated in holding-company-friendly states, such as New Jersey. Figure 1.28 shows no effect on incorporations by trusts after the enactment of a state antitrust law that is statistically distinguishable from zero. This null effect may reflect trusts' expectation that state antitrust laws posed a limited threat to their profitability. Alternatively, another possible reason for the null effect is that state of incorporation typically did not constrain state antitrust jurisdiction—states could generally bring antitrust suits if a firm conducted business within their borders regardless of where the firm was incorporated.

[52] I compute the labor share as the ratio of total wages to total revenue in industry i, state s, and year t. Results are similar if I instead use value added in the denominator, but I rely on revenue because value added is negative in some cases, complicating interpretation. Results using value added in the denominator are available in Figure 1.29.

states.[53] I rely on newspaper coverage as a proxy for enforcement intensity because records of cases brought by state attorneys general and district attorneys are not consistently available, and data on private enforcement are even scarcer. While imperfect, this approach provides the most consistent and broadly comparable proxy available for gauging state-level enforcement intensity during this period. The results in Table 1.7 suggest that stronger enforcement, as measured by this proxy, did not systematically generate stronger procompetitive effects. High- and low-enforcement states exhibit only minor differences in how enactment affected the number of establishments, average profits, labor demand, average wages, and the labor share. One notable result is that the average number of proprietors per establishment decreased by about 18 percent after enactment in low-enforcement states, pointing to a possible pattern of consolidation in the ownership of manufacturing establishments in an environment where regulatory oversight did relatively little to constrain market power. Overall, however, the evidence indicates that higher levels of newspaper-documented enforcement activity did not translate into a clearer pattern of improved competitive outcomes in manufacturing.

In Table 1.8, I extend the analysis to a continuous difference-in-differences framework to see if more stringent antitrust statutes produced stronger procompetitive effects. I do so using the Borusyak et al. (2024) estimator, which accommodates staggered treatment timing and continuous treatment.[54] I start by replicating the stacked difference-in-differences estimates using the Borusyak et al. (2024) estimator in column (1). The results are broadly consistent, though the Borusyak et al. (2024) estimator produces estimates with larger magnitudes than the stacked approach for the number of establishments and labor's share of income. Column (2) provides the continuous difference-in-differences specification that allows treatment effects to vary with the stringency of state antitrust laws. In this specification, the baseline ATT represents the projected effect of enactment when the index equals zero. Because an index of zero corresponds to having no antitrust statute at all, this estimate should be interpreted as an extrapolated intercept rather than a realized effect for untreated states. Moreover the "Antitrust Index" coefficient measures the marginal change in the treatment effect for a one-unit increase in the antitrust law index, allowing me to assess whether statutory intensity affects the magnitude of estimated treatment effects.

Overall, the results in Table 1.8 do not provide consistent evidence that greater statutory stringency trans-

[53] For each year a statute was in effect, I classify whether a state's newspaper mentions per capita were above or below the annual median among treated states. Importantly, I use newspaper mentions *per capita* to avoid conflating enforcement activity with population size, since larger states naturally have more and larger newspapers that produce more articles. Using a per capita measure ensures that the proxy reflects the relative importance of enforcement activity in a way that is comparable across states of very different sizes. Next, I classify a state as high enforcement if the share of its treated years spent above the annual median is greater than or equal to the median share across states.

[54] The Borusyak et al. (2024) estimator uses an "imputation" approach: it first estimates untreated potential outcomes using only not-yet-treated units, imputes counterfactual outcomes for treated units, and then averages the resulting treatment effects. Unlike the stacked difference-in-differences approach, which restructures the panel into cohort-specific event-time datasets, the Borusyak et al. (2024) approach works directly on the full panel and provides unbiased estimates under arbitrary treatment effect heterogeneity. The estimator cannot simultaneously accommodate both state-by-industry and industry-by-year fixed effects because imputation requires sufficient untreated variation within each fixed-effect cell. In this analysis I therefore include state-by-industry fixed effects only, which are more granular than industry-by-year in my setting and capture the key heterogeneity across industries within states.

lated into stronger procompetitive effects. For two outcomes, the coefficient on statutory stringency is insignificant, while for two others it is statistically significant but carries the opposite sign of what would be expected if stronger antitrust laws enhanced competition. Specifically, the coefficient is insignificant for the number of establishments (Panel A) and average wages (Panel D). In addition, Panel B and Panel E show that more stringent antitrust laws are actually associated with *higher* profits and a *lower* labor share, respectively. Results for average employment (Panel C) tell a somewhat different story. For this outcome, greater statutory stringency is associated with greater employment per manufacturing establishment. However, because the projected constant is negative, this positive association only emerges at relatively high levels of statutory stringency; hence, the result does not reflect a broad-based procompetitive effect. Together, the continuous difference-in-differences estimates provide little evidence that more stringent statutes systematically generated stronger procompetitive outcomes.

The absence of a discernible stock market reaction to the passage of state antitrust laws further supports the conclusion that the enactment of state antitrust laws did not represent a major competitive shock.[55] As shown in Figure 1.30, average daily returns to the Dow Jones composite portfolio remained close to zero in the days surrounding passage, with confidence intervals encompassing zero throughout the event window. If these laws had been expected to substantially erode incumbent firms' market power or profitability, one would anticipate negative abnormal returns upon enactment. It is worth noting that this stock market analysis captures short-run investor responses over a matter of days, whereas the manufacturing analysis examines outcomes that unfold over decades; the two therefore reflect very different dimensions of firms' reactions to antitrust legislation. Nonetheless, the lack of such a response suggests that investors did not view the legislation as a credible threat to established market structures, a finding consistent with the muted effects on profits and employment documented above.

The main results are robust to a range of alternative specifications and weighting strategies. As shown in Table 1.9, the positive effect of antitrust laws on the number of manufacturing establishments and the negative effect on the labor share remain statistically significant across all robustness checks. These include alternative weighting schemes based on the number of establishments (column (1)) and employment (column (2)). In column (3), I also control for the share of agricultural employment given that treated states were more agricultural than untreated states. In column (4), I control for whether the state constitution contained an anti-monopoly provision to account for other features of state competition policy that could have plausibly

[55]This analysis is descriptive rather than causal and should be interpreted cautiously. If the timing of antitrust law enactment coincided with other events that affected firms' profitability, examining changes in stock prices during the week of enactment may not capture the true expected effect of the laws. Moreover, the Dow Jones composite portfolio reflects large, nationally traded firms, many of which may have had limited exposure to specific state laws. Smaller, regionally focused firms, which were more likely to be directly affected, are not included in the index. Finally, this exercise does not capture reactions that occurred earlier (in anticipation of passage) or later (during enforcement), and averaging across all events may mask heterogeneous effects across states, industries, or firm types.

influenced the results. The consistency of results across these specifications indicates that the findings are not sensitive to modeling choices or to specific features of the data.

Figure 1.31 reveals heterogeneity in the effects of state antitrust laws across manufacturing industries. Across most sectors, the number of establishments increased significantly, consistent with incumbent firms expanding their operational footprints. By contrast, industries that were historically more concentrated—such as tobacco and petroleum—exhibited little to no change in establishment counts, suggesting that dominant firms in these sectors were less affected by state antitrust laws. At the same time, the labor share declined across a broad range of industries, with particularly pronounced reductions in sectors like food products and tobacco products. These patterns suggest that while state antitrust statutes may have prompted expansion in some parts of the manufacturing sector, they largely failed to shift the distribution of industrial output.

Finally, the enactment of state antitrust laws resulted in an increase in both assignee (firm-sponsored) and inventor patents. Specifically, assignee patents increased by about 7 percent and inventor patents increased by about 5 percent in the 40 years following the enactment of an antitrust statute, as shown in Table 1.10. Event study estimates in Figure 1.32 show that both types of patents started to increase in the two years prior to enactment, potentially reflecting anticipatory effects in states that would soon be treated.[56] The statistically significant increase in assignee patents was sustained for about half a decade after enactment before tapering off slightly and then continuing to increase slowly in later years. For inventor patents, the increase emerged more gradually, with statistically significant effects not appearing until about two decades after enactment. This divergence in timing and magnitude aligns with theoretical expectations about which type of patenting should be most responsive to changes in antitrust policy. Because assignee patents are generated within firms, they are more directly tied to product market conditions than inventor patents. The stronger and earlier increase in assignee patents supports the view that state antitrust laws reinforced firms' innovative capacity, and that this innovation was concentrated among incumbent firms rather than new entrants. Together with evidence that state antitrust laws entrenched rather than curtailed dominant firms' market power, this pattern accords with the view that "bigness"—rather than intensified competition—fuels innovation, insofar as large firms are able to invest considerable resources in research and development.

1.6 Conclusion

State antitrust statutes emerged amid growing concerns about monopoly power, but their economic impact reveals a more complex legacy. In this paper, I argue that state antitrust laws did not meaningfully curtail anticompetitive conduct and in fact may have entrenched the market power of dominant firms. Though these

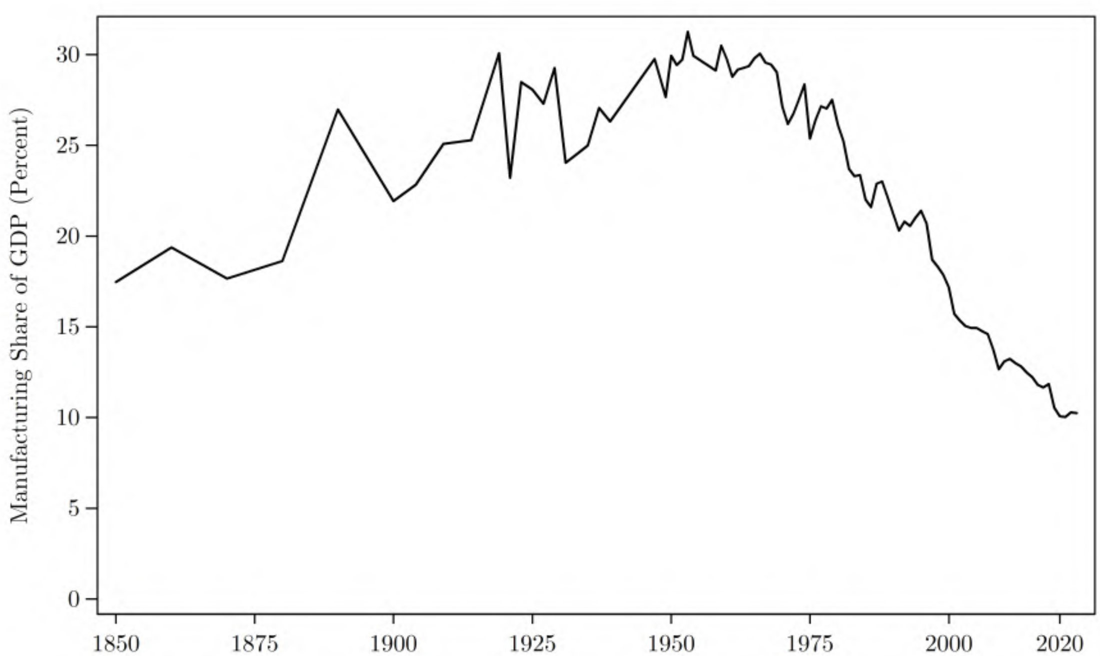
[56]This timing coincides with the increased newspaper coverage of antitrust policy I observe in states on the verge of enactment, as shown in Figure 1.22.

statutes increased the number of manufacturing establishments by about 10 percent, this growth appears to have been driven by the expansion of incumbent firms rather than new entry. Moreover, this expansion was accompanied by a decline in the labor share, especially in trust-affiliated industries, suggesting that dominant firms retained their market power and captured the gains from expanded activity. I find no evidence of increased employment or reduced profits, indicating that firms made few, if any, adjustments to their pricing or production strategies following the adoption of state antitrust laws. There is little indication that higher enforcement intensity or more stringent statutes moderated these effects. These findings are consistent with the idea that Progressive Era regulation was often promoted and shaped by large corporations themselves, rather than by reformers acting in the public interest to restrain business power (Kolko, 1963). Ultimately, the broader lesson from states' early forays into antitrust law is that legal reforms alone, absent effective enforcement and insulation from regulatory capture, are unlikely to shift markets towards competitive equilibria.

My findings also speak directly to recent critiques arguing that antitrust is "atomistic"—i.e., oriented around assessing the competitive effects of discrete acts in isolation even though market outcomes are the result of multiple acts by multiple parties (Feldman and Lemley, 2022). While these critiques focus on modern antitrust policy, the evidence I present suggests that the limitations of atomistic antitrust were evident in earlier periods as well. Though individual state cases may well have succeeded in curbing particular practices, my results suggest that state antitrust laws did not meaningfully restrain anticompetitive conduct in aggregate. Stated differently, the cumulative effect of case-by-case enforcement was insufficient to redirect industries toward competitive equilibria. These findings highlight the enduring challenge of designing antitrust regimes that can move beyond isolated acts to address the deeper structures of market power.

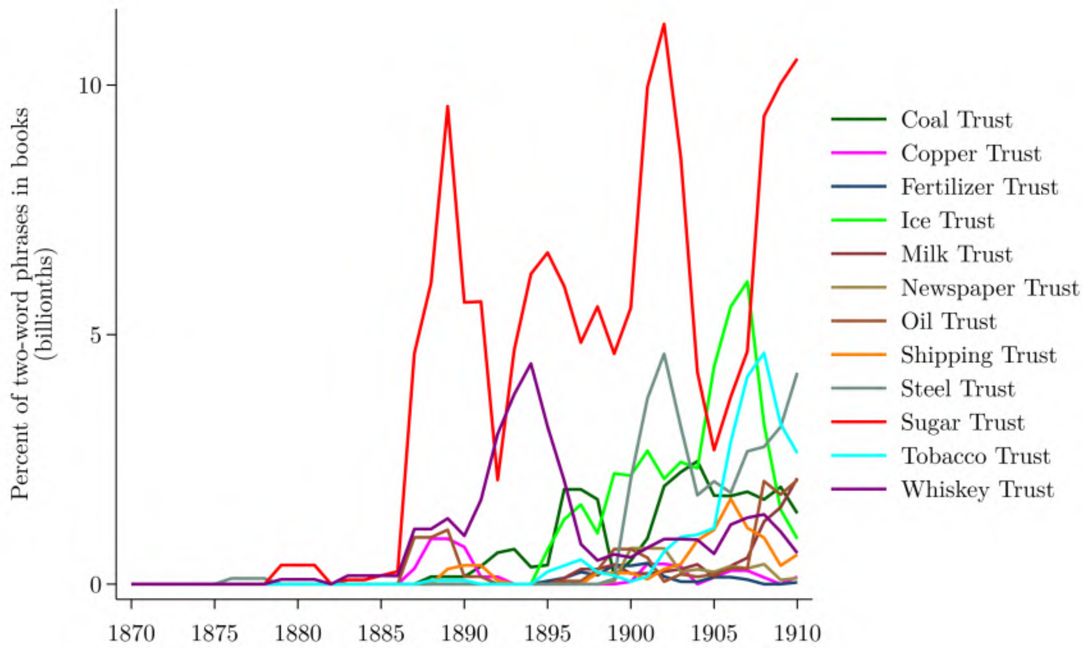
1.7 Figures

Figure 1.1: Manufacturing Share of Gross Domestic Product (GDP), 1850-2023



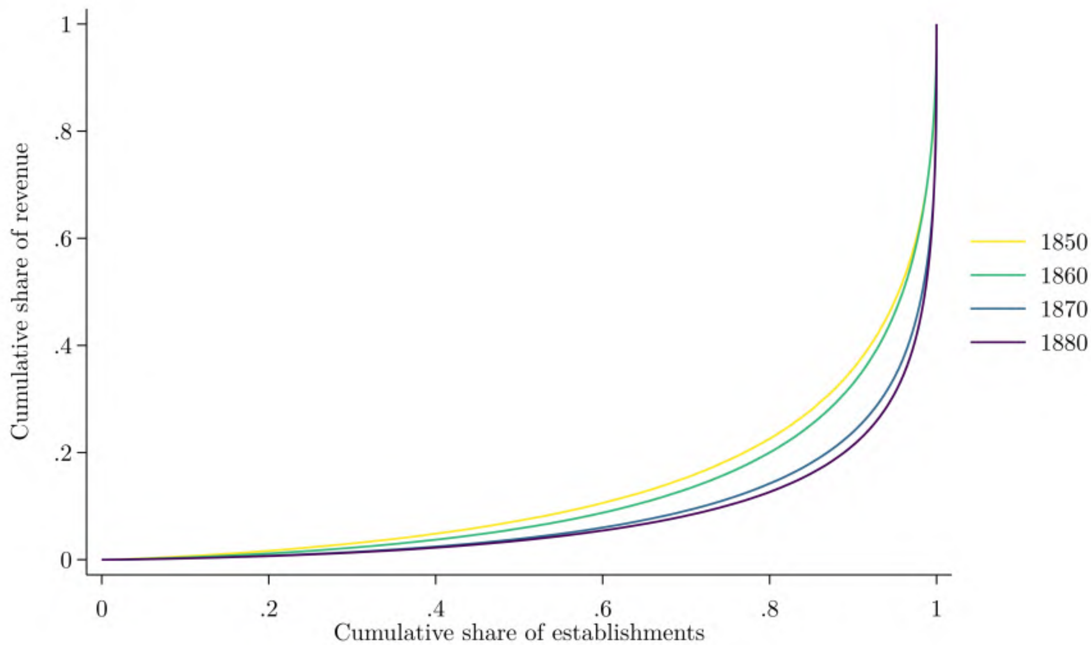
Notes: This figure plots the share of GDP in manufacturing, which is aggregate value added in the manufacturing sector over nominal GDP, for 1850 through 2023. I compiled data on value added in the manufacturing sector from several sources. The **?**, 19 provides data from past censuses of manufactures spanning 1850 through 1939 in Table 2. Between 1850 and 1890 so-called "hand and neighborhood" establishments are included in the data, but in 1900 and later years, the data exclude these smaller artisans and instead focus on larger factories. The **?**, 3 provides values for 1900 through 1954 in Table 1B. Becker et al. (2021) provide industry-level data for 1958 through 2018, which I aggregated by year. Finally, **?** provide data for 1997 through 2023. In years where these sources overlap, I averaged their values, though values generally only differed slightly. GDP data are from Johnston and Williamson (2024) for 1850 to 2022 and Federal Reserve Bank of St. Louis (2024) for 2023. As Baily and Bosworth (2014) show, trends in manufacturing's share of *real* GDP differ from trends in manufacturing's share of *nominal* GDP because the prices of manufactured goods have increased at a different rate than the prices of non-manufactured goods. However, a separate price index for manufactured goods is not currently available going as far back as 1850, so the figure must be interpreted with this caveat in mind.

Figure 1.2: Mentions of Well-Known Trusts in the *Google Books* Corpus



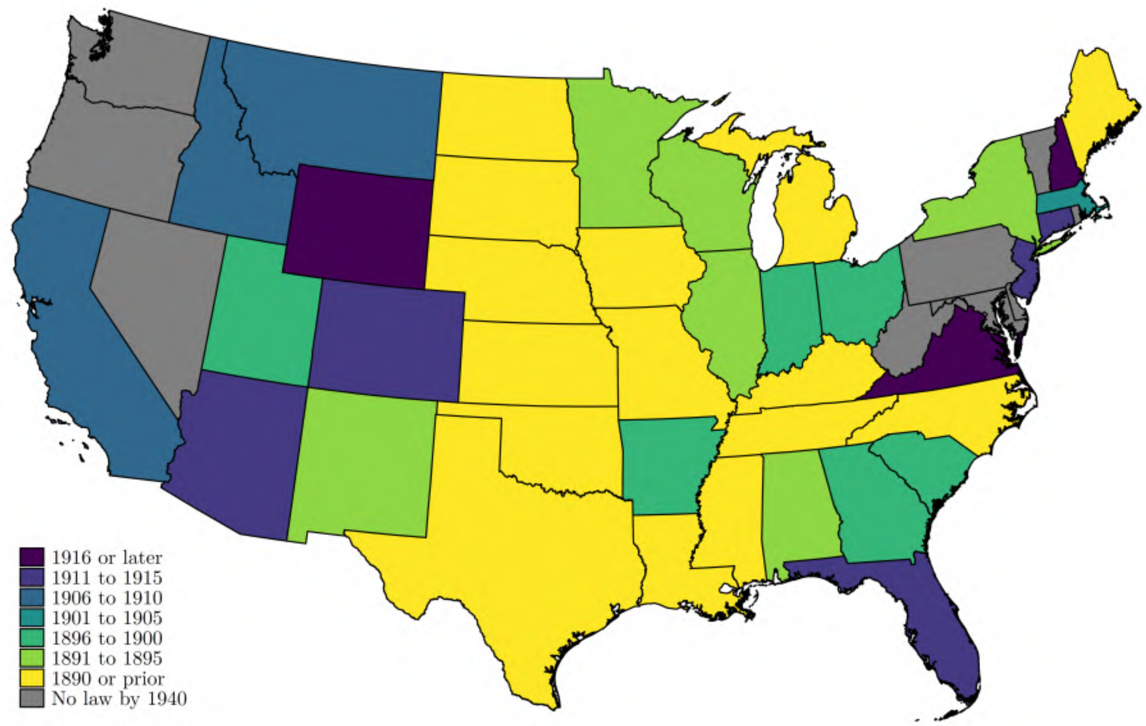
Notes: This graph shows the percentage of two-word phrases in the *Google Books* corpus equal to each of the two-word trust names listed in the legend, by publication year. I obtained the underlying data from the Google Ngram Viewer (Michel et al., 2011).

Figure 1.3: Concentration of Revenue in Manufacturing, 1850-1880



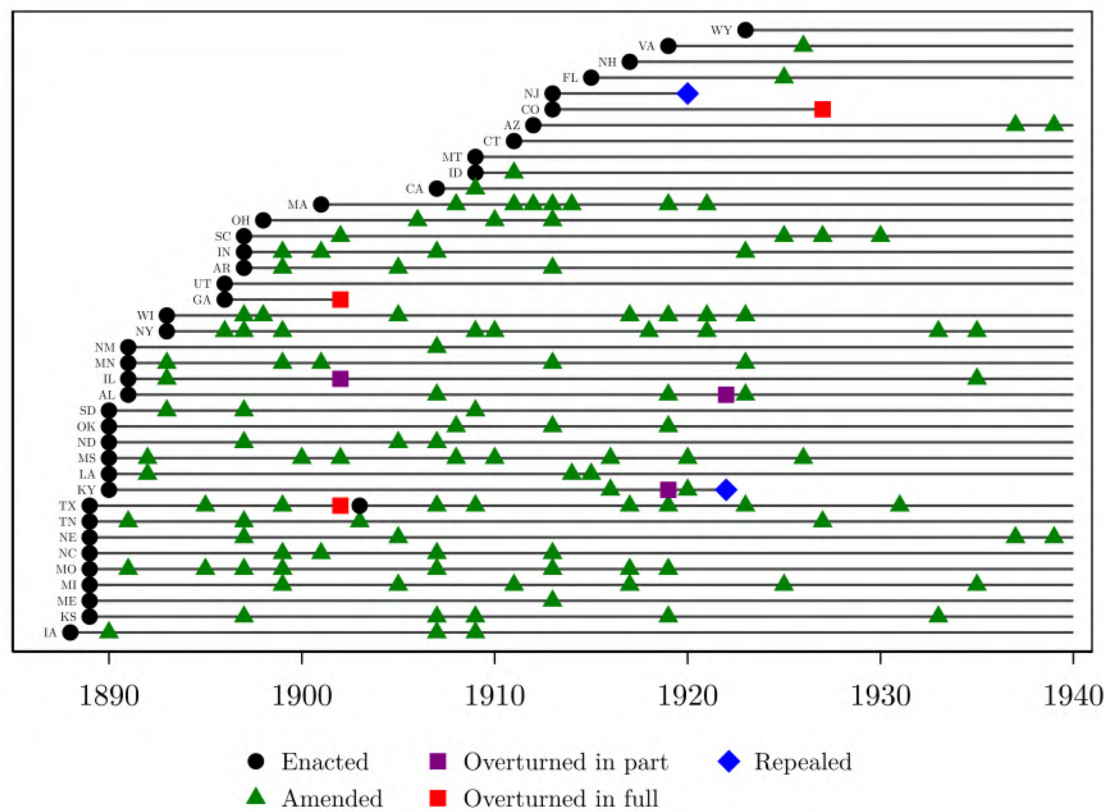
Notes: I estimated these Lorenz curves using the surviving schedules from the 1850-1880 censuses of manufactures (Hornbeck et al., 2024). Schedules for 1880 are missing from industries in which special agents were appointed to collect manufacturing data (Atack and Bateman, 1999; Delle Donne, 1973). These industries are iron and steel, cotton goods, woolen and worsted goods, silk and silk goods, chemical products and salt, coke and glass, shipbuilding, fisheries, all types of mining, coal, and petroleum (Wright, 1900, 63). For consistency over time, I removed these industries in estimating the Lorenz curves shown above.

Figure 1.4: Map of State Antitrust Law Adoption



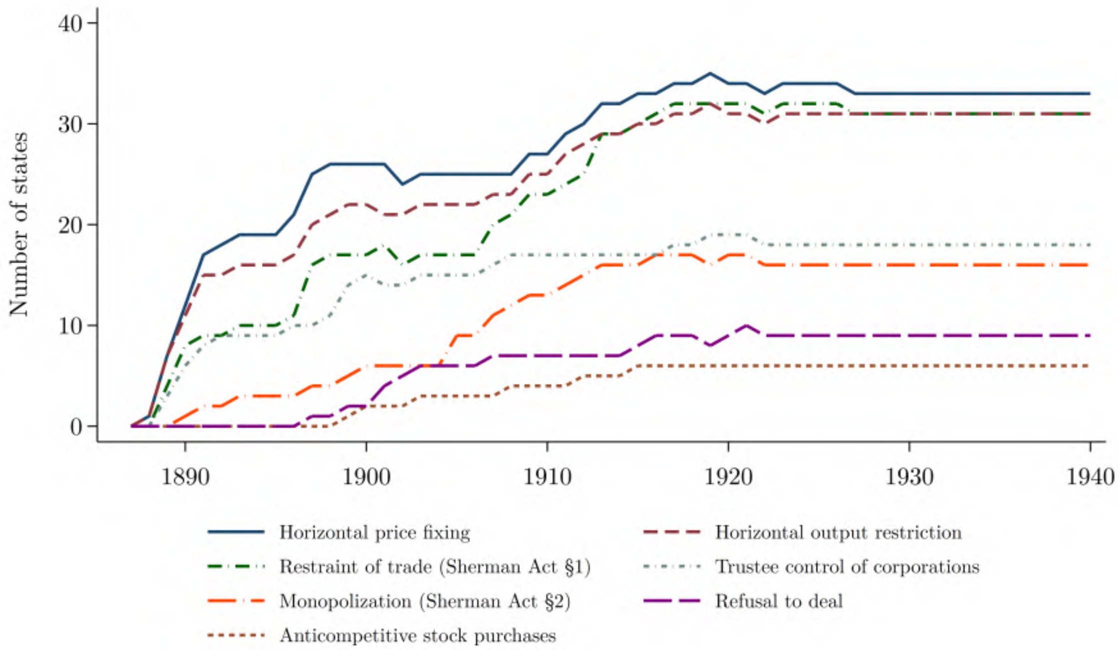
Notes: This map is shaded to show the year in which a state first adopted an antitrust law. In some cases, laws were later repealed by legislative act or overturned by court ruling, but this map does not reflect those changes. Thirteen states had already adopted an antitrust statute of their own by the time the Sherman Act—the first federal antitrust law—was enacted in 1890. These are the states shaded in yellow, excepting Oklahoma and Louisiana, which enacted antitrust statutes in 1890 after the passage of the Sherman Act on July 2, 1890. State boundaries in 1940 are shown.

Figure 1.5: State Antitrust Statutes, 1888-1940



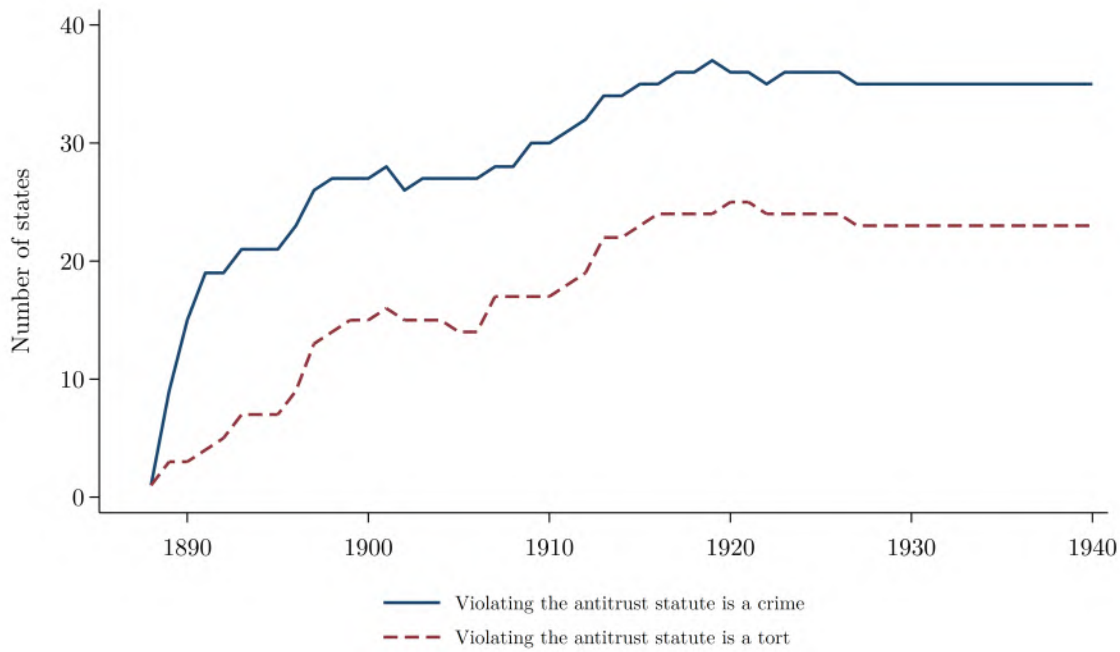
Notes: This figure illustrates the years in which state legislatures adopted and amended antitrust statutes, as well as the years in which antitrust statutes were repealed by legislative act or overturned in full or in part by a court. Delaware, Maryland, Nevada, Oregon, Pennsylvania, Rhode Island, Vermont, Washington, and West Virginia are not included in this figure because these states did not enact a general antitrust statute between 1860 and 1940. Alaska and Hawaii are not included in this figure because these states achieved statehood after 1940.

Figure 1.6: Anticompetitive Acts Made Illegal by State Antitrust Laws, 1888-1940



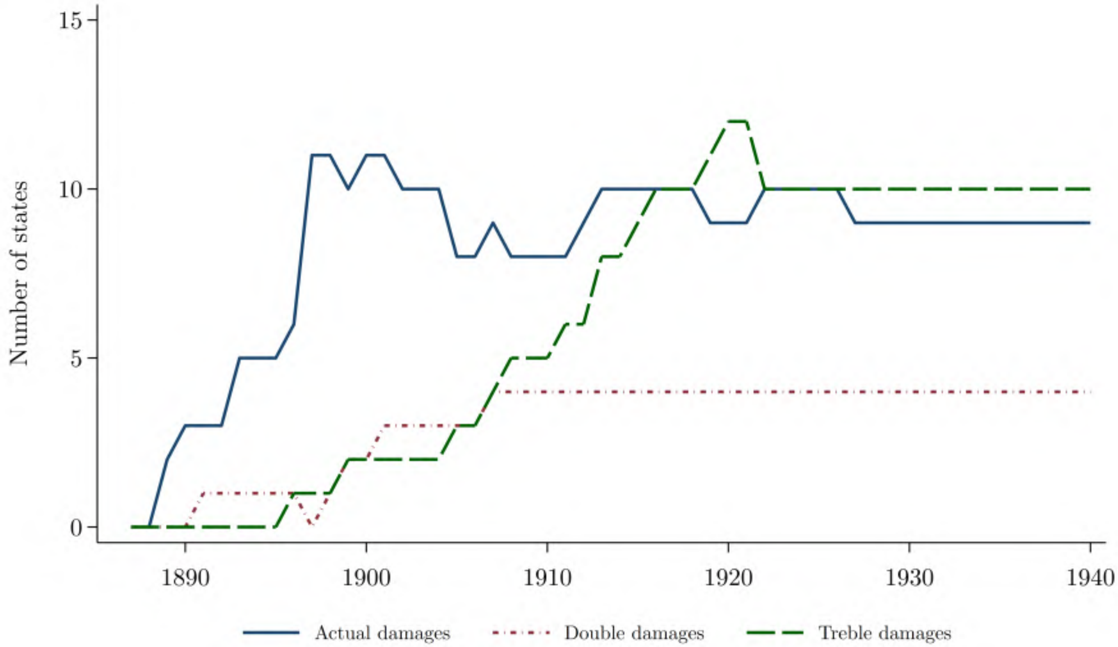
Notes: This figure illustrates the number of states that outlawed various anticompetitve acts between 1888 and 1940. See Table 1.11 for definitions of these acts.

Figure 1.7: States Declaring Antitrust Violations Crimes vs. Torts, 1888-1940



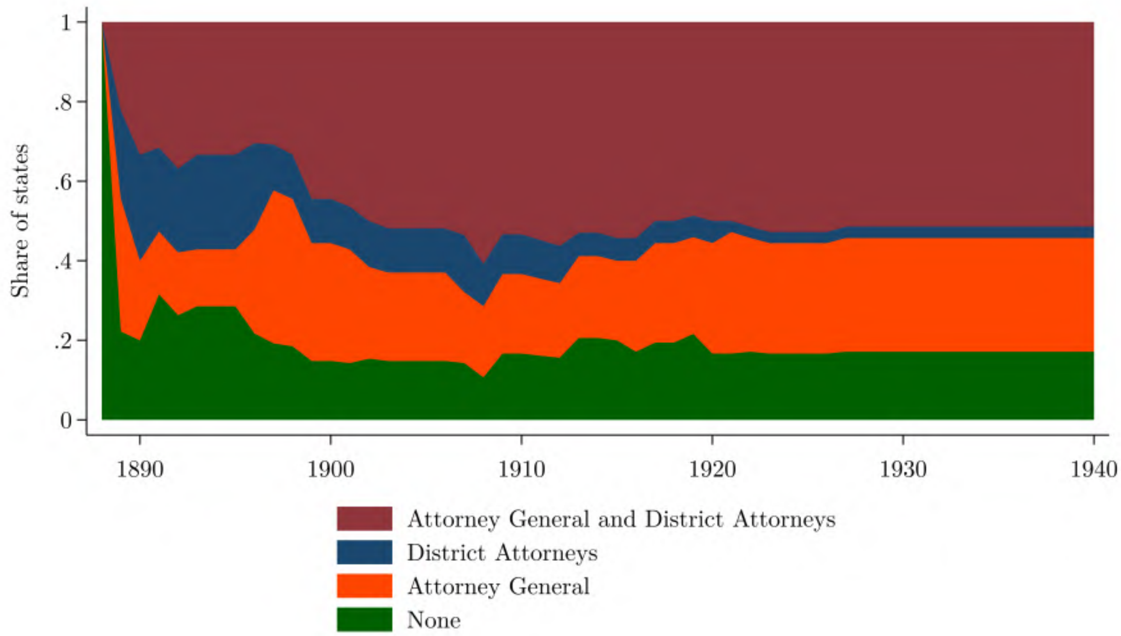
Notes: This figure illustrates the number of states that declared violations of their antitrust statute to be crimes and the number of states that declared violations of their antitrust statute to be torts between 1888 and 1940. A crime is a wrongful act that violates a law and is prosecuted by the government, while a tort is a civil wrong that causes harm to an individual, who may seek compensation through a lawsuit.

Figure 1.8: Civil Damages Under State Antitrust Laws, 1888-1940



Notes: This figure illustrates the number of states authorizing plaintiffs to seek actual damages, double damages, or treble damages in civil actions for injury due to unlawful conduct under state antitrust statutes.

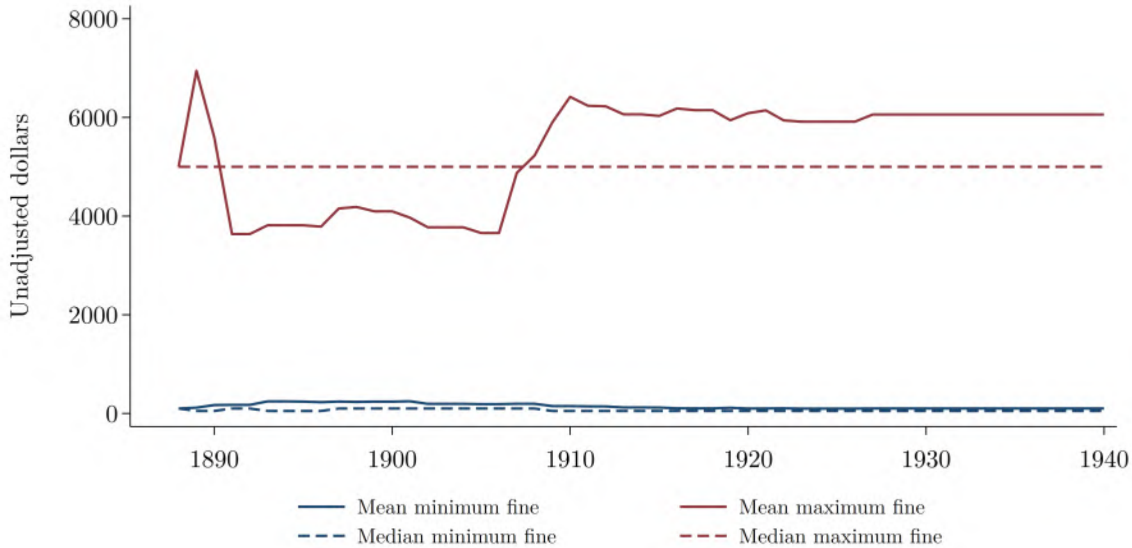
Figure 1.9: Enforcement Duties Under State Antitrust Laws, 1888-1940



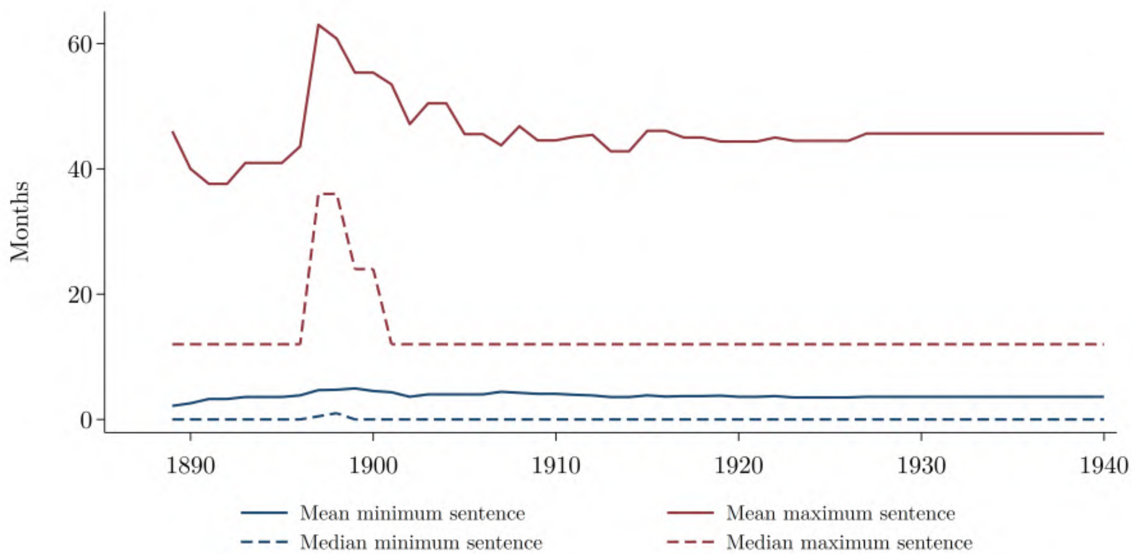
Notes: This figure illustrates the share of states designating the duty to enforce the antitrust statute to both the Attorney General and District Attorneys, District Attorneys only, the Attorney General only, or no authority at all. Shares are calculated among states with an antitrust statute in force in each year.

Figure 1.10: Fines and Prison Sentences Under State Antitrust Laws, 1888-1940

Panel A: Fines

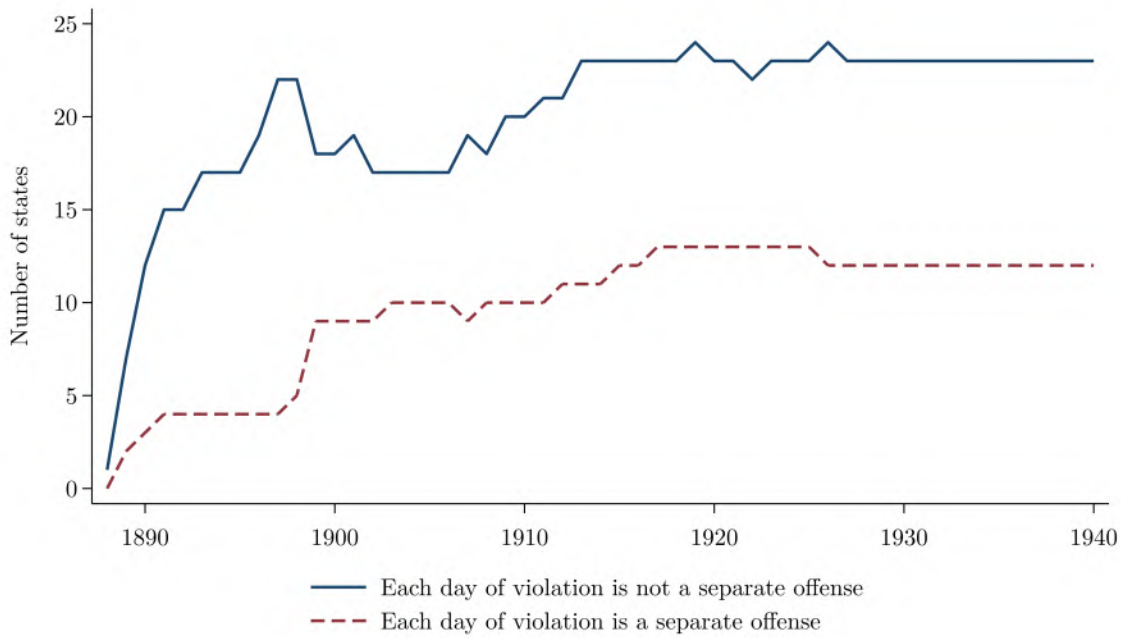


Panel B: Prison Sentences



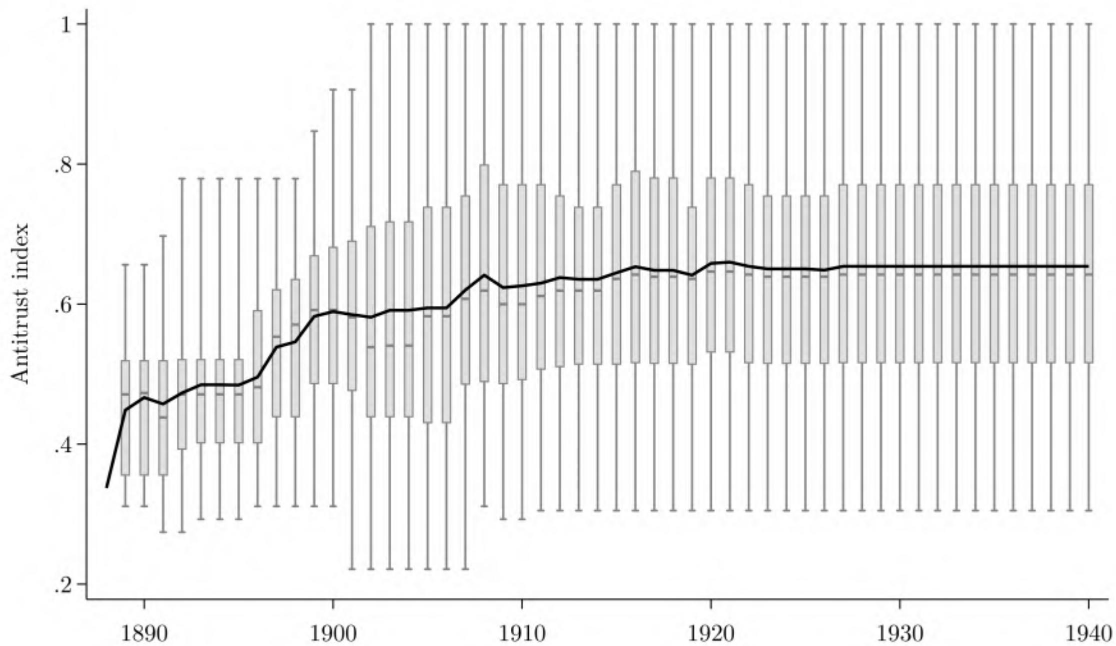
Notes: This figure illustrates the fines (Panel A) and prison sentences (Panel B) states described in their antitrust statutes. Among states with statutory minimum and maximum fines, mean and median values of these minimum and maximum fines are provided in unadjusted dollars in Panel A. Among states with statutory minimum and maximum prison sentences, mean and median values of these sentences are provided in months in Panel B.

Figure 1.11: State Antitrust Provisions Making Each Day of Violation a Separate Offense, 1888-1940



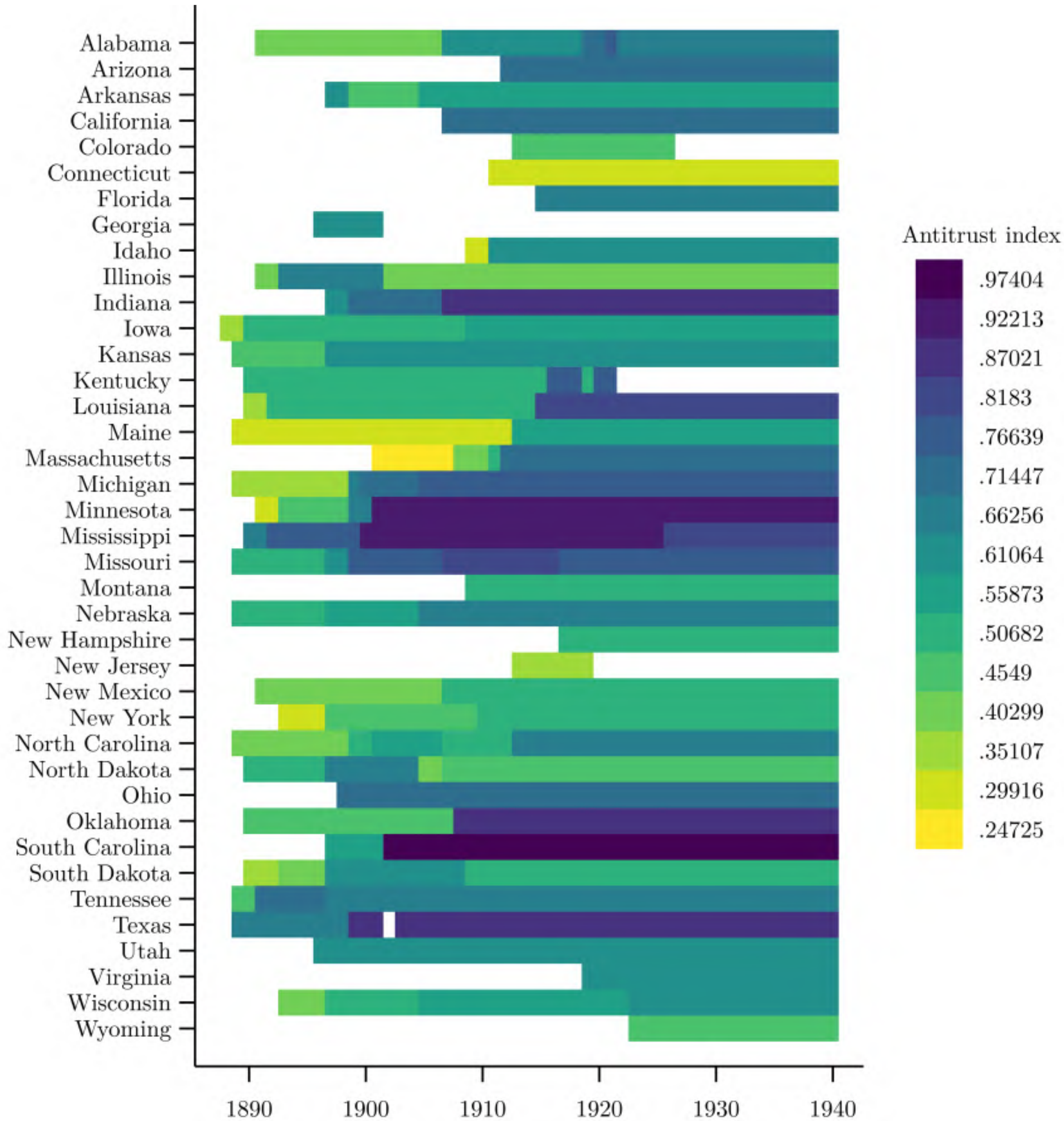
Notes: This figure illustrates the number of states with antitrust provisions specifying that each day of a continuing violation would constitute a separate offense.

Figure 1.12: Antitrust Law Index, 1888-1940



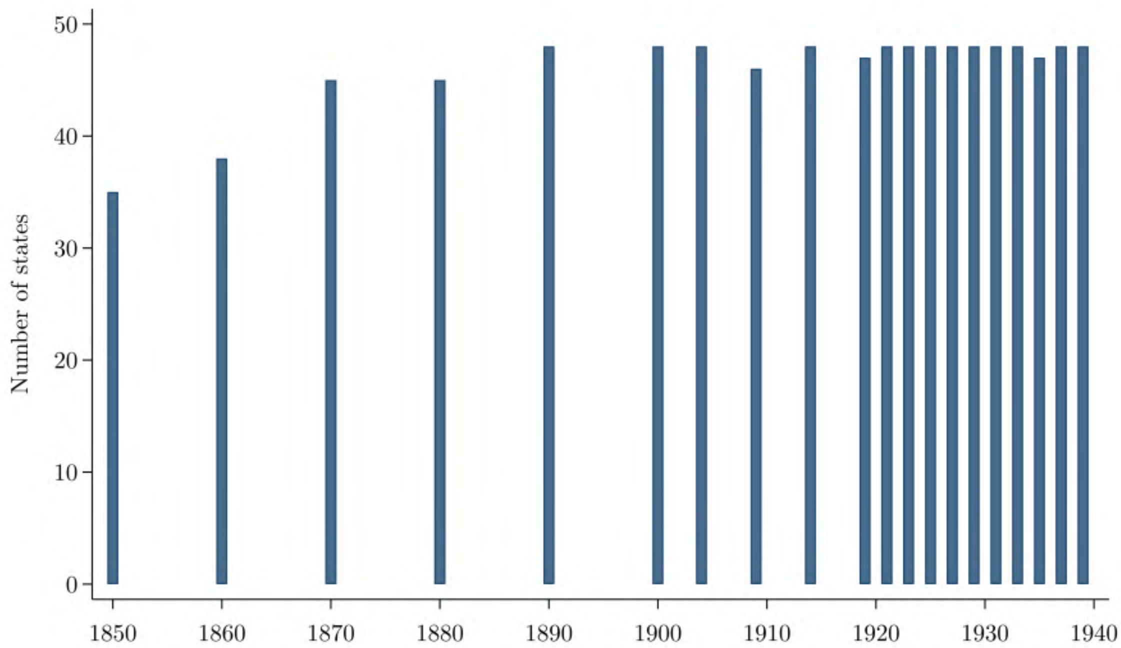
Notes: This figure illustrates the distribution of my antitrust law index over time, among states that had an antitrust statue by 1940. The index is constructed using the first principal component from a principal component analysis of statutory features, rescaled to range from 0 to 1. Section 1.3 describes the construction of the index in greater detail. Table 1.1 details the variables I used to construct my antitrust law index and the weight assigned to each one. The black line indicates the annual mean across states, while the boxplots show the distribution in each year (interquartile range in gray boxes, median denoted by a gray line within each gray box, and whiskers extending to the minimum and maximum values). The sample includes all U.S. states except Alaska and Hawaii. In 1888, only Iowa had an antitrust statute, so no distribution is shown for that year.

Figure 1.13: Antitrust Law Index by State, 1888-1940



Notes: This figure illustrates the antitrust law index over time, by state, among states that had an antitrust statue. The index is constructed using the first principal component from a principal component analysis of statutory features, rescaled to range from 0 to 1. Section 1.3 describes the construction of the index in greater detail. Table 1.1 details the variables I used to construct my antitrust law index and the weight assigned to each one. The sample includes all U.S. states except Alaska and Hawaii.

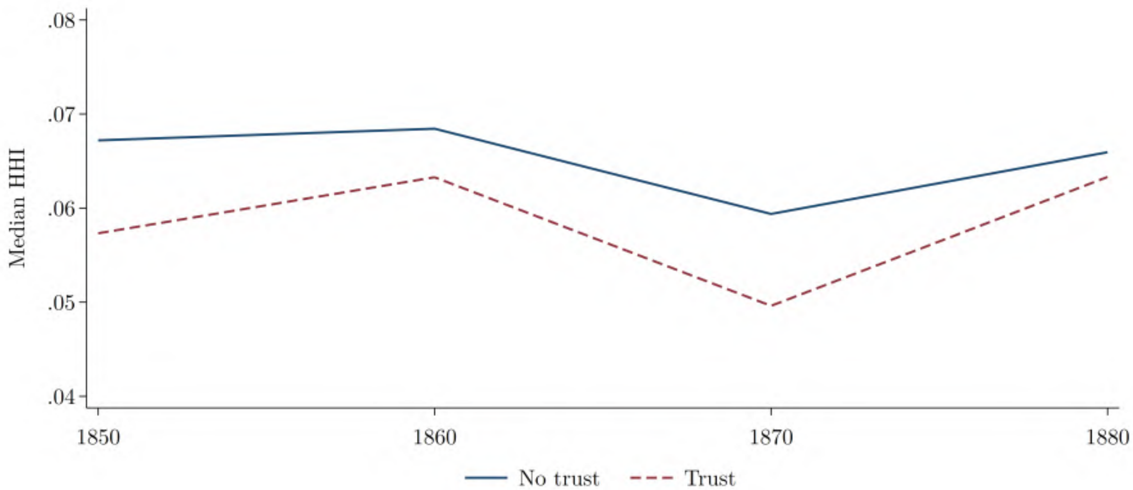
Figure 1.14: Coverage of State-by-Industry Census of Manufactures Data, 1850-1940



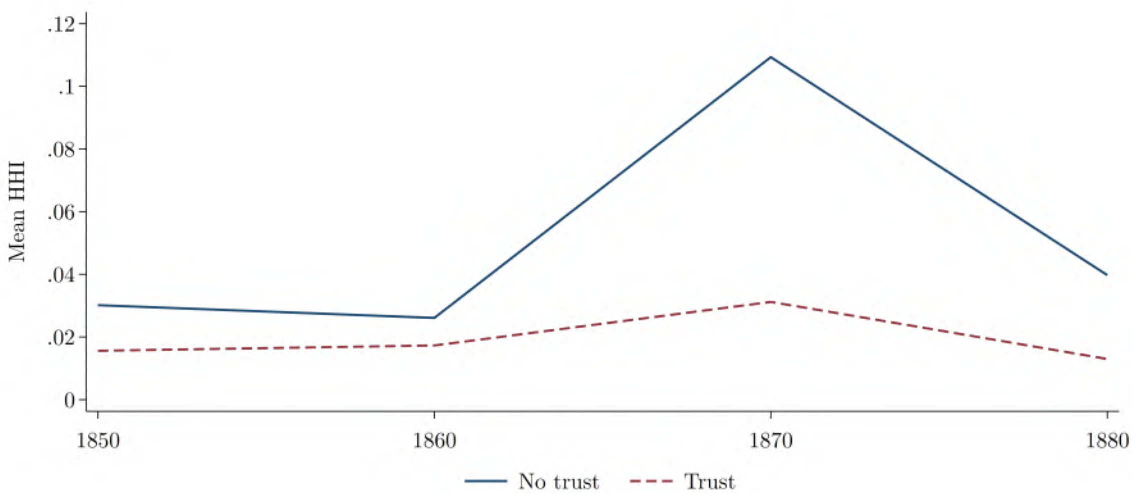
Notes: This figure illustrates the number of U.S. states included in the state-by-industry census of manufactures data from 1850 to 1940. Each bar corresponds to a census.

Figure 1.15: Concentration in Trust-Affiliated and Non-Trust Affiliated Industries, 1850-1880

Panel A: Unweighted Median HHI



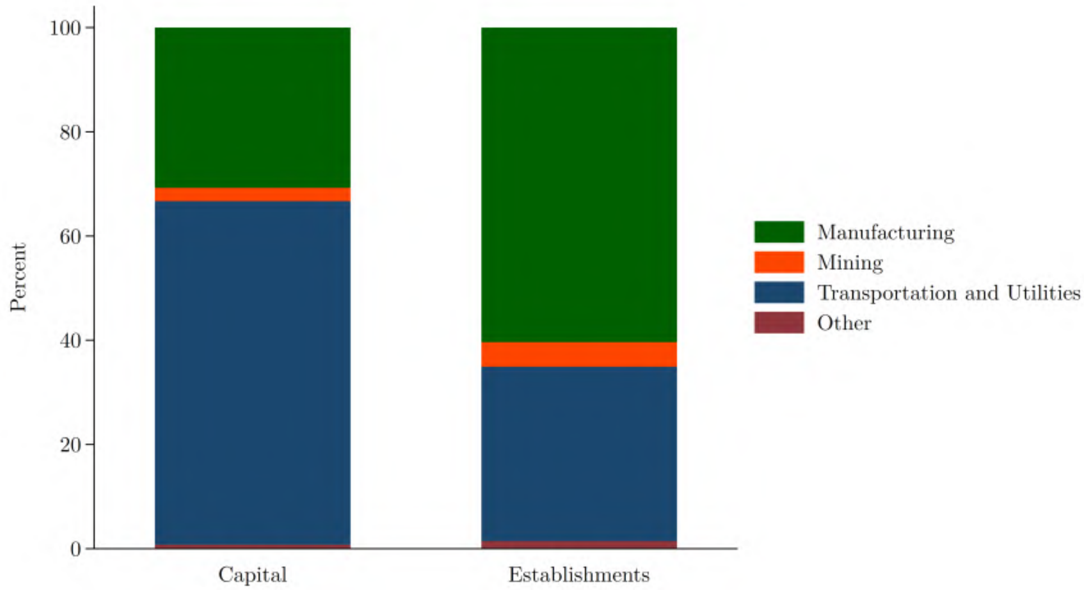
Panel B: Revenue-Weighted Mean HHI



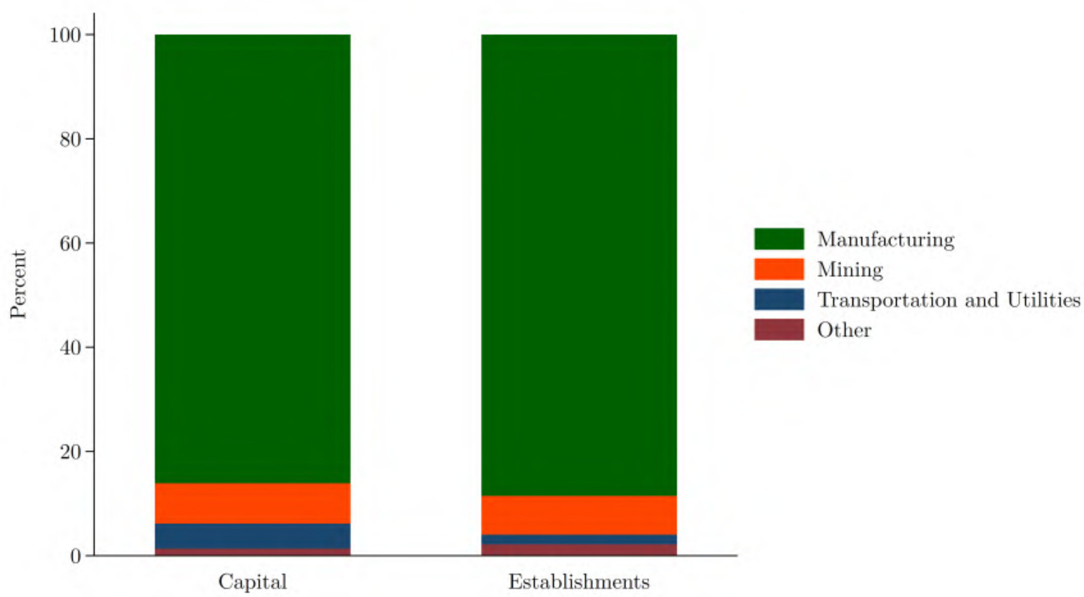
Notes: This figure illustrates the median (Panel A) and mean (Panel B) establishment-level Herfindahl–Hirschman Index (HHI) by trust status from 1850 through 1880. Panel A is unweighted, while Panel B weights industries by revenue. Trust status is measured in 1903 and is from Moody (1904). Schedules for 1880 are missing from industries in which special agents were appointed to collect manufacturing data (Atack and Bateman, 1999; Delle Donne, 1973). These industries are iron and steel, cotton goods, woolen and worsted goods, silk and silk goods, chemical products and salt, coke and glass, shipbuilding, fisheries, all types of mining, coal, and petroleum (Wright, 1900, 63). For consistency over time, I removed these industries in the graphs shown above.

Figure 1.16: Trusts by Sector, 1903

Panel A: All Trusts

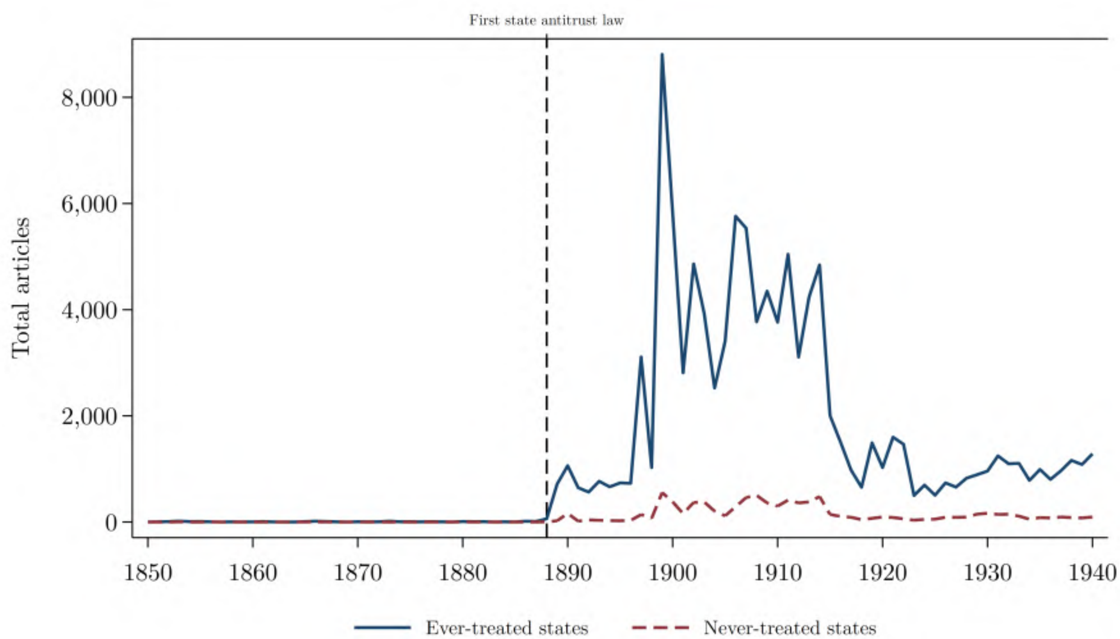


Panel B: "Industrial" Trusts



Notes: This figure shows the distribution of capital and establishments across the trusts listed in Moody (1904). These distributions are given for all 445 trusts in Panel A and for the 305 "industrial" trusts in Panel B.

Figure 1.17: Total Number of Newspaper Articles Mentioning State Antitrust Laws



Notes: "Ever-treated" states adopted a state antitrust statute at some point between 1888 and 1940. "Never-treated" states did not adopt a state antitrust statute at any point during this period.

Figure 1.18: Examples of Newspaper Articles Mentioning State Antitrust Laws

IOWA'S ANTI-TRUST LAW.

NO PROPER MACHINERY PROVIDED FOR ITS ENFORCEMENT.

DES MOINES, Iowa, June 6.—The Attorney General has given an important opinion in regard to the anti-trust law of the last General Assembly. That law provides that the Secretary of State shall address inquiries to all the corporations organized under the laws of or doing business in the State, as to whether they are violating the provisions of the act by unlawfully combining or merging their corporate existence in any other company. These inquiries the proper officer of the company is compelled to answer under oath. In event of their refusal to comply with the requirements of the statute, the Attorney General, through any County Attorneys of the State, shall direct proceedings, which shall have as their result on conviction, the revocation of the charter of the company.

The late Secretary Jackson sent the proper form of affidavit to the companies of the State. The results obtained can be tabulated: Corporations complying, 23; corporations as to which the Secretary had information that they had gone out of business, 370; corporations to which letters addressed were returned marked "Uncalled for," 951; corporations from which no reply was received, 1,991. These returns were properly certified to the Attorney General. Before instituting suits he concluded to examine the evidence obtained. As to the first three classes, there is no evidence that they have "refused" to comply with the demands of the State. Large numbers of them are agricultural, charitable, religious, and educational in character, and it is likely that many of the communications did not reach them or reached other parties who might formerly have been officers.

The Attorney General also maintains that it was hardly the intention of the General Assembly that summary measures should be instituted against corporations of this character. After discussing the difficulties of the situation, and what would be proper evidence to enable him to bring suit with any degree of expectancy of success, the Attorney General concludes: "In view of the insufficiency of the machinery provided for the execution of this section of this statute, I think the situation is of so grave a character as to warrant us in calling the attention of the Governor to it for such consideration as he may deem proper in the preparation of his biennial message to the General Assembly."

New York Times
New York, New York
June 7, 1891

THE GEORGIA ANTI-TRUST LAW.

(Atlanta Journal.)

The Journal has received from many States requests for copies of the Georgia anti-trust law. In some instances these requests have come from members of Legislatures who stated that they intended to introduce anti-trust bills. The Georgia law has attracted especial attention because it proved so speedily effective. A bill that will stop the operations of trusts in a State has naturally attracted very general attention. The Georgia bill has been introduced in four or five Legislatures, and it is probable that several other States will soon have anti-trust laws copied after ours. The Alabama House of Representatives last Saturday, without a dissenting voice, passed a bill which copies the Georgia law exactly. It is said that it wil certainly pass the Senate and become a law at the present session.

The Georgia Legislature seems to have set the anti-trust ball rolling at a lively rate. Let 'er roll.

News and Observer
Raleigh, North Carolina
February 10, 1897

STANDARD OIL FOUND GUILTY

Combine Convicted of Violating the Anti-Trust Law of Ohio.

Special Dispatch to the "Chronicle."

FINDLAY (O.), October 19.—By the verdict of a jury the Standard Oil Company of Ohio is guilty of conspiracy against trade, in violation of the Valentine anti-trust law of Ohio. The penalty is a fine of from $50 to $5000, which may be repeated for each day of the offense, or imprisonment of six to twelve months.

The Standard Oil Company of Ohio has given notice that it will file a motion for a new trial. Under the practice of the Court, the defendant has three days to put this motion in form. The next step will be for the Court to impose the penalty. The defense will then take their bill of exceptions to such rulings of Judge Banker as they objected to to the Circuit Court of the State. The appeal from this court is to the Supreme Court of the State, by which tribunal there is no doubt the issue will ultimately be decided.

To the State the suit, the verdict and the ultimate appeal is important, particularly because it initiates an entirely new method of proceeding against the alleged trade monopolies—that is, by information and affidavit, instead of by Grand Jury indictment.

The verdict was rendered at 4:30 o'clock this morning, and resulted from a continuous deliberation by the jury during thirty-two consecutive hours. The trial occupied seven days preceding this deliberation. When the case went to the jury at 8:30 o'clock Wednesday night the first ballot of the jurors stood nine for conviction and three for acquittal. As the result of continuous deliberations to 4 o'clock Thursday morning, one of the three for acquittal joined the majority. At 7 o'clock Thursday night one of the two remaining for acquittal went over to the other side, and at 4 o'clock this morning the last of the three gave his assent to the verdict of "guilty."

A touch of the dramatic marked the closing hours of the jury's deliberations. Hymns were sung during all but ten minutes of this time. This ten minutes came an the end and was occupied by the remaining juror who had stood out in explaining his position and surrender to the majority. There was not the slightest levity about this hymnal service. The jurors had then been many hours without sleep. The songs, which were started shortly after 2 o'clock in the morning by about three voices, echoed at first feebly through the spacious courthouse. After one familiar hymn after another was sung, it was evident that the spirit of fraternalism was gaining headway in the small chamber in which the twelve men were locked. The number of voices increased, the hymns gained in volume and enthusiasm. Then "Home, Sweet Home" was sung, the national anthem followed, then more hymns. Laughter was heard between the limited pauses. It bore no tone of derision, but of cordiality. A few minutes later came the announcement that the jury had reached an agreement and Judge Bankers hastily summoned.

San Francisco Chronicle
San Francisco, California
October 20, 1906

ANTI-TRUST SUIT SEEKS MILLIONS

LEXINGTON, Miss.—A suit has been entered by the Retail Lumber Dealers Association of Mississippi and Louisiana in the chancery court under the anti-trust statute for the recovery of penalties aggregating $14,184,000, the minimum under the statute, as the minmum penalty is $200 a day and the maximum $500.

Christian Science Monitor
Boston, Massachusetts
July 16, 1909

MISSOURI COURT OUSTS OPEN PRICE BOARDS

St. Louis Lumber Exchange's Members Fined $96,000 Under Anti-Trust Laws.

JEFFERSON CITY, Mo., Aug. 31 (Associated Press).—Missouri has scored a complete victory over the new form of business combinations known as "open price associations," in a suit brought by Attorney General Barrett against the St. Louis Lumber Trade Exchange. The Missouri Supreme Court ousted each of the nineteen St. Louis lumber companies composing the exchange, assessed fines totaling $96,000 and ousted the exchange itself.

The Court sustained the Attorney General in holding that the combination violated the State anti-trust laws, even though it did not directly fix prices. So far as is known it is the first decision of any State court on the points involved and follows closely the decision of the United States Supreme Court last year in the hardwood lumber cases, according to Attorney General Barrett.

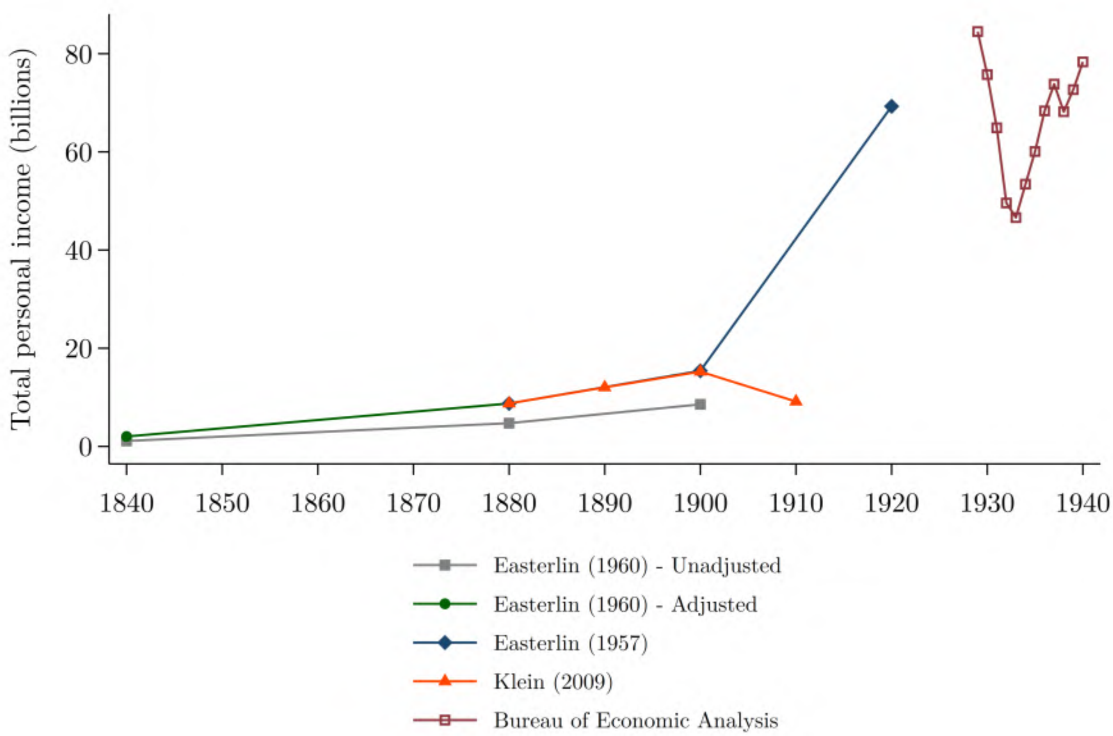
The association regulated the terms of credit and the general business practices of its members, and provided for cooperation that went far, but stopped short of the actual fixing of prices. Books were written describing the plan as "The New Competition," and it was believed, the Attorney General said, that the method would be proof against prosecution by the State or Federal authorities.

The Court in a sweeping decision declared that the Missouri statutes forbid not only agreements to lessen competition, but agreements which tend to lessen it. The decision said that the very creation of a system of machinery which can be improperly used is in itself a violation of the Missouri law.

The Court found that the increased cost of lumber was due to other factors as well, such as higher freight rates and greater labor costs, but condemned the lumber exchange in most positive terms.

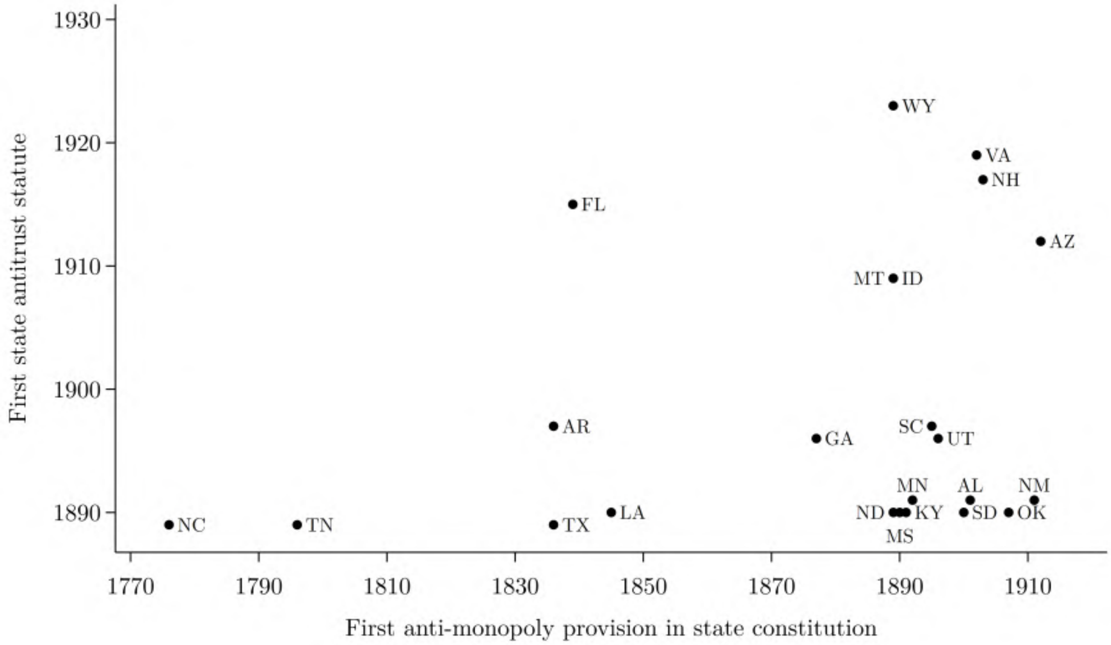
New York Times
New York, New York
September 1, 1923

Figure 1.19: State-Level Personal Income Data Sources, 1840-1940



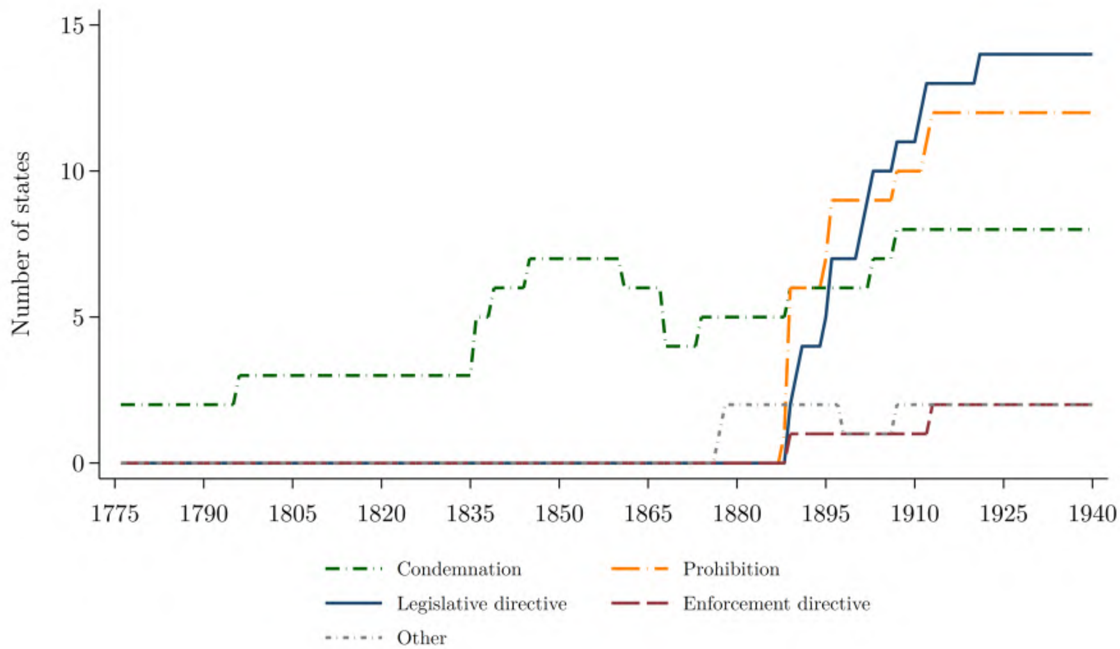
Notes: The composite series I construct on state-level personal income comes from four sources. Tables A-1, A-2, and A-3 in Easterlin (1960) provide data for 1840; Table 8 in Klein (2009) provides data for 1880 through 1910; Table Y-1 in Easterlin (1957, 753) provides data for 1920; and Bureau of Economic Analysis (2025) provides data for 1929 and later. The data Easterlin (1960) provides for 1840, 1880, and 1900 are adjusted upwards slightly to align with the estimates Easterlin (1957) and Klein (2009) provide for 1880 through 1900.

Figure 1.20: Adoption of State Antitrust Statutes and Anti-Monopoly Constitution Provisions, 1776-1930



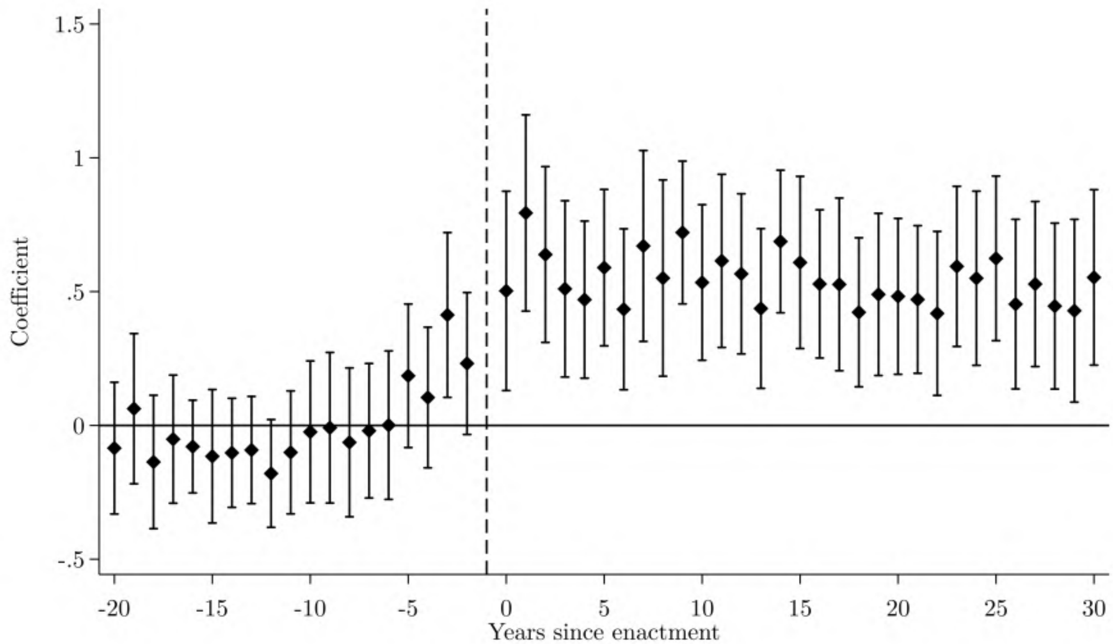
Notes: This figure illustrates the relationship between adoption of an antitrust statute and adoption of an anti-monopoly constitutional provision, among states that adopted both of these measures.

Figure 1.21: Contents of Anti-Monopoly Provisions in State Constitutions, 1776–1940



Notes: This figure illustrates the number of states with various kinds of anti-monopoly provisions in their state constitutions between 1776 and 1940. See Table 1.2 for definitions of these classes of provisions. Declines in the number of states with such provisions typically reflect the adoption of entirely new state constitutions that omitted the earlier language, rather than the targeted repeal of anti-monopoly provisions.

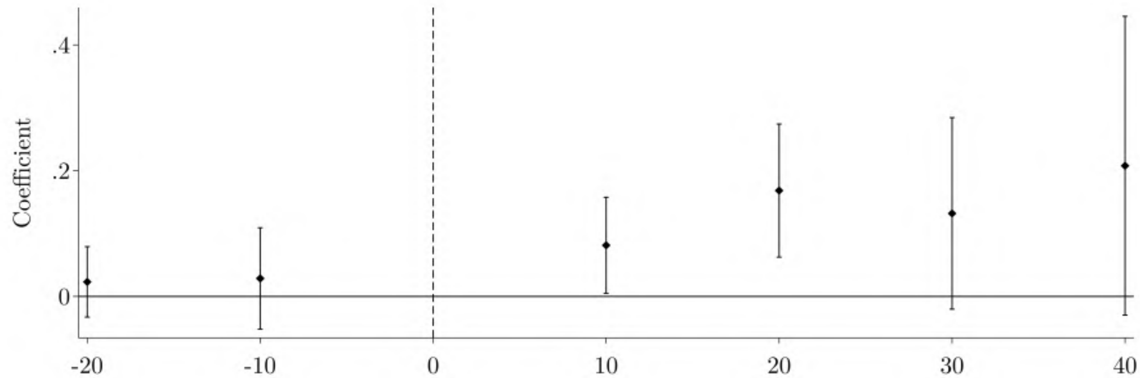
Figure 1.22: Effect of Enactment on the Number of Newspaper Articles Mentioning State Antitrust Laws



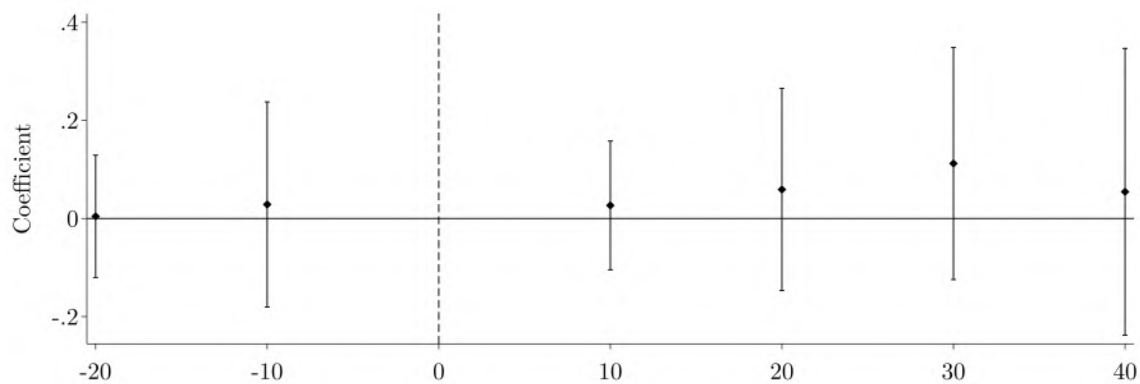
Notes: The dependent variable is the log number of newspaper articles mentioning each state's antitrust law. Estimates are obtained using the stacked difference-in-differences method. Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state level. States whose antitrust laws were overturned or repealed are excluded. This event study does not include controls.

Figure 1.23: Event Study Results for Establishment Outcomes

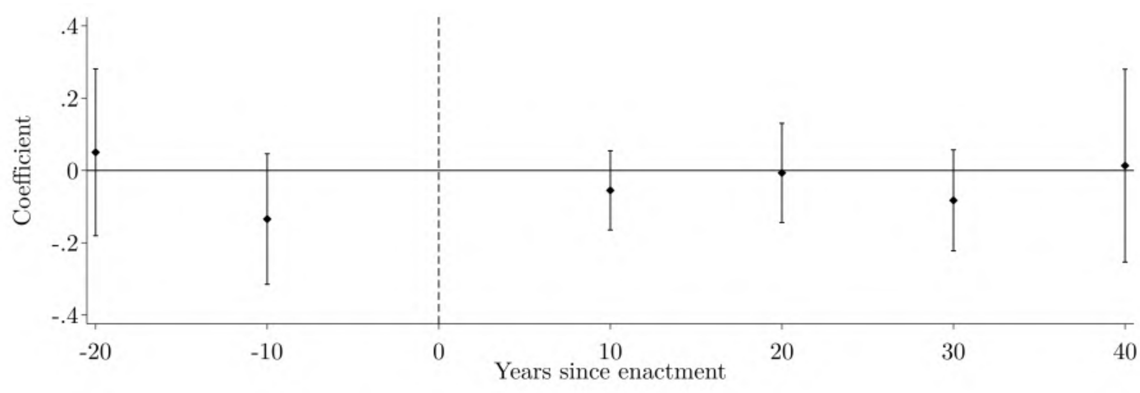
Panel A: Log Number of Establishments



Panel B: Log Average Profit per Establishment



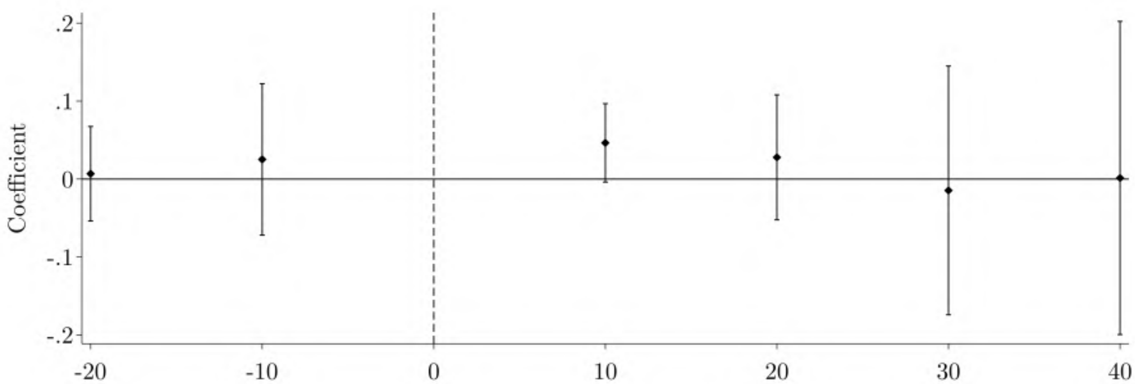
Panel C: Log Average Number of Proprietors per Establishment



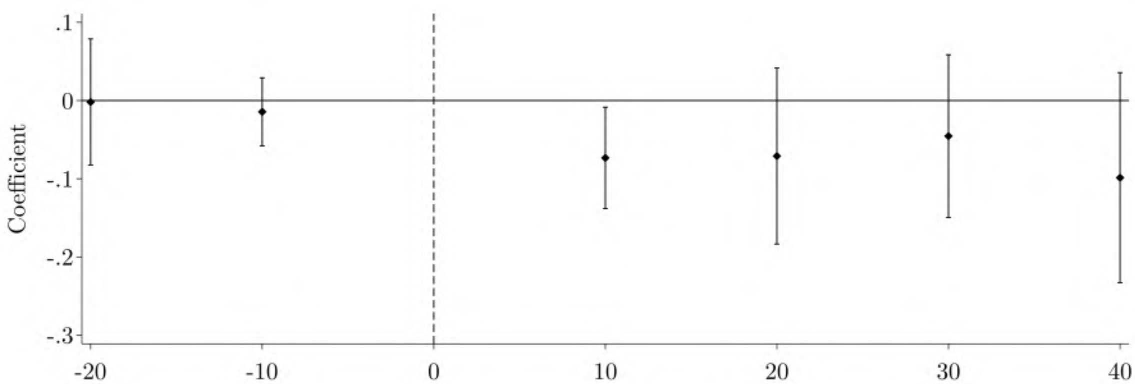
Notes: Estimates are obtained using the stacked difference-in-differences method. Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state level. States whose antitrust laws were overturned or repealed are excluded. Controls capture the population, median occupational score, estimated personal income, and literacy rate of each state. State-by-industry and year-by-industry fixed effects are also included.

Figure 1.24: Event Study Results for Labor Market Outcomes

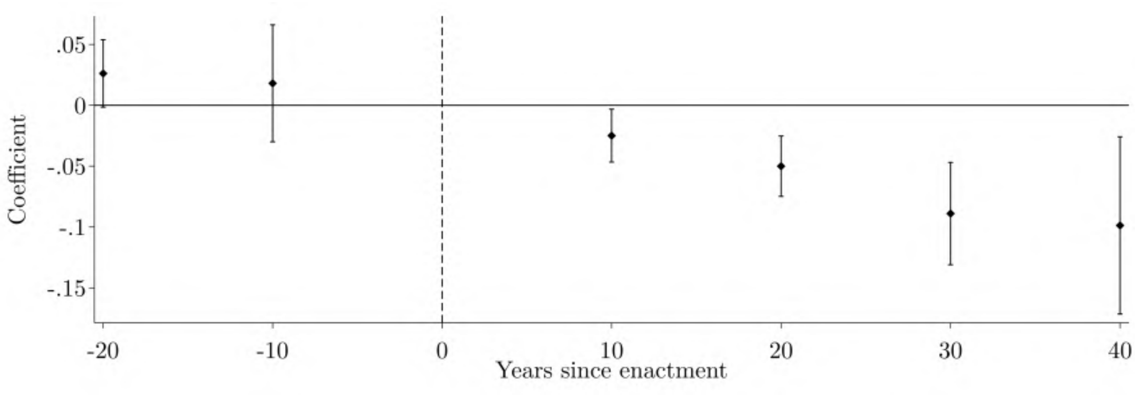
Panel A: Log Average Employment per Establishment



Panel B: Log Average Wage

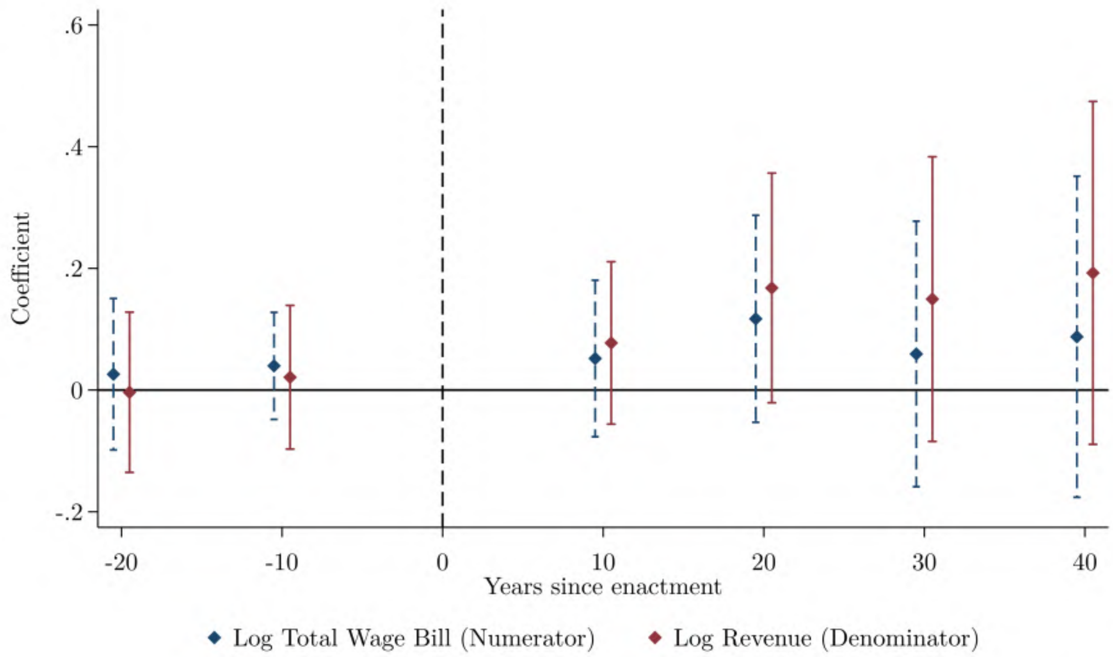


Panel C: Log Labor Share



Notes: Estimates are obtained using the stacked difference-in-differences method. Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state level. States whose antitrust laws were overturned or repealed are excluded. Controls capture the population, median occupational score, estimated personal income, and literacy rate of each state. State-by-industry and year-by-industry fixed effects are also included.

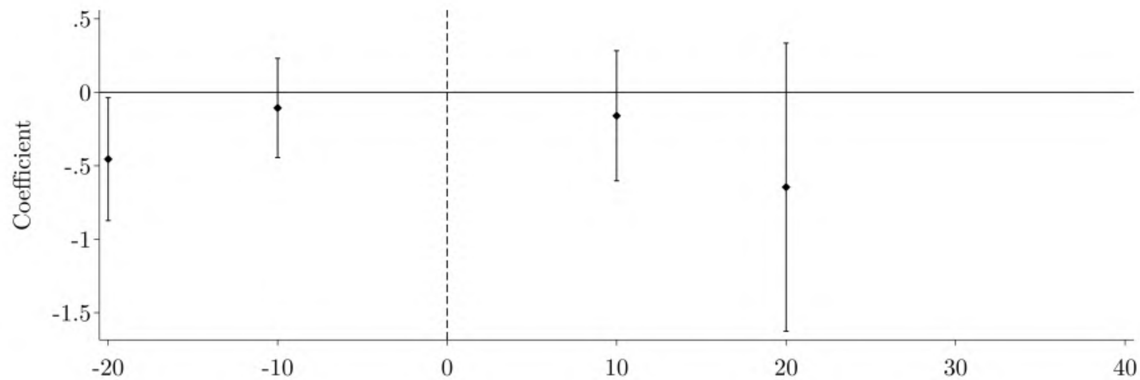
Figure 1.25: Event Study Results for Labor Share Components



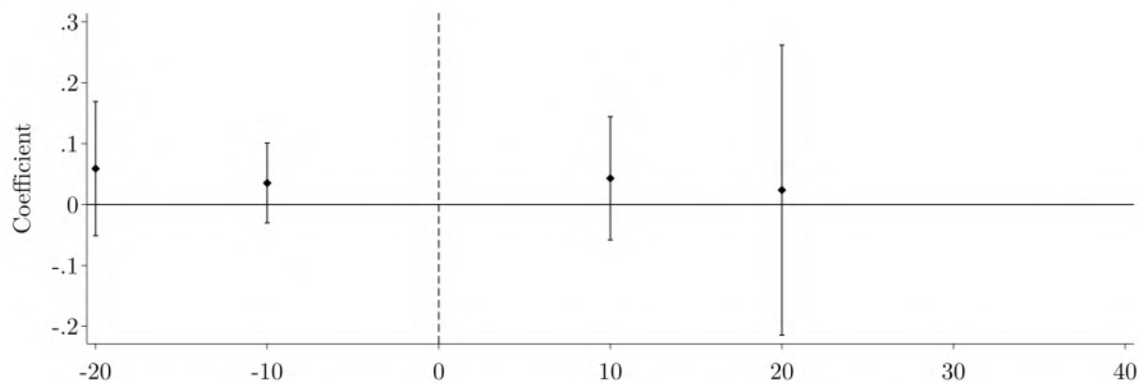
Notes: Estimates are obtained using the stacked difference-in-differences method. Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state level. States whose antitrust laws were overturned or repealed are excluded. Controls capture the population, median occupational score, estimated personal income, and literacy rate of each state. State-by-industry and year-by-industry fixed effects are also included.

Figure 1.26: Event Study Results for Productivity Outcomes

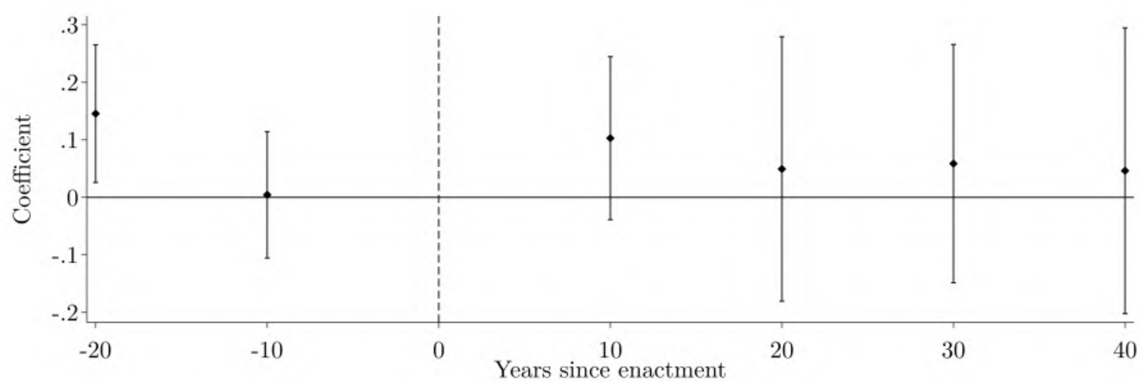
Panel A: Log Residual Value-Added TFP with Capital



Panel B: Log Levinsohn and Petrin (2003) TFP with Capital

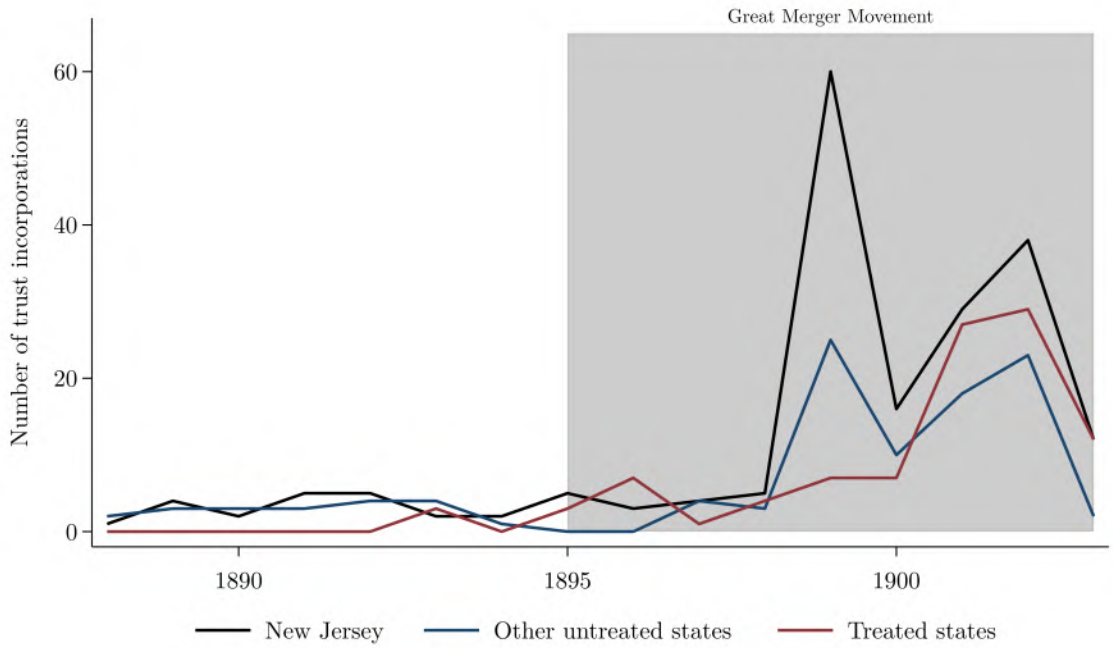


Panel C: Log Levinsohn and Petrin (2003) TFP with Establishments



Notes: Estimates are obtained using the stacked difference-in-differences method. Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state level. States whose laws were overturned or repealed are excluded. Controls capture the population, median occupational score, estimated personal income, and literacy rate of each state. State-by-industry and year-by-industry fixed effects are also included. TFP is A) the residual from a Cobb–Douglas production function; B) the first-stage residual from a Levinsohn and Petrin (2003) specification where labor is the free variable, materials is the unobserved productivity proxy, and capital is the state variable; and C) the same LP residual but using establishments as the state variable due to the limited availability of capital.

Figure 1.27: Trust Incorporations by Treatment Status, 1888-1903

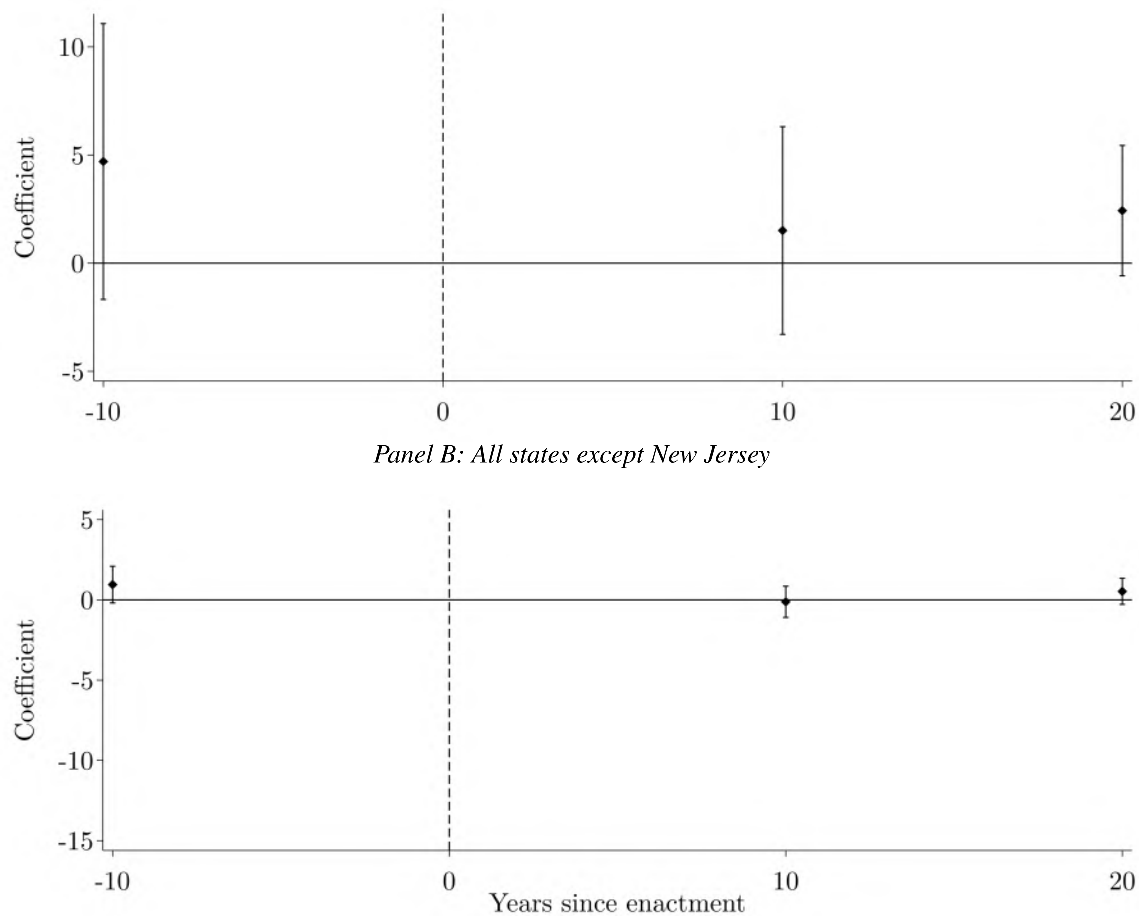


Notes: This figure illustrates the number of trusts incorporating over time, by the contemporaneous treatment status of the state where they chose to incorporate. Data are from Moody (1904).

Figure 1.28: Effect of Enactment on Number of Trust Incorporations

Panel A: All states

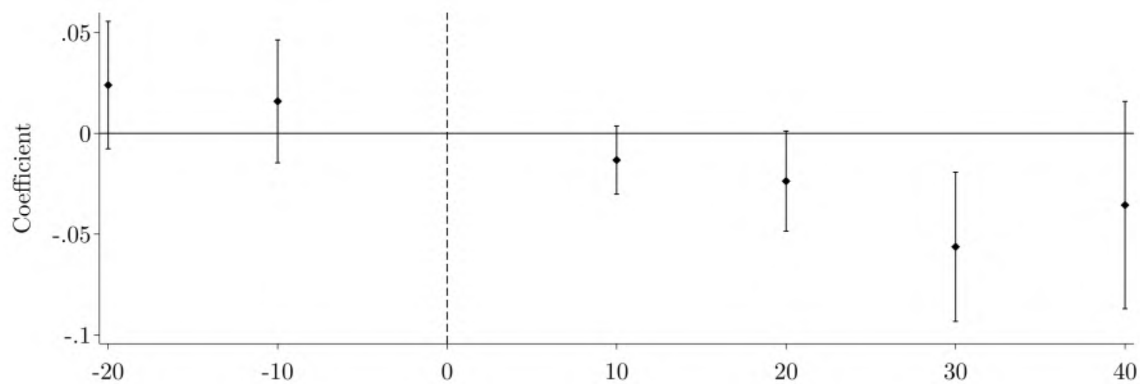
Panel B: All states except New Jersey



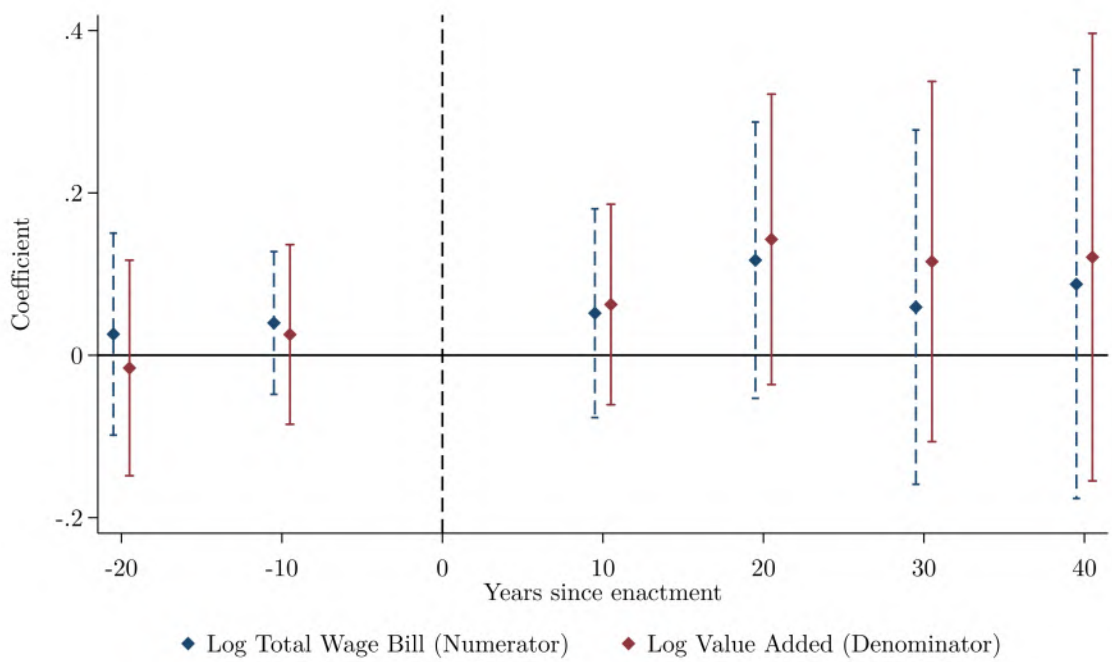
Notes: The dependent variable is the number of trust incorporations by state and decade. Data are from Moody (1904). Estimates are obtained using the stacked difference-in-differences method. Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state level. States whose antitrust laws were overturned or repealed are excluded. Controls capture the population, median occupational score, estimated personal income, and literacy rate of each state.

Figure 1.29: Event Study Results for Labor Share of Value Added and Components

Panel A: Log Labor Share (of Value Added)

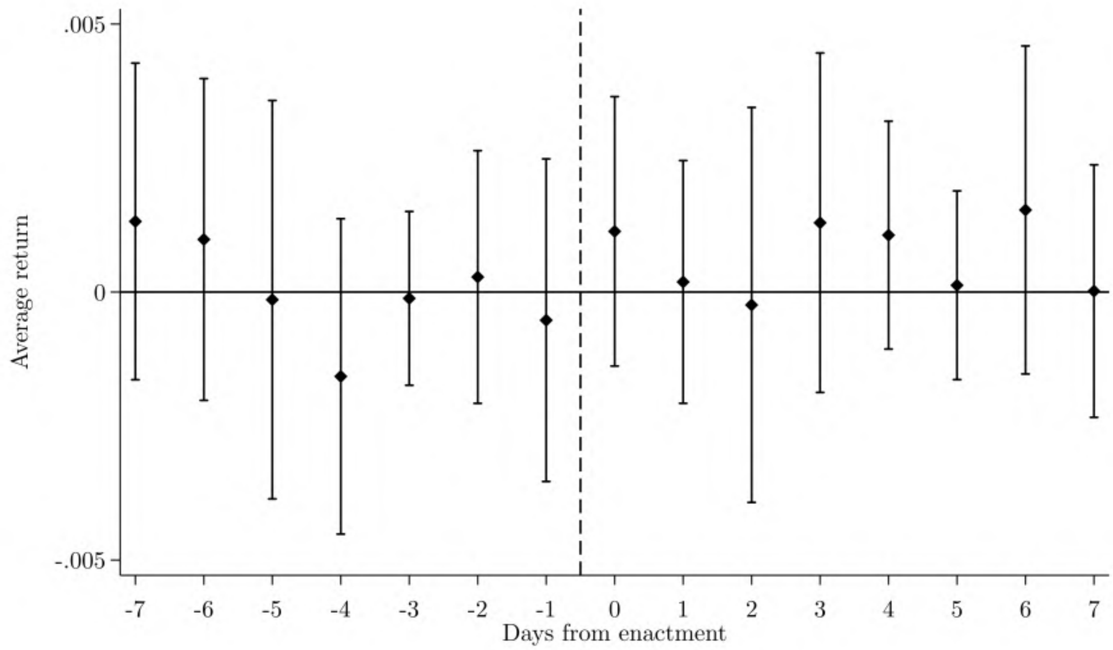


Panel B: Components of Labor Share (of Value Added)



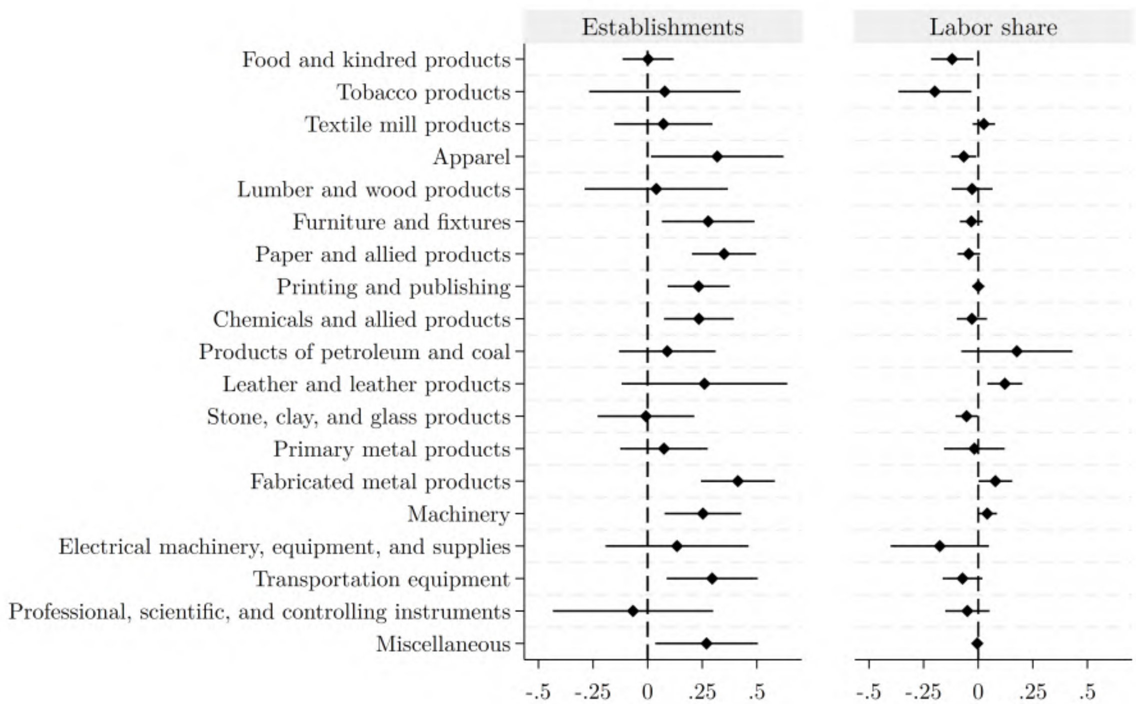
Notes: Estimates are obtained using the stacked difference-in-differences method. Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state level. States whose antitrust laws were overturned or repealed are excluded. Controls capture the population, median occupational score, estimated personal income, and literacy rate of each state. State-by-industry and year-by-industry fixed effects are also included.

Figure 1.30: Average Daily Stock Returns Around Enactment



Notes: This figure shows the average daily return to the Dow Jones composite portfolio in the seven trading days before and after the passage of a state antitrust law. For each passage date, I extracted a 15-day event window (date of passage ± seven days) using historical daily return data (Schwert, 1990). To account for non-trading days (i.e., weekends and holidays), I selected the 15 closest trading days around each date of passage. I then stacked the data across all dates of passage and computed the mean return and its standard error for each relative day. Days appearing in multiple states' windows were thus included multiple times. Bars indicate 95 percent confidence intervals.

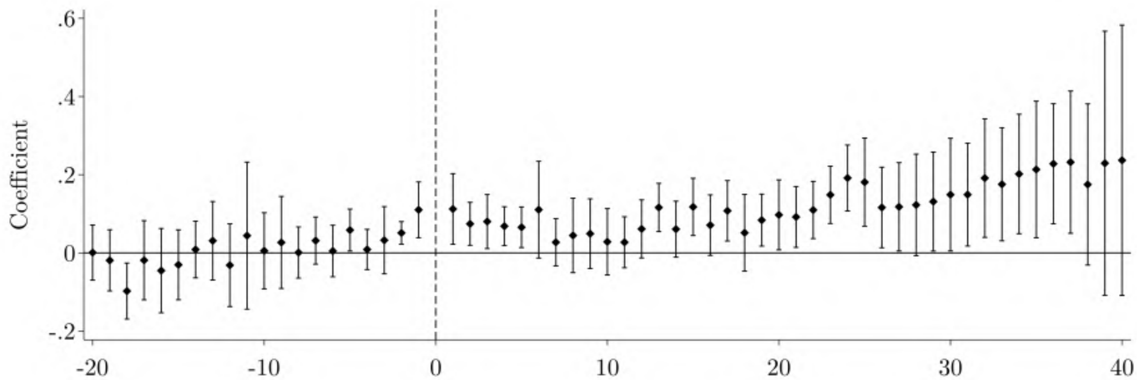
Figure 1.31: Difference-in-Differences Results by Industry Category



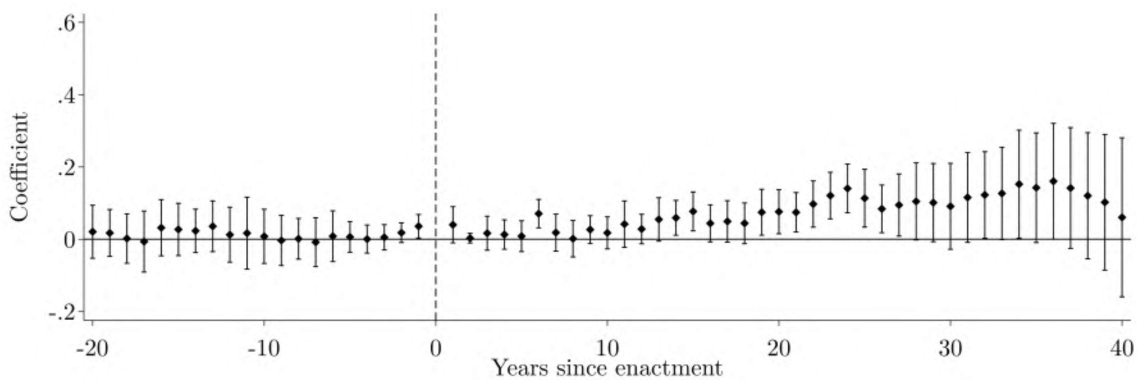
Notes: The dependent variable is the log number of manufacturing establishments in the first column and the log labor share in the second column. Estimates are obtained using the stacked difference-in-differences method. Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state level. States whose antitrust laws were overturned or repealed are excluded. Controls capture the population, median occupational score, estimated personal income, and literacy rate of each state. State-by-industry and year-by-industry fixed effects are also included.

Figure 1.32: Event Study Results for Patenting Behavior

Panel A: Log Number of Assignee Patents



Panel B: Log Number of Inventor Patents



Notes: Estimates are obtained using the stacked difference-in-differences method. Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state level. The dependent variable is the number of patents, aggregated to the county-by–USPC–by-year level, where USPC denotes the U.S. Patent Classification (USPC) categories. This method of aggregation maximizes statistical power by retaining geographic and technological variation and enables the use of USPC category fixed effects, which allow me to account for systematic differences in patenting intensity across technologies. Regressions are weighted by the number of patents. States whose antitrust laws were overturned or repealed are excluded. Controls capture the population, median occupational score, estimated personal income, and literacy rate of each state. USPC category fixed effects are also included.

1.8 Tables

Table 1.1: Antitrust Law Index Components

	Variable	First PC	First PC Squared
1	Violating the statute is a crime[1]	0.298	0.089
2	Violating the statute is a tort[1]	0.253	0.064
3	Minimum prison sentence for violating the statute[2]	0.148	0.022
4	Maximum prison sentence for violating the statute[2]	0.225	0.050
5	Minimum fine for violating the statute[2]	0.201	0.040
6	Maximum fine for violating the statute[2]	0.142	0.020
7	Level of damages that can be sought for violations[2]	0.231	0.053
8	The statute outlaws horizontal price fixing for goods[1]	0.291	0.085
9	The statute outlaws horizontal price fixing for services[1]	0.046	0.002
10	The statute outlaws horizontal output restriction[1]	0.275	0.075
11	The statute outlaws trustee control of corporations[1]	0.223	0.050
12	The statute outlaws restraints of trade[1]	0.278	0.077
13	The statute outlaws monopolization[1]	0.186	0.035
14	The statute outlaws anticompetitive stock purchases[1]	0.148	0.022
15	The statute outlaws refusals to deal[1]	0.167	0.028
16	The statute allows the state to revoke the charters of offending corporations[1]	0.275	0.076
17	The statute allows both a fine and a prison sentence to be imposed[2]	0.297	0.088
18	How the statute assigns enforcement responsibilities[2]	0.291	0.085
19	The statute stipulates that each day of violation is a separate offense[1]	0.198	0.039
	Total		1.000

Notes: "PC" abbreviates principal component. "1" denotes an indicator variable and "2" denotes a continuous or ordinal variable. Minimum and maximum prison sentence variables (variables 3 and 4, respectively) record the minimum and maximum incarceration periods authorized for violating the antitrust statute. When no minimum prison sentence is set, I recode this variable to 0. I recode statutes with no statutory maximum (i.e., theoretically unlimited) to the longest maximum observed in the data. Minimum and maximum fine variables (variables 5 and 6, respectively) are constructed analogously. When no minimum fine is set, I recode this variable to 0, and when there is no statutory maximum, I recode to the highest maximum fine observed. The level of damages variable (variable 7) is coded as 1 for statutes authorizing actual damages, 2 for statutes authorizing double damages, and 3 for statutes authorizing treble damages. Variable 17 takes four values: 1 if the statute does not authorize either a fine or prison sentence, 2 if it authorizes one of the two but not both, 3 if it authorizes either or both, and 4 if it requires both a fine and a prison sentence. The enforcement responsibility variable (variable 18) takes four values: 1 if no official is assigned enforcement duties, 2 if only the Attorney General is assigned enforcement duties, 3 if all District Attorneys are assigned enforcement duties, and 4 if both the Attorney General and all District Attorneys are assigned enforcement duties.

Table 1.2: Contents of Anti-Monopoly Provisions in State Constitutions

Type	Definition
Condemnation	Decries the existence of monopolies.
Prohibition	Declares that combinations, trusts, and/or other restraints of trade are illegal.
Legislative directive	Orders the state's legislature to pass laws to regulate monopolies.
Enforcement directive	Designates a state officer, such as the Attorney General, to enforce the constitutional provision against monopolies.
Other	Deals with competition policy in a manner not covered by the above categories.

Table 1.3: Descriptive Statistics for Ever- and Never-Treated States, 1880

	(1) Ever-Treated	(2) Never-Treated	(3) Difference
Population (thousands)	1192.75	767.10	425.657
	(1038.23)	(1348.72)	(410.928)
Urban share of population	0.216	0.301	-0.086
	(0.166)	(0.236)	(0.067)
Foreign-born share of population	0.151	0.168	-0.016
	(0.120)	(0.117)	(0.045)
Agricultural revenue per capita	42.80	43.66	-0.859
	(16.83)	(19.77)	(6.489)
School expenditures per capita	1.768	1.935	-0.167
	(1.153)	(0.973)	(0.418)
Railroad miles per 1,000 square miles	57.33	77.70	-20.371
	(62.07)	(70.45)	(23.745)
Average value of farmland (millions)	18.39	26.00	-7.611
	(15.36)	(15.99)	(5.768)
Newspapers per 10,000 population	2.43	2.83	-0.393
	(1.19)	(1.53)	(0.471)
Personal income per capita	187.87	241.61	-53.737
	(115.03)	(142.49)	(44.951)
Grange chapters per 1,000,000 population (1887)	591.47	398.73	192.745
	(369.20)	(448.36)	(145.280)
Greenback vote share (1884)	0.033	0.011	0.021
	(0.030)	(0.009)	(0.018)
Share of manufacturing employment in	0.785	0.727	0.058
industries with a trust by 1904	(0.169)	(0.200)	(0.066)
Observations	36	9	45

Notes: Columns (1) and (2) report means and standard deviations for ever-treated and never-treated states, respectively. Column (3) shows the difference in means, with standard errors in parentheses. Some variables are only consistently available at the state level in 1880, so each observation represents a single state-level mean; the sample size thus differs from that in the regression analysis, which uses disaggregated data. The source for the first eight measures is the National Historical Geographic Information System (Manson et al. (2022)). State-level income estimates are from Easterlin (1960). Grange chapter data are courtesy of the National Grange. Greenback vote shares are from Inter-University Consortium for Political and Social Research (1999). I computed the last measure using data from Moody (1904) and the 1880 census of manufactures. Oklahoma, North Dakota, and South Dakota are excluded due to data limitations. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels, respectively.

Table 1.4: Descriptive Statistics for Early- and Late-Treated States, 1880

	(1) Early-Treated	(2) Late-Treated	(3) Difference
Population (thousands)	1523.36	862.14	661.220*
	(1108.22)	(871.57)	(332.315)
Urban share of population	0.176	0.255	-0.078
	(0.130)	(0.191)	(0.054)
Foreign-born share of population	0.125	0.177	-0.052
	(0.107)	(0.130)	(0.040)
Agricultural revenue per capita	48.88	36.73	12.147**
	(15.85)	(15.92)	(5.296)
School expenditures per capita	1.597	1.940	-0.342
	(1.111)	(1.200)	(0.385)
Railroad miles per 1,000 square miles	47.39	67.28	-19.889
	(36.24)	(80.06)	(20.714)
Average value of farmland (millions)	15.78	21.00	-5.222
	(11.47)	(18.43)	(5.116)
Newspapers per 10,000 population	2.217	2.649	-0.432
	(1.016)	(1.344)	(0.397)
Personal income per capita	139.10	236.64	-97.533**
	(53.00)	(139.28)	(35.124)
Grange chapters per 1,000,000 population (1887)	725.13	449.46	275.669**
	(402.58)	(275.96)	(120.911)
Greenback vote share (1884)	0.034	0.031	0.003
	(0.033)	(0.028)	(0.018)
Share of manufacturing employment in industries with a trust by 1904	0.766	0.806	-0.041
	(0.188)	(0.150)	(0.058)
Observations	18	18	36

Notes: Columns (1) and (2) report means and standard deviations for “early-treated” states, which adopted a statute before 1895, and “late-treated” states, which adopted a statute in 1895 or later, respectively. Column (3) shows the difference in means, with standard errors in parentheses. Some variables are only consistently available at the state level in 1880, so each observation represents a single state-level mean; the sample size thus differs from that in the regression analysis, which uses disaggregated data. The source for the first eight measures is the National Historical Geographic Information System (Manson et al. (2022)). State-level income estimates are from Easterlin (1960). Grange chapter data are courtesy of the National Grange. Greenback vote shares are from Inter-University Consortium for Political and Social Research (1999). I computed the last measure using data from Moody (1904) and the 1880 census of manufactures. Oklahoma, North Dakota, and South Dakota are excluded due to data limitations. ***, **, and * denote statistical significance at the 1, 5, and 10 percent levels, respectively.

Table 1.5: Main Difference-in-Differences Results

	(1)	(2)	(3)	(4)
Panel A: Log Number of Establishments				
ATT	0.187	0.083	0.181**	0.102**
	(0.125)	(0.072)	(0.088)	(0.050)
Observations	70,322	70,322	70,322	70,322
Panel B: Log Average Profit per Establishment				
ATT	-0.051	-0.074	-0.040	-0.012
	(0.093)	(0.095)	(0.060)	(0.058)
Observations	68,801	68,801	68,801	68,801
Panel C: Log Average Number of Proprietors per Establishment				
ATT	-0.091	-0.111	-0.060	-0.098
	(0.150)	(0.146)	(0.124)	(0.121)
Observations	29,878	29,878	29,878	29,878
Panel D: Log Average Employment per Establishment				
ATT	-0.077	-0.089	-0.059	-0.002
	(0.065)	(0.069)	(0.037)	(0.037)
Observations	70,115	70,115	70,115	70,115
Panel E: Log Average Wage				
ATT	-0.026	-0.026	-0.040*	-0.034
	(0.024)	(0.024)	(0.022)	(0.021)
Observations	69,160	69,160	69,160	69,160
Panel F: Log Labor Share				
ATT	-0.048**	-0.039**	-0.053***	-0.042***
	(0.019)	(0.019)	(0.018)	(0.014)
Observations	69,158	69,158	69,158	69,158
Baseline controls	✓	✓	✓	✓
Industry FE	✓			
State-by-industry FE		✓		✓
Year-by-industry FE			✓	✓

Notes: Estimates are obtained using the stacked difference-in-differences method. Robust standard errors clustered at the state level are reported in parentheses. *** denotes statistical significance at the 1 percent level, ** denotes statistical significance at the 5 percent level, and * denotes statistical significance at the 10 percent level. States whose antitrust laws were overturned or repealed are excluded. All models include controls for the population, median occupational score, estimated personal income, and literacy rate of each state.

Table 1.6: Difference-in-Differences Results by Trust Status

	Industries with a Trust		Industries with No Trust	
	(1)	(2)	(3)	(4)
Panel A: Log Number of Establishments				
ATT	0.127	0.061	0.211*	0.155**
	(0.088)	(0.053)	(0.105)	(0.066)
Observations	30,807	30,807	39,515	39,515
Panel B: Log Average Profit per Establishment				
ATT	0.014	0.003	-0.069	-0.020
	(0.081)	(0.077)	(0.050)	(0.039)
Observations	30,144	30,144	38,657	38,657
Panel C: Log Average Number of Proprietors per Establishment				
ATT	0.028	-0.011	-0.147*	-0.167*
	(0.156)	(0.139)	(0.086)	(0.088)
Observations	12,586	12,586	17,292	17,292
Panel D: Log Average Employment per Establishment				
ATT	-0.019	0.001	-0.062	0.011
	(0.046)	(0.050)	(0.053)	(0.033)
Observations	30,712	30,712	39,403	39,403
Panel E: Log Average Wage				
ATT	-0.047*	-0.033	-0.027	-0.031
	(0.024)	(0.022)	(0.024)	(0.025)
Observations	30,324	30,324	38,836	38,836
Panel F: Log Labor Share				
ATT	-0.071***	-0.053***	-0.025	-0.022
	(0.025)	(0.019)	(0.018)	(0.017)
Observations	30,322	30,322	38,836	38,836
Baseline controls	✓	✓	✓	✓
State-by-industry FE		✓		✓
Year-by-industry FE	✓	✓	✓	✓

Notes: Trust status is measured in 1903 and is from Moody (1904). Estimates are obtained using the stacked difference-in-differences method. Robust standard errors clustered at the state level are reported in parentheses. *** denotes statistical significance at the 1 percent level, ** denotes statistical significance at the 5 percent level, and * denotes statistical significance at the 10 percent level. States whose antitrust laws were overturned or repealed are excluded. All models include controls for the population, median occupational score, estimated personal income, and literacy rate of each state.

Table 1.7: Difference-in-Differences Results by Level of Enforcement

	Low-Enforcement States		High-Enforcement States	
	(1)	(2)	(3)	(4)
Panel A: Log Number of Establishments				
ATT	0.203**	0.137*	0.220**	0.093
	(0.098)	(0.071)	(0.100)	(0.065)
Observations	43,033	43,033	43,272	43,272
Panel B: Log Average Profit per Establishment				
ATT	-0.040	-0.004	-0.025	0.002
	(0.124)	(0.123)	(0.077)	(0.079)
Observations	42,019	42,019	42,420	42,420
Panel C: Log Average Number of Proprietors per Establishment				
ATT	-0.116*	-0.175***	0.144	0.147
	(0.067)	(0.057)	(0.111)	(0.098)
Observations	17,787	17,787	18,176	18,176
Panel D: Log Average Employment per Establishment				
ATT	-0.030	0.048	-0.067	-0.017
	(0.085)	(0.085)	(0.043)	(0.055)
Observations	42,905	42,905	43,143	43,143
Panel E: Log Average Wage				
ATT	-0.061	-0.051	-0.048	-0.040
	(0.042)	(0.041)	(0.031)	(0.032)
Observations	42,268	42,268	42,614	42,614
Panel F: Log Labor Share				
ATT	-0.060**	-0.049*	-0.067**	-0.054***
	(0.025)	(0.025)	(0.027)	(0.019)
Observations	42,266	42,266	42,612	42,612
Baseline controls	✓	✓	✓	✓
State-by-industry FE		✓		✓
Year-by-industry FE	✓	✓	✓	✓

Notes: Enforcement is measured using mentions of state antitrust policy in contemporaneous newspapers. A state is designated as high enforcement if, during the years its antitrust statute was in effect, the share of years with above-median newspaper mentions per capita (relative to other treated states in that year) is at least as large as the median share across states. Estimates are obtained using the stacked difference-in-differences method. Robust standard errors clustered at the state level are reported in parentheses. *** denotes statistical significance at the 1 percent level, ** denotes statistical significance at the 5 percent level, and * denotes statistical significance at the 10 percent level. States whose antitrust laws were overturned or repealed are excluded. All models include controls for the population, median occupational score, estimated personal income, and literacy rate of each state.

Table 1.8: Difference-in-Differences Results with Continuous Treatment

	(1)	(2)	Observations
Panel A: Log Number of Establishments			64,080
ATT	0.641***		
	(0.061)		
Baseline ATT (Projected Constant)		0.663***	
		(0.160)	
Antitrust Index		-0.036	
		(0.305)	
Panel B: Log Average Profit per Establishment			63,508
ATT	0.153		
	(0.133)		
Baseline ATT (Projected Constant)		-1.021**	
		(0.405)	
Antitrust Index		1.975***	
		(0.735)	
Panel C: Log Average Employment per Establishment			63,932
ATT	0.119		
	(0.111)		
Baseline ATT (Projected Constant)		-0.715**	
		(0.330)	
Antitrust Index		1.404***	
		(0.531)	
Panel D: Log Average Wage			63,766
ATT	-0.030		
	(0.034)		
Baseline ATT (Projected Constant)		-0.111	
		(0.080)	
Antitrust Index		0.136	
		(0.150)	
Panel E: Log Labor Share			63,770
ATT	-0.128***		
	(0.024)		
Baseline ATT (Projected Constant)		0.158**	
		(0.073)	
Antitrust Index		-0.480***	
		(0.124)	
Baseline controls	✓	✓	
State-by-industry FE	✓	✓	

Notes: Estimates are obtained using the Borusyak et al. (2024) method. Robust standard errors clustered at the state level are reported in parentheses. *** denotes statistical significance at the 1 percent level, ** denotes statistical significance at the 5 percent level, and * denotes statistical significance at the 10 percent level. States whose antitrust laws were overturned or repealed are excluded. All models include controls for the population, median occupational score, estimated personal income, and literacy rate of each state.

Table 1.9: Robustness Checks on Main Difference-in-Differences Results

	(1)	(2)	(3)	(4)
Panel A: Log Number of Establishments				
ATT	0.148**	0.148**	0.100*	0.092*
	(0.065)	(0.062)	(0.051)	(0.051)
Observations	70,322	70,322	70,322	70,322
Panel B: Log Average Profit per Establishment				
ATT	-0.010	0.008	-0.015	-0.061
	(0.040)	(0.049)	(0.058)	(0.062)
Observations	68,801	68,801	68,801	68,801
Panel C: Log Average Number of Proprietors per Establishment				
ATT	-0.023	-0.140	-0.098	-0.130
	(0.073)	(0.103)	(0.121)	(0.112)
Observations	29,878	29,878	29,878	29,878
Panel D: Log Average Employment per Establishment				
ATT	-0.001	-0.013	-0.006	-0.026
	(0.035)	(0.041)	(0.037)	(0.038)
Observations	70,115	70,115	70,115	70,115
Panel E: Log Average Wage				
ATT	-0.018	-0.014	-0.032	-0.053*
	(0.020)	(0.020)	(0.021)	(0.027)
Observations	69,160	69,160	69,160	69,160
Panel F: Log Labor Share				
ATT	-0.030**	-0.024*	-0.040***	-0.033**
	(0.014)	(0.012)	(0.014)	(0.013)
Observations	69,158	69,158	69,158	69,158
Baseline controls	✓	✓	✓	✓
State-by-industry FE	✓	✓	✓	✓
Year-by-industry FE	✓	✓	✓	✓
Establishment weighting	✓			
Employment weighting		✓		
Const. provision control			✓	
Ag. share control				✓

Notes: Estimates are obtained using the stacked difference-in-differences method. Robust standard errors clustered at the state level are reported in parentheses. *** denotes statistical significance at the 1 percent level, ** denotes statistical significance at the 5 percent level, and * denotes statistical significance at the 10 percent level. States whose antitrust laws were overturned or repealed are excluded. All models include controls for the population, median occupational score, estimated personal income, and literacy rate of each state. Additionally, column (3) controls for whether the state constitution contained an anti-monopoly provision, and column (4) controls for the share of employment in agriculture.

Table 1.10: Difference-in-Differences Results for Patenting Behavior

	(1)	(2)	(3)	(4)
Panel A: Log Number of Assignee Patents				
ATT	0.076***	0.085***	0.088***	0.074***
	(0.026)	(0.028)	(0.031)	(0.025)
Observations	272,496	272,496	272,496	272,496
Panel B: Log Number of Inventor Patents				
ATT	0.052***	0.045***	0.058***	0.051***
	(0.014)	(0.012)	(0.018)	(0.014)
Observations	747,934	747,934	747,934	747,934
Baseline controls	✓	✓	✓	✓
USPC FE	✓	✓		✓
USPC-by-year FE			✓	
Const. provision control	✓			
Ag. share control		✓		

Notes: Estimates are obtained using the stacked difference-in-differences method. Robust standard errors clustered at the state level are reported in parentheses. *** denotes statistical significance at the 1 percent level, ** denotes statistical significance at the 5 percent level, and * denotes statistical significance at the 10 percent level. The dependent variable is the number of patents, aggregated to the county-by–USPC–by-year level, where USPC denotes the U.S. Patent Classification (USPC) categories. This method of aggregation maximizes statistical power by retaining geographic and technological variation and enables the use of USPC category fixed effects, which allow me to account for systematic differences in patenting intensity across technologies. Regressions are weighted by the number of patents. States whose antitrust laws were overturned or repealed are excluded. All models include controls for the population, median occupational score, estimated personal income, and literacy rate of each state. USPC category fixed effects are also included in columns (1), (2), and (4), and column (3) includes USPC-by-year fixed effects. Additionally, column (1) controls for whether the state constitution contained an anti-monopoly provision, and column (2) controls for the share of employment in agriculture.

1.9 Methodology for Identifying and Coding State Antitrust Laws

To identify all state antitrust statutes in effect between 1860 and 1940, I developed a list of common words and phrases appearing in state antitrust statutes other scholars had already identified.[57] Then, I used HeinOnline, a popular online database for legal research, to search state session laws and state legal codes for these terms. The specific search terms I used are `anti-trust`, `antitrust`, `unfair trade practices`, `restraint of trade`, `pools`, `pooling`, `monopoly`, `monopolies`, `combination`, `fix the price`, `price fixing`, `substantially lessen competition`, `unreasonably restrain trade`, and `conspiracy against trade`. I used the Boolean operator "or" in my searches to ensure that session laws and historical codes containing *any* of these terms would be returned. Despite returning many false positives, which I tossed out upon inspection, this approach allowed me to cast a wide net and reduce the chance of missing any relevant laws.

I considered any state statute outlawing restraints against trade, monopolization, horizontal price fixing, horizontal output restrictions, trustee control of corporations, anticompetitve stock purchases, refusals to deal, or some combination thereof to be a state antitrust statute. Table 1.11 defines each of these acts. Price discrimination, tying, predatory pricing, and retail price maintenance are notably excluded from the list of infractions I counted. In my review, I found that state statutes prohibiting these forms of anticompetitive conduct were often enacted independently of a general antitrust statute prohibiting more basic forms of anticompetitive conduct, such as price fixing. As a result, I excluded price discrimination, tying, predatory pricing, and retail price maintenance from my analysis to avoid characterizing, for example, a state that did not prohibit price fixing—but did prohibit price discrimination—as having an antitrust statute. Further, to avoid counting some sections of a chapter or title but not others, I regarded whole chapters or titles as antitrust statutes as long as they contained at least one eligible element (e.g., a prohibition of monopolization), even if they contained one or more sections with ineligible elements (e.g., a prohibition of price discrimination). I also did not consider statutes prohibiting price gouging or profiteering to be antitrust laws. Further, I did not count statutes clarifying that certain groups (e.g., labor unions or agricultural associations) did not constitute trusts. As described in Section 1.3, I do not regard statutes applying to a single industry, rather than to commerce more generally, as antitrust statutes in this paper. However, I do consider statutes that apply to "necessities of life" to be antitrust statutes given the breadth of this term.[58] I coded laws as in effect the year they were enacted, regardless of the exact enactment date. Further, in cases where a law was repealed or overturned, I coded the law as inactive in the year it was repealed or overturned.[59]

[57] Forrest (1896) and von Halle (1899) are two early works that served as a helpful starting point.

[58] Connecticut's 1911 antitrust law is one example of a law containing this language. Act of August 15, 1911, Chapter 185, 1911 Connecticut Acts 1461.

[59] For example, if a law had been passed in 1900 and overturned in 1903, I would have coded the law as in effect in 1900, 1901, and 1902. In 1903, I would have again coded the law as inactive.

Table 1.11: Illegal Acts Under State Antitrust Laws

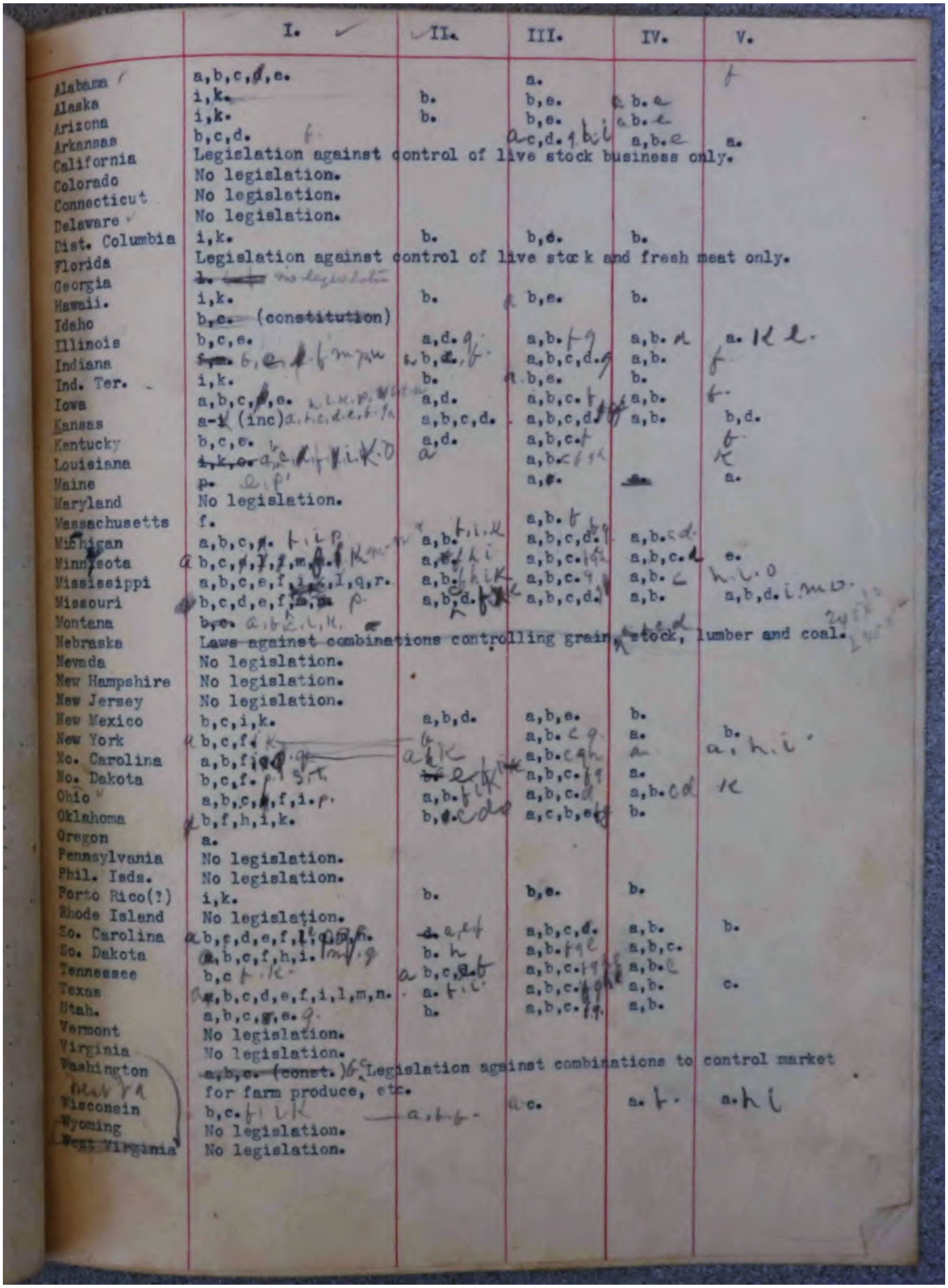
Illegal Act	Definition
Restraint of trade	To restrict competition or restrain trade in the production, manufacture, or sale of an article or commodity.
Monopolization	To monopolize or to attempt to monopolize the production, manufacture, or sale of an article or commodity.
Horizontal price fixing	To form a combination to control the price of an article or commodity.
Horizontal output restriction	To form a combination to limit the quantity of any article or commodity to be produced, manufactured, or sold.
Trustee control of corporations	To sell trust certificates establishing trustee control of several firms that would otherwise compete.
Anticompetitive stock purchases	When one firm buys shares or stock in another firm for the purpose of restricting competition.
Refusal to deal	To refuse to sell because purchaser is not a combination member or to refuse to deal with customers or suppliers who transact with a competitor.

I also compared my database to other scholars' compilations of state antitrust laws to ensure accuracy. For example, Forrest (1896), von Halle (1899), Bureau of Corporations (1915), and Works Progress Administration (1940) are contemporaneous works on state and federal antitrust laws. Any time I encountered an accounting of state antitrust statutes that disagreed with my own, I reviewed the differences in detail to confirm my understanding was correct. I also compared my database to internal memoranda on state antitrust laws from the now-defunct Bureau of Corporations. Figure 1.33 provides an example of one such memorandum. These documents were particularly helpful because they illustrated how contemporaneous regulators at the Bureau of Corporations viewed state antitrust law. I obtained copies of these documents during a visit to the National Archives in August 2022.

After identifying state antitrust statutes in effect between 1860 and 1940, I analyzed the content of these laws. For example, I coded damages authorized by state antitrust statutes as actual damages, double damages, or treble damages. In cases where the damages plaintiffs could seek differed according to the antitrust violation, I coded the larger allowable damages.[60] I also only coded statutes as declaring antitrust violations to be torts when the statute empowered individual people or corporations harmed by anticompetitive conduct to sue under the antitrust statute. If the statute, for example, only empowered the state's Attorney General to sue, I did not count the statute as declaring antitrust violations to be torts. One special case comes from Al-

[60]For example, a 1907 amendment to Indiana's antitrust statute allowed treble damages to be sought for injury due to restraint of trade, horizontal price fixing, horizontal output restriction, or monopolization. Meanwhile, an 1899 Indiana law continued to set maximum damages equal to the "costs [accrued to injured persons] and a reasonable attorney's fee" for injury due to refusals to deal. In 1907 and later, I code the damages authorized by Indiana's antitrust statute as treble damages, even though the 1899 statute continued to set damages for refusals to deal at a lower amount. Act of March 3, 1899, Chapter 148, §4, 1899 Indiana Acts 257. Act of March 11, 1907, Chapter 243, §7, 1907 Indiana Acts 490.

Figure 1.33: Bureau of Corporations Internal Memorandum on State Antitrust Laws



	I.	II.	III.	IV.	V.
Alabama	a,b,c,d,e.		a.		f
Alaska	i,k.	b.	b,e.	b.	
Arizona	i,k.	b.	b,e.	b.	
Arkansas	b,c,d.		c,d.	a,b.	a.
California	Legislation against control of live stock business only.				
Colorado	No legislation.				
Connecticut	No legislation.				
Delaware	No legislation.				
Dist. Columbia	i,k.	b.	b,e.	b.	
Florida	Legislation against control of live stock and fresh meat only.				
Georgia	~~b.~~ no legislation				
Hawaii.	i,k.	b.	b,e.	b.	
Idaho	b,~~e.~~ (constitution)				
Illinois	b,c,e.	a,d.	a,b.	a,b.	a.
Indiana		b,d.	a,b,c,d.	a,b.	
Ind. Ter.	i,k.	b.	b,e.	b.	
Iowa	a,b,c,d,e.	a,d.	a,b,c.	a,b.	
Kansas	a-k (inc)	a,b,c,d.	a,b,c,d.	a,b.	b,d.
Kentucky	b,c,e.	a,d.	a,b,c.		
Louisiana	~~i,k,o.~~		a,b.		
Maine	p.		a,e.		a.
Maryland	No legislation.				
Massachusetts	f.		a,b.		
Michigan	a,b,c,e.	a,b.	a,b,c,d.	a,b.	
Minnesota	b,c,d,f,i,m,n.	a,b.	a,b,c.	a,b,c.	e.
Mississippi	a,b,c,e,f,i,k,l,q,r.	a,b.	a,b,c.	a,b.	
Missouri	b,c,d,e,f,i,m.	a,b,d.	a,b,c,d.	a,b.	a,b,d.
Montana	b,e.				
Nebraska	~~Laws against~~ combinations controlling grain, stock, lumber and coal.				
Nevada	No legislation.				
New Hampshire	No legislation.				
New Jersey	No legislation.				
New Mexico	b,c,i,k.	a,b,d.	a,b,e.	b.	
New York	b,c,f.		a,b.	a.	b.
No. Carolina	a,b,f.		a,b.		a.
No. Dakota	b,c,f.		a,b,c.	a.	
Ohio	a,b,c,d,f,i.	a,b.	a,b,c.	a,b.	
Oklahoma	b,f,h,i,k.	b,d.	a,c,b,e.	b.	
Oregon	a.				
Pennsylvania	No legislation.				
Phil. Isds.	No legislation.				
Porto Rico(?)	i,k.	b.	b,e.	b.	
Rhode Island	No legislation.				
So. Carolina	b,c,d,e,f,i,...		a,b,c,d.	a,b.	b.
So. Dakota	a,b,c,f,h,i.	b.	a,b.	a,b,c.	
Tennessee	b,c	b,c.	a,b,c.	a,b.	
Texas	a,b,c,d,e,f,i,l,m,n.	a.	a,b,c.	a,b.	c.
Utah.	a,b,c,g,e.	b.	a,b,c.	a,b.	
Vermont	No legislation.				
Virginia	No legislation.				
Washington	~~a,b,e. (const.)~~ Legislation against combinations to control market for farm produce, etc.				
Wisconsin	b,c.		c.	a.	a.
Wyoming	No legislation.				
West Virginia	No legislation.				

Notes: This photograph depicts a page from a 1903 internal memorandum on state antitrust laws authored by staff at the Bureau of Corporations. Each column represents some aspect of the state's antitrust law. For example, column I tallies the acts declared illegal under each state's antitrust law, where each letter represents a different illegal act and the meaning of each letter is defined elsewhere in the document. I took this photograph during a visit to the National Archives.

abama's 1919 antitrust law, which allowed "such damages [to] be recovered as the jury [saw] fit to assess."[61] This language suggests damages could theoretically attain any value. However, since I encountered no other instances of states allowing effectively unlimited damages, I coded this statute as allowing the next-highest level of damages I coded (treble damages).

Coding the criminal penalties described in state antitrust statutes was mostly straightforward, but some rules I applied are as follows. In cases where a law described one set of minimum and maximum fines for an offender's first violation, and a different set of minimum and maximum fines for an offender's second violation, I counted the first set of fines in my coding. Further, though some states set different minimum and maximum fines for corporate offenders and individual offenders, I did not distinguish between the two fine types in my coding. Instead, I coded minima as $\min\{\text{minimum fine for corporations, minimum fine for individuals}\}$ and maxima as $\max\{\text{maximum fine for corporations, maximum fine for individuals}\}$.[62] Similarly, in cases where fines differed according to the antitrust violation, I determined minima and maxima across all violations.[63] Since I counted prison sentences in months, I coded 30-day minimum prison sentences as one month and one-year minimum prison sentences as 12 months. I also coded 24-hour minimum prison sentences as 1/30 or 0.0333 months.[64]

[61] Act of September 30, 1919, Chapter 741, 1919 Alabama Acts 1088.

[62] One exception to this rule comes from an 1890 Iowa law. In this case, I coded the individual fines rather than the corporate fines because the statute sets corporate fines as between 1 percent and 20 percent of the firm's capital stock. Whether this amounted to more or less than the individual fine would have depended on the firm. Act of May 6, 1890, Chapter 28, 1890 Iowa Acts 41.

[63] For example, an 1897 Indiana law established a minimum fine of \$100 for price fixing, while an 1899 Indiana law established a minimum fine of \$50 for refusing to deal. I thus coded Indiana's minimum fine for antitrust violations as \$100 in 1897 and 1898 and \$50 in 1899 and the years thereafter (until 1907, when the Indiana General Assembly adopted a wholly new antitrust statute). Act of March 5, 1897, Chapter 104, 1897 Indiana Acts 159. Act of March 3, 1899, Chapter 148, 1899 Indiana Acts 257. Act of March 11, 1907, Chapter 243, 1907 Indiana Acts 490.

[64] For example, Montana's 1909 antitrust law describes a 24-hour minimum prison sentence. Act of March 6, 1909, Chapter 97, 1909 Montana Acts 127.

1.10 Food Retail Price Analysis

Although not the central focus of the paper, this appendix presents a supplementary analysis of how state antitrust laws affected retail food prices. To facilitate this analysis, I digitized and assembled a panel of data on the prices consumers paid at retail stores for various foods in cities across the United States. The data span 1851 to 1911, with some gaps, and are drawn from Weeks (1886) and three subsequent reports published by the Department of Commerce and Labor.[65] The foods covered in this dataset (see Figure 1.34) reflect the consumption patterns of typical households during the study period and serve as a meaningful proxy for the cost of living in the late nineteenth and early twentieth centuries; Weeks (1886, 1) aptly describes them as "the chief necessaries of life."

To estimate the average effect of state antitrust law enactment on food prices, I use a stacked difference-in-differences design that accommodates variation in treatment timing across states.[66] For each of the 30 food items, I construct a separate dataset that stacks treated states around their year of adoption and includes both never-treated and not-yet-treated states as controls. The estimating equation is:

$$\ln(P_{ast}) = \beta_0 + \beta_1(statelaw_s \times post_{st}) + \Gamma X_{st} + \delta_{as} + \delta_{at} + \varepsilon_{ast} \tag{1.5}$$

where s denotes a state, t denotes a year, and a denotes a stack. The log retail price of a given food item is denoted by $\ln(P_{st})$, whether state s adopted an antitrust law is indicated by $statelaw_s$, and the time period after which a state antitrust law was adopted is denoted by $post_{st}$. The model also includes state fixed effects δ_s and year fixed effects δ_t, as well as a vector of time-varying controls X_{st}. Specifically, X_{st} includes state-level personal income (to account for changes in consumer purchasing power), and county-level population (to capture demand and urbanization), median occupational score (a proxy for local economic development), and the share of employment in agriculture (to account for variation in food production and consumption patterns). Standard errors are clustered at the state level. The coefficient β_1 captures the average treatment effect on the treated for each food item, netting out time-invariant state characteristics, national trends, and covariates.

The estimated effects of state antitrust laws on retail food prices are mixed overall, but the results suggest meaningful price declines for meat products and point to a substantial reduction in the cost of overall household expenditures on food. Of the 30 foods shown in Figure 1.34, six exhibit statistically significant effects at the 95 percent confidence level. Four of these six—bacon, fresh beef roast, fresh lamb or mutton, and fresh veal—show negative and significant coefficients, suggesting that the enactment of state antitrust

[65] The sources for this dataset are Weeks (1886) and Department of Commerce and Labor (1904, 1906, 1912). These reports provide annual price data for the years 1851–1880, 1890–1903, 1904–1905, and 1907–1911, respectively. I digitized the latter three reports myself and thank Michael Haines for sharing digitized data from Weeks (1886).

[66] See Section 1.4 for more information about the stacked difference-in-differences design I employ.

laws was associated with lower prices for these products. Notably, all four are meat products. Two additional meats—fresh beef steak and fresh chicken—also have negative coefficients that fall just short of statistical significance. Taken together, these results suggest that state antitrust laws may have been effective in reducing the prices of meat. This effect is plausibly linked to the intense public and political scrutiny of the Chicago meat packers in the late nineteenth century, which may have placed pressure on the leading firms to scale back price-fixing and other anticompetitive practices in response to the enactment of state antitrust laws.[67] Consistent with this pattern, a "representative food basket"—i.e., a weighted index of food retail prices based on household expenditure shares circa 1890, as documented by Haines (1989)—became approximately 27 percent cheaper following the enactment of a state antitrust law. While most individual food price effects are imprecisely estimated, the index-level result points to a potentially broad decline in consumer food spending.

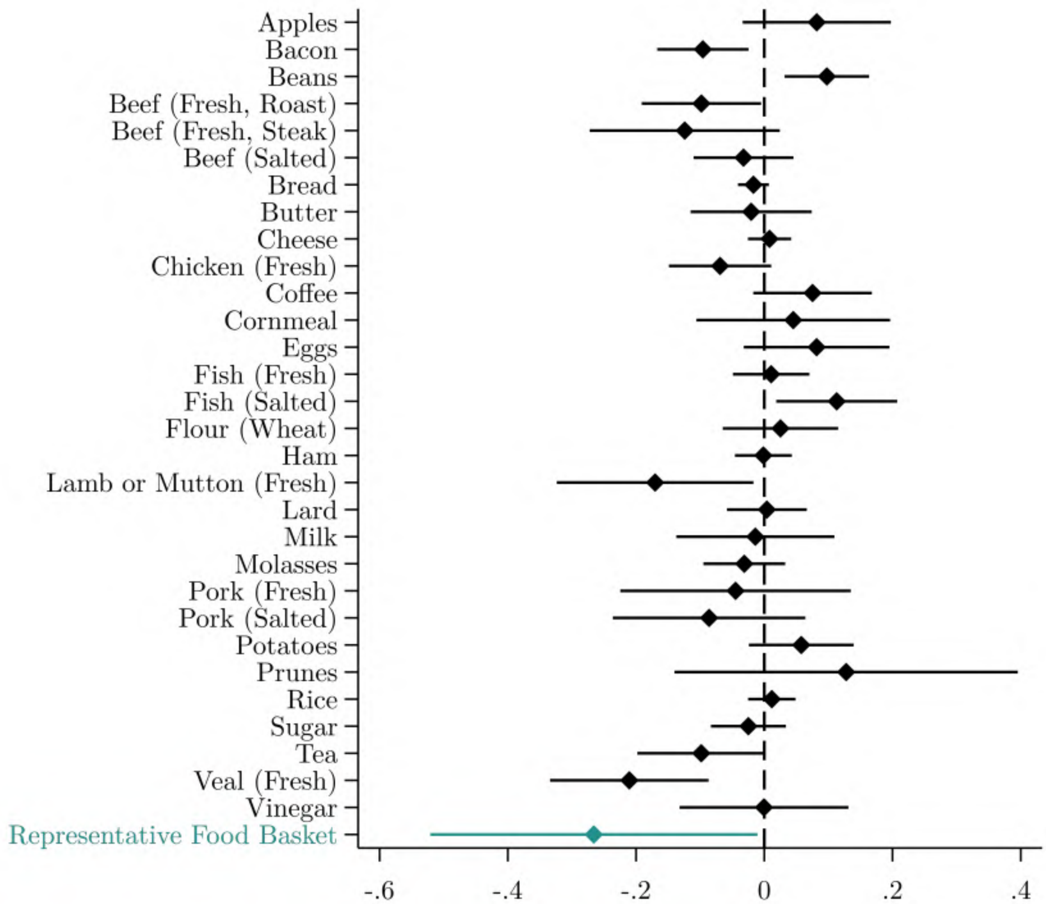
These results should be interpreted with some caution. As discussed in Section 1.2.3, the impact of an antitrust law depends in part on pre-existing market conditions. If the market for a given food was already competitive before the enactment of a state antitrust law, substantial price effects would not be expected. Accordingly, the many null results in Figure 1.34 is in line with the view that antitrust laws had limited scope to reduce prices in already competitive markets. Two results—positive coefficients for beans and for salted fish—run counter to this pattern, though a small number of counterintuitive results may arise by chance given the number of outcomes tested. On the other hand, laws introduced in markets that were initially cartelized or otherwise uncompetitive are more likely to generate price reductions through increased competition. One notable case is sugar, a product from a notoriously concentrated industry. The Sugar Trust, formed in 1887, controlled a dominant share of national refining capacity and was a frequent target of criticism and regulatory scrutiny.[68] Despite its notoriety, no state successfully sued the Sugar Trust under their antitrust statute.[69] Although the coefficient for sugar is negative, it is not statistically significant, offering a cautionary example of a highly concentrated industry where antitrust law may not have translated into measurable benefits for consumers—perhaps due to a lack of enforcement.

[67] In the late nineteenth century, meatpacking was dominated by a few large Chicago-based firms. By 1890, the four largest—Armour, Swift, Morris, and Hammond—accounted for about 89 percent of the national supply of dressed beef (Yeager, 1981, 67). Known collectively as the "Beef Trust," these firms processed a wide range of meats, not just beef, and were often accused of price-fixing, market division, and collusion with railroads. Public outrage over the Beef Trust made meatpacking a target of early state antitrust enforcement efforts. For example, in 1903, the Missouri Supreme Court found five major meat packers guilty of violating the state's antitrust statute and ordered them to cease operations in the state, pay $5,000 each in fines, and cover court costs. *State ex rel. Crow v. Armour Packing Co.*, 173 Mo. 356, 73 S.W. 645 (1903). Arkansas also brought several suits of a similar nature against the Beef Trust, including a notable case against Hammond that the U.S. Supreme Court decided in the state's favor. *Hammond Packing Company v. State of Arkansas*, 212 U.S. 322 (1909). See Willis (2021) for a detailed account of Arkansas's antitrust enforcement efforts during the Progressive Era. State-level enforcement actions, such as those by Missouri and Arkansas, may help explain the strong price effects observed for meat products.

[68] Henry Osborne Havemeyer, founder of the Sugar Trust, testified that 17 or 18 of 21 refineries belonged to the Trust, with the remainder accounting for about 30 percent of the national market (Industrial Commission, 1900).

[69] New York did famously dissolve a sugar refining company for joining the Sugar Trust. This action, upheld in *People v. North River Sugar Refining Company*, was brought as a *quo warranto* proceeding under corporate law rather than using antitrust law. 121 N.Y. 582 (1890).

Figure 1.34: Difference-in-Differences Results for Food Retail Prices



Notes: Each coefficient in this figure represents a separate estimation of equation 1.5, where the dependent variable is the log retail price of the listed good. Estimated values of β_1 from equation 1.5 are shown, and bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state level. Estimates are obtained using the stacked difference-in-differences method. The "representative food basket" refers to a price index constructed using the items consumed by a typical household around 1890, as documented by Haines (1989), where each food item is weighted by its share of household food expenditures. States whose antitrust laws were overturned or repealed are excluded. Controls capture state-level personal income and the county-level population, median occupational score, and share of employment in agriculture.

1.11 List of State Antitrust Statutes

In alphabetical order by state name, citations for the antitrust statutes I analyze in this paper are as follows:

1. Act of February 7, 1891, Chapter 202, 1891 Alabama Acts 438
2. 1907 Alabama Code, §§ 2487-2488, § 7581
3. Act of September 30, 1919, Chapter 741, 1919 Alabama Acts 1088
4. Act of May 18, 1912, Chapter 73, 1912 Arizona Acts 354
5. Act of March 16, 1897, Chapter 46, 1897 Arkansas Acts 60
6. Act of March 6, 1899, Chapter 41, 1899 Arkansas Acts 50
7. Act of January 23, 1905, Chapter 1, 1905 Arkansas Acts 1
8. Act of March 12, 1913, Chapter 161, 1913 Arkansas Acts 680
9. Act of March 23, 1907, Chapter 530, 1907 California Acts 984
10. Act of March 20, 1909, Chapter 362, 1909 California Acts 593
11. Act of April 7, 1913, Chapter 161, 1913 Colorado Acts 613
12. Act of August 15, 1911, Chapter 185, 1911 Connecticut Acts 1461
13. Act of June 4, 1915, Chapter 6933, 1915 Florida Acts 281
14. Act of May 13, 1925, Chapter 10283, 1925 Florida Acts 517
15. Act of December 23, 1896, Chapter 122, 1896 Georgia Acts 68
16. Act of March 11, 1909, 1909 Idaho Acts 297
17. Act of March 10, 1911, Chapter 215, 1911 Idaho Acts 688
18. Act of June 11, 1891, 1891 Illinois Acts 206
19. Act of June 20, 1893, 1893 Illinois Acts 182
20. Act of June 28, 1935, 1935 Illinois Acts 718
21. Act of March 5, 1897, Chapter 104, 1897 Indiana Acts 159
22. Act of March 3, 1899, Chapter 148, 1899 Indiana Acts 257
23. Act of March 8, 1901, Chapter 107, 1901 Indiana Acts 178
24. Act of March 11, 1907, Chapter 243, 1907 Indiana Acts 490
25. Act of March 6, 1923, Chapter 82, 1923 Indiana Acts 252
26. Act of April 26, 1888, Chapter 84, 1888 Iowa Acts 124
27. Act of May 6, 1890, Chapter 28, 1890 Iowa Acts 41
28. Act of April 5, 1907, Chapter 187, 1907 Iowa Acts 185
29. Act of April 12, 1909, Chapter 225, 1909 Iowa Acts 204
30. Act of March 9, 1889, Chapter 257, 1889 Kansas Acts 389
31. Act of March 8, 1897, Chapter 265, 1897 Kansas Acts 481
32. Act of March 14, 1907, Chapter 259, 1907 Kansas Acts 410
33. Act of April 2, 1909, Chapter 261, 1909 Kansas Acts 628
34. Act of February 24, 1919, Chapter 316, 1919 Kansas Acts 442
35. Act of November 20, 1933, Chapter 78, 1933 Kansas Acts 96
36. Act of May 20, 1890, Chapter 1621, 1890 Kentucky Acts 143
37. Act of March 15, 1916, Chapter 17, 1916 Kentucky Acts 74
38. Act of March 22, 1920, Chapter 51, 1920 Kentucky Acts 211

39. Act of March 23, 1922, Chapter 71, 1922 Kentucky Acts 217
40. Act of July 5, 1890, Chapter 86, 1890 Louisiana Acts 90
41. Act of July 7, 1892, Chapter 90, 1892 Louisiana Acts 120
42. Act of July 9, 1914, Chapter 288, 1914 Louisiana Acts 589
43. Act of June 10, 1915, Chapter 11, 1915 Louisiana Acts 23
44. Act of June 10, 1915, Chapter 12, 1915 Louisiana Acts 30
45. Act of March 7, 1889, Chapter 266, 1889 Maine Acts 235
46. Act of March 25, 1913, Chapter 106, 1913 Maine Acts 107
47. Act of June 7, 1901, Chapter 478, 1901 Massachusetts Acts 411
48. Act of April 28, 1908, Chapter 454, 1908 Massachusetts Acts 409
49. Act of May 27, 1911, Chapter 503, 1911 Massachusetts Acts 491
50. Act of May 27, 1912, Chapter 651, 1912 Massachusetts Acts 709
51. Act of May 28, 1913, Chapter 709, 1913 Massachusetts Acts 670
52. Act of July 1, 1914, Chapter 728, 1914 Massachusetts Acts 759
53. Act of July 10, 1919, Chapter 298, 1919 Massachusetts Acts 279
54. Act of May 27, 1921, Chapter 486, 1921 Massachusetts Acts 577
55. Act of July 1, 1889, Chapter 225, 1889 Michigan Acts 331
56. Act of June 23, 1899, Chapter 255, 1899 Michigan Acts 409
57. Act of June 16, 1905, Chapter 229, 1905 Michigan Acts 331
58. Act of June 20, 1905, Chapter 329, 1905 Michigan Acts 507
59. Act of February 25, 1911, Chapter 2, 1911 Michigan Acts 4
60. Act of May 2, 1917, Chapter 171, 1917 Michigan Acts 346
61. Act of April 20, 1925, Chapter 60, 1925 Michigan Acts 79
62. Act of June 6, 1935, Chapter 202, 1935 Michigan Acts 329
63. Act of April 20, 1891, Chapter 10, 1891 Minnesota Acts 82
64. Act of April 8, 1893, Chapter 125, 1893 Minnesota Acts 251
65. Act of April 21, 1899, Chapter 359, 1899 Minnesota Acts 487
66. Act of April 10, 1901, Chapter 194, 1901 Minnesota Acts 269
67. Act of April 22, 1913, Chapter 378, 1913 Minnesota Acts 527
68. Act of April 13, 1923, Chapter 251, 1923 Minnesota Acts 310
69. Act of February 22, 1890, Chapter 36, 1890 Mississippi Acts 36
70. 1892 Mississippi Code, §§ 4437-4442
71. Act of March 12, 1900, Chapter 88, 1900 Mississippi Acts 125
72. Act of March 5, 1902, Chapter 62, 1902 Mississippi Acts 100
73. Act of February 28, 1908, Chapter 119, 1908 Mississippi Acts 124
74. Act of April 2, 1910, Chapter 222, 1910 Mississippi Acts 221
75. Act of April 2, 1910, Chapter 223, 1910 Mississippi Acts 222
76. Act of March 30, 1916, Chapter 123, 1916 Mississippi Acts 180
77. Act of March 9, 1920, Chapter 313, 1920 Mississippi Acts 441
78. Act of March 19, 1926, Chapter 182, 1926 Mississippi Acts 285
79. Act of May 18, 1889, 1889 Missouri Acts 96

80. Act of April 2, 1891, 1891 Missouri Acts 186
81. Act of April 11, 1895, 1895 Missouri Acts 237
82. Act of March 24, 1897, 1897 Missouri Acts 208
83. 1899 Missouri Code, §§ 8978-8983, 8989
84. 1899 Missouri Code, §§ 8984-8988, §§ 8990-8992
85. Act of March 20 1907, 1907 Missouri Acts 374
86. Act of March 19, 1907, 1907 Missouri Acts 377
87. Act of March 19, 1907, 1907 Missouri Acts 382
88. Act of March 19, 1907, 1907 Missouri Acts 383
89. Act of March 29, 1913, 1913 Missouri Acts 549
90. Act of April 10, 1917, 1917 Missouri Acts 426
91. Act of May 22, 1919, 1919 Missouri Acts 618
92. Act of March 6, 1909, Chapter 97, 1909 Montana Acts 127
93. Act of March 29, 1889, Chapter 69, 1889 Nebraska Acts 516
94. Chapter 79, 1897 Nebraska Acts 347
95. Act of April 4, 1905, Chapter 162, 1905 Nebraska Acts 636
96. Act of May 29, 1939, Chapter 76, 1939 Nebraska Acts 312
97. Act of April 18, 1917, Chapter 177, 1917 New Hampshire Acts 698
98. Act of February 19, 1913, Chapter 13, 1913 New Jersey Acts 25
99. Act of April 9, 1920, Chapter 143, 1920 New Jersey Acts 285
100. Act of February 4, 1891, Chapter 10, 1891 New Mexico Acts 27
101. Act of March 14, 1907, Chapter 18, 1907 New Mexico Acts 19
102. Act of May 17, 1893, Chapter 716, 1893 New York Acts 1782
103. Act of April 15, 1896, Chapter 267, 1896 New York Acts 212
104. Act of May 7, 1897, Chapter 383, 1897 New York Acts 310
105. Act of May 25, 1899, Chapter 690, 1899 New York Acts 1514
106. Act of June 23, 1910, Chapter 633, 1910 New York Acts 1672
107. Act of May 6, 1918, Chapter 490, 1918 New York Acts 1561
108. Act of May 13, 1921, Chapter 712, 1921 New York Acts 2501
109. Act of August 26, 1933, Chapter 804, 1933 New York Acts 1635
110. Act of January 25, 1935, Chapter 12, 1935 New York Acts 20
111. Act of March 11, 1889, Chapter 374, 1889 North Carolina Acts 372
112. Act of March 8, 1899, Chapter 666, 1899 North Carolina Acts 852
113. Act of March 11, 1901, Chapter 586, 1901 North Carolina Acts 820
114. Act of March 11, 1907, Chapter 218, 1907 North Carolina Acts 254
115. Act of March 3, 1913, Chapter 41, 1913 North Carolina Acts 66
116. Act of March 3, 1890, Chapter 174, 1890 North Dakota Acts 503
117. Act of March 9, 1897, Chapter 141, 1897 North Dakota Acts 318
118. Act of March 13, 1905, Chapter 188, 1905 North Dakota Acts 336
119. Act of March 8, 1907, Chapter 258, 1907 North Dakota Acts 412
120. Act of April 19, 1898, 1898 Ohio Acts 143

121. Act of April 2, 1906, 1906 Ohio Acts 313
122. Act of May 10, 1910, 1910 Ohio Acts 274
123. Act of April 18, 1913, 1913 Ohio Acts 254
124. Act of April 29, 1913, 1913 Ohio Acts 405
125. Act of December 25, 1890, Chapter 84, 1890 Oklahoma Acts 1168
126. Act of June 10, 1908, Chapter 83, 1908 Oklahoma Acts 750
127. Act of March 29, 1913, Chapter 114, 1913 Oklahoma Acts 211
128. Act of March 29, 1919, Chapter 52, 1919 Oklahoma Acts 86
129. Act of February 25, 1897, Chapter 265, 1897 South Carolina Acts 434
130. Act of February 26, 1902, Chapter 574, 1902 South Carolina Acts 1057
131. Act of April 3, 1930, Chapter 837, 1930 South Carolina Acts 1396
132. Act of March 7, 1890, Chapter 154, 1890 South Dakota Acts 323
133. Act of March 6, 1893, Chapter 171, 1893 South Dakota Acts 282
134. Act of March 1, 1897, Chapter 94, 1897 South Dakota Acts 249
135. Act of March 8, 1909, Chapter 224, 1909 South Dakota Acts 348
136. Act of April 4, 1889, Chapter 250, 1889 Tennessee Acts 475
137. Act of March 30, 1891, Chapter 218, 1891 Tennessee Acts 428
138. Act of April 30, 1897, Chapter 94, 1897 Tennessee Acts 241
139. Act of March 23, 1903, Chapter 140, 1903 Tennessee Acts 268
140. Act of April 20, 1927, Chapter 60, 1927 Tennessee Acts 201
141. Act of March 30, 1889, Chapter 117, 1889 Texas Acts 141
142. Act of April 30, 1895, Chapter 83, 1895 Texas Acts 112
143. Act of May 25, 1899, Chapter 146, 1899 Texas Acts 246
144. Act of March 31, 1903, Chapter 94, 1903 Texas Acts 119
145. Act of April 11, 1907, Chapter 12, 1907 Texas Acts 16
146. Act of April 11, 1907, Chapter 87, 1907 Texas Acts 175
147. Act of April 15, 1907, Chapter 97, 1907 Texas Acts 194
148. Act of April 16, 1907, Chapter 120, 1907 Texas Acts 221
149. Act of April 25, 1907, Chapter 173, 1907 Texas Acts 322
150. Act of May 16, 1907, Chapter 10, 1907 Texas Acts 456
151. Act of April 7, 1909, Chapter 4, 1909 Texas Acts 266
152. Act of April 13, 1909, Chapter 11, 1909 Texas Acts 281
153. Act of February 23, 1917, Chapter 37, 1917 Texas Acts 63
154. Act of February 25, 1919, Chapter 30, 1919 Texas Acts 48
155. Act of January 30, 1923, Chapter 5, 1923 Texas Acts 12
156. Act of September 1, 1931, Chapter 44, 1931 Texas Acts 90
157. Act of March 9, 1896, Chapter 39, 1896 Utah Acts 125
158. Act of September 9, 1919, Chapter 54, 1919 Virginia Acts 82
159. Act of March 17, 1926, Chapter 171, 1926 Virginia Acts 314
160. Act of April 17, 1893, Chapter 219, 1893 Wisconsin Acts 264
161. Act of April 27, 1897, Chapter 357, 1897 Wisconsin Acts 909

162. 1898 Wisconsin Code, Chapter 84a, §§ 1747e-1747h
163. Act of June 20, 1905, Chapter 507 §7, 1905 Wisconsin Acts 944
164. Act of July 11, 1917, Chapter 646, 1917 Wisconsin Acts 1173
165. Act of July 23, 1919, Chapter 628, 1919 Wisconsin Acts 1097
166. Act of July 6, 1921, Chapter 458, 1921 Wisconsin Acts 761
167. Act of July 13, 1923, Chapter 406, 1923 Wisconsin Acts 693
168. Act of March 1, 1923, Chapter 86, 1923 Wyoming Acts 154

CHAPTER 2

A Fresh Look at the Origins and Impetus for Antitrust

This project is joint work with John Mayo.

2.1 Introduction

Economists' and lawyers' understanding of the origins and impetus of antitrust has historically been focused on the introduction and passage of the Sherman Act of 1890 (Scherer and Ross, 1990; Viscusi et al., 2018). This understanding has largely rested upon historical accounts (e.g., Thorelli, 1955) and analysis of the legislative history of the Sherman Act (e.g., Bork, 1966; Grandy, 1993; Hazlett, 1992). This focus has produced three alternative interpretations of the origins and drivers for the passage of the Sherman Act: (1) that the antitrust movement was driven by agrarian discontent, which was largely the product of farmers' antipathy toward railroads' monopolistic practices; (2) that the emergence of antitrust legislation was driven by small, non-agricultural businesses that became increasingly threatened by the growth of larger, more efficient businesses in the late 1800s; and (3) that never mind pressures from special interest groups, legislators acted in the public interest to craft legislation that would act unwaveringly to promote competition. The perpetuation of these alternative explanations is unsettling not only because they disagree, but also because they evoke quite different implications regarding the actual intent of the Sherman Act. Did the Act arise to protect competitors from competition or, alternatively, to more narrowly focus on protecting consumers?

Largely unnoticed in the development of this understanding, however, is the fact that a number of states within the United States adopted antitrust laws prior to the 1890 passage of the Sherman Act. This state-level heterogeneity introduces the potential for a significantly improved analysis and understanding of the origins and impetus of the antitrust movement. Accordingly, in this study, we turn away from the singular focus on the passage of federal legislation to instead redirect fresh attention to developing an understanding of the origins and impetus for state-level antitrust legislation, which occurred prior to 1890 for a number of states.

Our approach is generated principally by three features: First, we eschew the traditional path of reviews of the legislative history of the passage of the single initial federal legislation, instead employing the richness of state-level antitrust efforts that preceded the passage of the Sherman Act. Second, we employ the power of modern search methods—namely, scraping local newspaper accounts—to identify contemporaneous, and therefore more credible, explanations of the drivers of the original antitrust legislation. Third, rather than narrative explanations, we rely to the extent possible on quantitative analysis of contemporaneous state-level data to speak to the observed pattern of early antitrust in the United States.

This paper contributes to the existing literature in several ways. It first provides a fresh and systematic examination of newspaper accounts that highlight the tenor and type of concerns the public had toward monopolies in the late 1880s. Rather than hand-collecting newspaper accounts as has been done in the past (e.g., Gordon, 1963), we utilize a digitized collection of thousands of newspaper articles to provide a significantly more comprehensive account of the public's sentiment than was previously available in the literature. This examination sheds considerable light on the key concerns of the public regarding monopoly on the eve of the introduction of the country's first antitrust statutes.

Second, the paper examines the political and economic forces that gave rise to early antitrust legislation in the United States. As discussed in detail in Section 2.2.2, competing explanations attribute the origins of antitrust to interest-group lobbying by agricultural organizations seeking protection from railroads (e.g., Dilorenzo, 1985; Boudreaux and DiLorenzo, 1993; Boudreaux et al., 1995; Dilorenzo and High, 1988); the defensive efforts of small, non-agricultural businesses threatened by more efficient competitors (Baxter, 1980; Stigler, 1985; John, 2017); or popular outrage over trusts such as Standard Oil and the Whiskey Trust (Gordon, 1963; Lande, 1982; Millon, 1988; Werden, 2020; Letwin, 1965; Flynn, 1988; Lande, 1988). We advance this literature in several ways.

To begin, we assemble a harmonized and transparent dataset of pre-1890 state antitrust statutes, resolving long-standing inconsistencies in how prior scholars have defined and dated "antitrust" laws. We then move beyond narrative explanations by treating these early enactments as quasi-natural experiments, allowing us to reconcile conflicting historical claims with modern econometric methods. This approach allows us to extend and modernize Stigler's (1985) pioneering empirical framework, transforming his descriptive statistical exercise into a more comprehensive econometric analysis that quantifies the determinants of states' legislative behavior. Finally, we shift the principal unit of analysis from federal to state law, using cross-state variation in the passage of pre-Sherman statutes to uncover the conditions and interests that shaped early antitrust policy.

Another contribution of the paper lies in narrowing modern legal debates. Specifically, understanding the motivations behind antitrust laws is crucial given that courts continue to rely on legislative intent to guide their interpretation of these statutes. Our findings provide what we believe is the most systematic assessment of the forces that drove—and, therefore, the intent behind—the adoption of the first antitrust laws in the United States. That is, this analysis has the potential to more than simply clarify the historical account. Rather, it also has the prospect of informing modern debates that have emerged regarding the "original intent" of antitrust legislation in the United States.

The paper is organized as follows. Section 2.2 provides an introductory discussion of state-level antitrust activity prior to the passage of the Sherman Act. Of particular importance, Section 2.2.1 addresses the foundational question of what constitutional or statutory activities qualify as "antitrust" in the period of time prior to the Sherman Act. Section 2.2.2 also identifies various theories of the precipitators of state-level antitrust. In Section 2.3, we draw upon a robust digitized database of newspaper articles published in the decade prior to the passage of the Sherman Act to provide insights into the origins and impetus for state-level antitrust activities. Section 2.4 provides a quantitative analysis of pre-Sherman Act state-level antitrust activity with an eye toward providing additional clarity into the observed variation in states' adoption of antitrust prior to the passage of the Sherman Act. Section 2.5 concludes.

2.2 The Origins of Pre-Sherman Act State-level Antitrust

2.2.1 State-Level Antitrust

As noted by numerous scholars, the latter half of the nineteenth century was laden with both substantial changes in the industrial structure of the United States as well as the emergence of public policies that we today refer to as "antitrust." The goal of this paper is to provide additional insights into the causes and origins of antitrust in the United States. To do so, however, a preliminary question must be addressed; namely, what is meant by the term "antitrust?" For instance, does a simple declaration in a state's constitution that monopolies "ought not to be suffered" keynote antitrust law? An expansive interpretation of U.S. states' first antitrust activity would seem to include declarations, such as in Maryland, "[t]hat monopolies are odious, contrary to the spirit of a free government and the principles of commerce, and ought not to be suffered." After all, antitrust laws are often referred to as "anti-monopoly" laws.

The caveat to this interpretation, however, is that historically "monopolies" were often interpreted as stemming solely from governmental grants of exclusivity (See, e.g., Calabresi and Leibowitz, 2013), which would seem to exclude monopolies achieved through collusion, mergers and acquisitions, predatory pricing or other anticompetitive practices that are widely recognized to be the heart of modern "anti-monopoly" or antitrust laws. As observed by Millon (1990), "[t]hese [constitutional provisions] were typically little more than broad denunciations to the effect that monopolies are repugnant to a free people and should not be allowed. None of these, however, was concerned with the particular problems that the newly emergent trusts presented during the 1880s because the trusts were not state-franchised monopolies."[1] Thus, it can reasonably be inferred that constitutional provisions targeting "monopolies" (without other language that more broadly is aimed at practices that "defeat or lessen competition") did not, in fact, keynote antitrust activity.[2]

[1] There are exceptions; some states established granular prohibitions in their constitutions. For instance, the Idaho Constitution, adopted in 1889, made it quite clear that "combinations" established "for the purpose of fixing the price or regulating the production of any article of commerce or of produce of the soil, or of consumption by the people" were prohibited. See Idaho Const. art. 11, §18 (1889).

[2] The presence of simple, state-level provisions stating that monopolies "ought not to be suffered" may have relieved political pressure for more detailed prohibitions on anticompetitive practices. Conversely, the presence of a simple constitutional prohibition of "monopolies" may have signaled a keen interest by the state in preventing a wide array of anticompetitive practices, leading to early adoption of granular state antitrust legislation. In our empirical model developed in Section 2.4 below, we consider whether and how states' simple or more granular constitutional provisions affected the timing of the introduction of states' first antitrust statute.

Previous scholars that have chronicled state-level antitrust activity have offered conflicting accounts of early state antitrust legislation. Stigler (1985, 6) identifies 20 states that passed antitrust laws by the end of 1890, the year of the passage of the Sherman Act. In contrast, Collins (2013) identifies 13 states that had adopted their own antitrust law prior to the passage of the Sherman Act. Chen (1988, 94) and Millon (1990, 146) each identify 12 states that passed antitrust statutes prior to the passage of the Sherman Act. Seager and Gulick (1929, 341-343) state that "at least thirteen states" had antitrust statutes by this time. Others blend state antitrust statutes and antitrust-esque provisions in state constitutions. Dameron (2016) identifies 13 state antitrust statutes and five constitutional provisions adopted between 1888 and 1890. In Appendix 2.8, we systematically evaluate the existing studies that have produced these conflicting accounts of pre-Sherman Act state-level antitrust activity. This analysis indicates that the states identified in Table 2.1 passed an antitrust statute prior to the passage of the federal Sherman Act on July 2, 1990.

2.2.2 Alternative Theories of the Precipitators of State-Level Antitrust

A close examination of the initial impetus for antitrust statutes in the United States produces conflicting explanations. Bork (1966, 16) argues that Sherman himself "wanted the courts not merely to be influenced by the consumer interest but to be controlled completely by it." In a similar vein, Collins (2013, 2342) concludes that "[t]he state and federal antitrust statutes passed between 1888 and 1890 were enacted to deal with the increasing number of horizontal combinations—what would be called price-fixing cartels today—that had emerged throughout the economy in the 1870s and 1880s." Collins specifically contends that the emergence of state antitrust laws was a reaction to the liberalization of state corporation laws that increasingly made the formation of combinations in restraint of trade (heretofore proscribed, though not criminally, under common law) legal.

Also in this vein, Dameron (2016) indicates that "[t]he plain language of the Sherman Act's state predecessors shows that most were designed to promote what we now call consumer welfare. These statutes prohibited only those business combinations that harmed consumers by artificially constraining the supply of consumer goods." Dameron (2016, 1095) does not equivocate:

> "This Note departs from past approaches by examining the contents of the state antitrust statutes themselves. By and large, the statutory language does not support the view that the agricultural states of the South and West adopted antitrust legislation as protectionist or as 'anti-bigness' weapons against the advances of powerful corporations. The prairie populists—or at least the pieces of legislation they enacted—were doing something else: they were prohibiting only those combinations and business practices that harmed consumers through restrictions on the production of a given commodity or article of commerce. In other words, they were working toward a policy of consumer welfare."

In short, Dameron (2016) concludes that most states that moved to adopt antitrust statutes prior to the passage of the Sherman Act did so with the intent of promoting consumer welfare, only acting to proscribe business combinations that harmed consumers by restricting the supply of goods.

In contrast to the "consumer welfare" interpretation, the seminal work of Thorelli (1955) provides a story of agrarian discontent, with farmers turning into a political force as a consequence of railroads' monopolistic pricing practices. He concludes that "[p]erhaps the main root of western discontent was the permanent agricultural depression throughout this period." Stigler (1985), however, finds this explanation puzzling because, by his calculations, farm incomes were not particularly sensitive to railroads' financial returns. Moreover, Stigler (1985) argues that the price of freight rail services fell significantly in the decade prior to the introduction of the antitrust laws. Thus, Stigler (1985, 3) concludes that it was "perverse behavior" for farmers to lay their woes on the rail industry.[3] Stigler (1985) consequently investigates the potential driving force of two groups: (1) small, non-agricultural, businesses who may have stood to gain from the adoption broad antitrust laws capable of hamstringing not only monopolists but also large, efficient competitors; and (2) monopolists, who certainly stood to lose with the enactment of either narrow legislation that only targeted monopolization or broader legislation that enveloped legitimate business practices carried out by large firms.[4]

Boudreaux and DiLorenzo (1993) also argue that pre-Sherman Act antitrust legislation was driven by special interest groups and designed for their benefits. In particular, they identify farmers as a driving force. Note though that while Boudreaux et al. (1995) point to special interest groups as driving the law in Missouri, Dameron (2016, 1099), quoting the exact language of the Missouri statute, argues that:

> "First, agreements and combinations that did not reduce output and raise prices did not run afoul of the law. Second, Missouri lawmakers cared about price and output effects on particular commodities and articles of merchandise, not overall social output or price effects. Third, under the antimonopoly provision of section 2, a business organization had to act with the 'intent' to control output or prices. Monopolies achieved through productive efficiency, without any intention of controlling prices or output, did not violate the terms of the terms of the statute."

In summary, several competing theories have emerged over the years regarding the original impetus for the antitrust statutes. These include, most prominently, that the antitrust statutes arose: (1) in furtherance of consumers, consumer welfare, and competition; (2) as a result of discontent among the agricultural sector; and (3) as a populist-driven backlash against the emergence of large firms. Despite their outstanding differences, systematic scrutiny that would permit us to differentiate among them has largely been absent. It is to that effort that we now turn.

[3] Beyond the concerns of Stigler (1985), Gilligan et al. (1989) also observe that in the context of the consideration of the Interstate Commerce Act, farmers "were clearly not of one voice" with some farmers in the Midwest facing a single rail carrier, while those farmers west of Chicago enjoyed the benefits of long-haul competitive rail pricing. This divergence of interests may have similarly acted to diminish the political strength of the farm lobby in considerations of state antitrust legislation.

[4] In a similar vein, Baxter (1980) conjectures that antitrust law in general may be driven by the self-interest of small businesses, the antitrust bar, or public policy officials whose power would increase with the growth of administrative bureaucracy afforded by the passage and expansion of antitrust statutes.

2.3 Newspaper and Book Evidence

Attempts to sharpen the lens of after-the-fact sense-making of economic phenomena has, over the years, been immensely aided by research that examines contemporaneous accounts of historical events. As helpful as this archival research has been in shedding light on various economic phenomena, two features of this research methodology have been limiting. First, the process of identifying relevant contemporaneous press accounts was, until recently, a massively laborious effort. Consequently, the research necessarily was targeted at the accounts of a relatively small set of daily and weekly publications. Second, the research only anecdotally identifies geographic variations in public opinions. An article in the *Albany Journal* was likely to reflect the opinions of New Yorkers while an article from the *Cincinnati Commercial Gazette* was more likely to reflect the opinions of Ohioans. In recent years, however, technological advances have made it possible to overcome both of these limitations, enabling both searches of a substantially wider set of publications as well as the ability to geographically disaggregate variations in public opinion across the United States.

To develop greater coverage of contemporaneous newspaper articles addressing the growing presence and effects of monopoly in the United States in the 1880s, and to examine this with a geographically disaggregated lens, we rely upon two prominent sources of digital newspaper articles from the historical United States: Newspapers.com and American Stories.[5] Newspapers.com is a widely-used website that is part of the Ancestry portfolio, while American Stories is the product of an extensive programming effort to simplify access to the millions of newspaper scans that are part of the Library of Congress' *Chronicling America* collection (Dell et al., 2023).

Consistent with Gentzkow et al. (2011); Hopkins et al. (2017); Perlman and Schuster (2016); and Wang (2025), we interpret temporal and geographic variation in the intensity of newspaper accounts as reflective of public opinion, which, in turn, may drive legislative changes.[6] This interpretation finds specific support in the case of newspaper accounts of monopoly in the pre-Sherman Act period. Consider, for instance, Cook (1888, 53) describes the role of newspaper accounts of monopolistic trusts during the pre-Sherman Act period:

> "From the daily and weekly press of the country there will come a stormy advocacy of the people's rights which cannot be defied or suppressed or conquered. It is the reflector of public opinion. It is the medium of the voice of the people, and the unerring index of the thoughts and feelings and will of the masses. And it is a formidable indictment which the people, through the press, have brought against the Trust."

While the data we employ permit a more robust examination of public opinion toward monopoly in the 1880s than was previously available in the literature, it is not the first such examination. Thorelli (1955) offers a general summary of newspapers' treatment of monopoly in the period leading up to the passage of

[5]For a thorough review of other research that relies upon archival newspaper data, see Beach and Hanlon (2022).

[6]In a similar vein, (Thorelli, 1955, 138) argues that "[n]ewspapers perhaps more faithfully than any other media reflect the interests of the public."

the Sherman Act. He does so while moderating alternative viewpoints. On the one hand, Thorelli (1955, 136) writes of an impression that there was a "mighty tidal wave of discussion and agitation concerning the trust problem [that] surged through the nation's popular magazines before 1890." On the other hand, Clark's (1931) examiniation of newspaper coverage convinced him that there was little widespread interest in industrial consolidation, monopoly, and trusts in the years leading up to the passage of the Sherman Act. Thorelli (1955, 141) concludes that "[t]he persistence of this striking difference of views is easily explained by the fact that no exhaustive, or even systematic, survey of primary materials was ever undertaken by a representative of either one of these 'schools.'"

Akin to Thorelli (1955), Gordon (1963) evaluates both the contemporaneous literature on the emergence of industrial consolidations and trusts, as well as the attitudes of the public in the decade preceding the passage of the Sherman Act, by examining primary source materials. Gordon (1963, 157) concludes that the "evidence would seem to indicate that there were but few industrial combinations before 1887, and that from that date the movement for consolidations accelerated rapidly." To get a sense of public opinion regarding the emerging industrial consolidation, Gordon examines several sources: (1) the sources listed in a bibliography prepared by the Division of Bibliography of the U.S. Library of Congress relating to trusts; (2) newspaper articles in *The New York Times* and the *New York Daily Tribune*; (3) the *Public Opinion*[7]; (4) pamphlets and journals of special interest groups such as farmers, workers, and business people; and (5) relevant petitions and memorials entered into the *Congressional Record*. Gordon (1963, 157-158) ultimately concludes that:

> "Of all the sources examined, 1,647 in all, less than ten percent were written before 1887. Of the more than one thousand articles appearing in *The New York Times* and the *New York Daily Tribune*, less than five percent appeared before 1887. The magazine *Public Opinion* started publication in April of 1886 and carried eighty-two articles on combinations. All but eight of these articles were published in the two years preceding the passage of the Sherman Act."

Conversely, Letwin (1956, 224) argues that:

> "Those who have questioned this standard opinion have pointed out few books or journal articles were published on the trust problem before 1890, fewer, at least, than were printed afterwards. This fact led a leading historian of the Sherman Act, John Davidson Clark, to conclude that the public was not 'hysterical' but rather indifferent to the trusts. Unfortunately Clark exaggerated the evidence. Far more was printed than he realized, and especially in the newspapers. For instance, in February, 1888, *The New York Times* published articles and editorials about trusts on every day but one. If the *Times* was never again so persistent, neither did it become silent on subject, and it certainly did not—as Clark asserted—ignore the hearings of the congressional committee investigating the trusts. The most widely distributed newspaper in the Midwest, the *Chicago Tribune*, was at least assiduously interested. Moreover, there is evidence that other newspapers throughout the country were not much less attentive to the problem." [footnotes omitted]

While suggesting that press accounts of the emergence of antitrust sentiment in the 1880s were more prominent than Clark (1931) indicated, Letwin (1956, 225) offers an even more pointed criticism: "[a]ntitrust

[7] This was a weekly periodical that sought to provide a comprehensive summary of "the press throughout the world on all important current topics."

sentiment, in any case, cannot be measured merely by counting the number of articles concerned exclusively with trusts that appeared within the three years prior to the passage of the Sherman Act." He is certainly correct that counting articles on antitrust may not robustly capture the public's sentiment. Letwin (1956, 226) instead argues that the antitrust issue must be seen to reflect the broader concerns of the public, prominent among them the problem of "monopoly":

> "The pervasive antitrust sentiment did not spring up overnight. Hatred of monopoly is one of the oldest American political habits and like most profound traditions, it consisted of an essentially permanent idea expressed at different times. 'Monopoly,' as the word was used in America, meant at first a special legal privilege granted by the state; later it came more often to mean exclusive control that a few persons achieved by their own efforts; but it always meant some sort of unjustified power, especially one that raised obstacles to equality of opportunity. The trust was popularly regarded as nothing but a new form of monopoly, and the whole force of the tradition was focused against it immediately."

Following the admonition of Letwin (1956), we broaden our keyword search of newspaper articles to include "monopoly" for both conceptual and practical reasons. First, as a conceptual matter, the word "antitrust" was an outgrowth of the formation of the Standard Oil Trust in 1882 and the subsequent growth of trusts around the country. The more fundamental concern was with monopolistic exploitation, produced by the presence of monopoly. Second, as a practical matter, searching digitized newspapers for "antitrust" or "anti-trust" is fraught. In particular, "antitrust" and "anti-trust" keyword searches produce many false positives, indicating that an article addressed the topic of the keyword when, in fact, it did not.[8] Similarly, keyword searches can produce false negatives, indicating that there was no newspaper article addressing the topic of the keyword when, in fact, there was such an article.[9] Searches may also occasionally produce duplicate articles.[10] Finally, searches for a particular time range may sometimes produce articles that were published outside of the designated time range.[11]

Turning to the data, Figure 2.1 presents the annual count of the keyword "monopoly" in U.S. newspapers, based on data from Newspapers.com, for the period 1880 to 1889.[12] While the absolute number of matches rises over the decade, this trend coincides with an increase in the number of newspaper pages archived in the database. Notably, when adjusted for coverage, the frequency of "monopoly" mentions does not exhibit a clear upward trend leading up to the passage of the Sherman Antitrust Act in 1890.

However, this aggregate pattern conceals substantial variation across states. As Figure 2.2 shows, when

[8]For example, a Newspapers.com search for articles containing "anti-trust" in Ohio in 1887 produces a February 24, 1887 article in the *Cleveland Leader* that refers to a "Savings and Trust Company."

[9]For example, a Newspapers.com search for articles containing "anti-trust" in Kansas in 1887 identifies a December 3, 1887 edition of the *Bronson Pilot* that purports to have no matches, yet an article titled "The Farmer and the 'Trusts"' appeared in that newspaper on that day.

[10]For example, a Newspapers.com search for articles containing "anti-trust" in New York in 1887 identifies three articles published on December 5, 1887 in *The Evening World*. The articles are, however, duplicates.

[11]For example, a Newspapers.com search for articles containing "anti-trust" between 1880 and 1886 identifies several articles that were published in 1889 or 1890.

[12]Each appearance of the keyword is counted separately; a single article can contain several keyword matches.

the data are disaggregated by whether a state enacted antitrust legislation prior to the Sherman Act, a striking pattern emerges: states with earlier antitrust laws consistently display higher mentions of "monopoly" per newspaper page throughout the 1880s. Figure 2.3 provides a corresponding state-level map for 1887, the year before the pre-Sherman Act states began to adopt antitrust legislation. As the map indicates, public discussion of monopoly in 1887 was concentrated in the Midwest, where states were also among the earliest to adopt antitrust legislation.

Yet another lens emerges from an examination of newspaper articles published in the period immediately preceding the passage of state-level antitrust statutes. Specifically, granular scrutiny of contemporaneous newspaper articles discussing "monopoly" allows us to explore what specific practices or attributes of monopolies were most in the public discussion and would, therefore, be the most significant drivers of the pre-Sherman Act adoption of antitrust statutes by some states. So, for instance, was it a general antipathy toward monopoly that appeared most prominently in the contemporaneous press? Was it concern over the formation and consequences of specific trusts? Was it concerns that monopoly was leading to undesirable wealth redistribution? Was it concern about the impact of monopolies on farmers? Or, was it something else altogether?

To address these questions, we first identify all articles in American Stories which contain the word "monopoly" (or its derivatives) in 1887, the year prior to the passage of the first state antitrust statute.[13] This process identified 9,841 articles drawn from newspapers across 48 states, including those that adopted antitrust statutes before and after the passage of the Sherman Act.[14] Next, from these articles, we drew a sample of 150 articles. Reading these articles allowed us to craft a classification system into which each article could be assigned. The classification system, presented in Table 2.2, is designed to capture the concern expressed in the article regarding monopoly.

To extend our ability to engage in the widest possible examination of articles, we next drew upon ChatGPT-4-o, a generative artificial intelligence (AI) algorithm with the capacity to understand and interpret complex language well.[15] Specifically, we assigned the generative AI model the task of reading and classifying all of the articles identified by American Stories containing "monopoly" (or a derivative thereof) in 1887. This initial classification produced a quite low Cohen's Kappa when compared with our classification. Consequently, we next iterated in two ways. First, we re-read each of the 150 sample articles to explore why our assignment of a particular article differed from that of the AI algorithm. In most cases, we determined that the AI application failed to identify nuanced elements of the article. For example, a discussion in a Kansas newspaper indicating that the "Sante Fe Road" had "practically monopolized" business to Salina

[13] For this portion of our research, we were unable to secure the rights to download these data from Newspapers.com.

[14] Note that our filter does not exclude articles that express concerns regarding monopoly outside the United States.

[15] See Ochieng et al. (2024).

and Minneapolis, Kansas was rated as a concern over a land monopoly, when a more nuanced understanding of the prose reveals that this is not a concern about monopolization of a "Road" but rather by the "Sante Fe Railroad."[16] Similarly, an article the *Lambertville Record* criticized the term "lady doctors" while allowing "scrubladies, the wash ladies, and chamberladies, the salesladies, and the foreladies continue to monopolize the professional use of the word lady." Although this is clearly a non-commercial use of the term "monopolize" (i.e., Category 5) ChatGPT interpreted the article to be more consistent with "a general, nonspecific antipathy for monopoly" (Category 1), with a rationale that the article "reflect[ed] a broader critique of societal labels rather than a specific concern about monopoly in a particular industry." In a smaller number of cases, we found that the classification produced by AI more accurately captured the concerns of the article than did our initial classification. After our re-classification, there was substantial (though as described, not perfect) agreement between our classification and that of ChatGPT (Cohen's Kappa, $\kappa = 0.63$).

With our classification in hand, we next asked ChatGPT to read all articles published in 1887 in American Stories that contained the keyword "monopoly" (or derivatives thereof). In total, the AI algorithm read and classified 9,841 articles. The histogram for those categories, excluding the "not elsewhere classified" category, is shown in Figure 2.4.[17] Figure 2.4 provides several interesting insights. First, concerns over harm wrought by monopolies in specific industries represented the largest share of newspaper articles. Second, as conjectured by Letwin (1956), these concerns about monopoly in specific industries was layered on top of a prominent general antipathy for monopoly. If we consider general and specific monopoly-related concerns to approximate a "consumer welfare" perspective, a majority of articles (56 percent) are represented. Populist-related concerns represented a not-inconsequential 30 percent of monopoly-related articles. Farm-related monopoly concerns represented only a small fraction (13 percent) of monopoly-related articles. Informed by this inquiry, we turn in the next section to a quantitative analysis of the leading determinants of the drivers of states' early adoption of antitrust laws.

2.4 Quantitative Analysis of Pre-Sherman Act State-level Antitrust

Although scholars have produced several explanations for the causes of the antitrust movement, empirical scrutiny of these explanations has been noticeably thin. In his seminal study, Stigler (1985) identifies two potential interest groups with interest in the development and design of antitrust. First, small businesses, who—fearing the growing presence of larger and more efficient competitors—desired to advance an antitrust

[16] See *The Salina Journal*, November 24, 1887.

[17] A number of researchers have documented the potential for LLMs to produce different results in situations where the AI is asked to simply repeat the same exercise (e.g., Barrie et al., 2024). To ensure that our analysis does not suffer from this lack of replicability, we repeated our exercise, with the same instructions, allowing ChatGPT to produce another classification using the same data. We then calculated the Cohen's Kappa, κ, for these two classifications, finding that κ between the two AI-generated classifications was 0.95, indicating very substantial agreement. Although not *perfectly* replicable, the high κ provides a very high degree of qualitative replicability and confidence in the ability of ChatGPT to accurately classify our set of newspaper articles.

agenda that would challenge larger companies. Importantly, small businesses desired statutory language that would not only halt monopolistic practices that would harm consumers, but—more self-servingly—could be used to protect them from legitimate competition that harmed them. Antitrust was seen as a tool that could be designed as an "anti-bigness" weapon to protect not only competition but also competitors.

Second, firms endowed with sufficient monopoly power that they not only had the incentive, but also the wherewithal, to engage in monopolistic practices that harmed consumers had a vested interest in forestalling the passage of antitrust legislation. Under the broad statutory language favored by small businesses, large firms would have been constrained not only from engaging in genuinely monopolistic practices but also from competing aggressively on their merits—for instance, in cases where vigorous competition harmed competitors but not competition itself. Consequently, these firms desired either no statutory antitrust language or, in the alternative, language that granted state antitrust authorities only the power to protect consumers.

Although desirable to separately identify whether, and the extent to which, these two interest groups affected the design and implementation of state-level antitrust, Stigler (1985) finds that this separate identification is not possible due to a high negative correlation ($\rho = -0.86$) in his measures of the intensity of these two groups interests. He then concludes that if he finds an interest group influence he is only able to infer that it was one—or the other—of the two that was responsible. Stigler (1985, 4) then draws upon data on the intensity of "actual and potential" monopolies in states to shed light on whether they may have, in fact, delayed the passage of antitrust measures in states where their presence was more intensive.[18]

To test whether the pattern of states adopting state antitrust measures before the adoption of the Sherman Act is consistent with an interest group explanation, Stigler (1985) focuses on examining how the intensity of monopolies, by state, may have influenced the timing of states' adoption of antitrust legislation. In particular, he sorts states into those that had either an anti-monopoly or antitrust statute or constitutional amendment in place prior to 1890 and those that did not, and then further sorts these states into those that had a below average share of potential monopolists and those with an above average share of potential monopolists. Stigler (1985, 7) then performs a Chi-squared test of these groups, finding that "the evidence suggests that potential monopolists' opposition to the Act, or small business support for the Act, was operative in the manner predicted by the self-interest theory."

Our quantitative analysis consists of two principal steps. First, we reconsider Stigler's (1985) Chi-square test, replicating his analysis as closely as possible. Once this replication is complete, we re-perform the Chi-squared test, using those states identified in Table 2.1 as having passed antitrust statutes prior to the passage of the Sherman Act. In both cases, the Chi-squared tests produce the same qualitative results as found by

[18]His choice of the adjective "potential" is purposeful, as his measure draws on data on the intensity on monopolies in 1899, rather than the monopoly intensity at the time when these firms may have plausibly brought political pressure on state legislators to delay the adoption of anti-monopoly and/or antitrust measures.

Stigler.

Second, given the significant body of popular discussion of monopoly in the late 1880s, we extend Stigler's (1985) analysis to allow for the potential that early state-level adoption of antitrust legislation was driven by public opinion and motivated by a multitude of factors, including a widespread antipathy for monopolies and/or anger over monopoly abuses in specific industries, as expressed by greater amplification in contemporaneous newspapers. In particular, we specify a model of state-level adoption of antitrust statutes as:

$$PreSherman_s = \beta_0 + \beta_1 NPMon_s + \beta_2 SBInt_s + \beta_3 MonInt_s + \beta_4 AgInt_s + \beta_5 ArtCap_s + \varepsilon_s \tag{2.1}$$

where *PreSherman* equals 1 if state *s* adopted an antitrust statute prior to the passage of the Sherman Act, and 0 otherwise. Note here that the measurement of the dependent variable differs from Stigler (1985), as is described in detail in Appendix 2.8. *NPMon* measures the number of times "monopoly" is mentioned in the newspapers of state *s* during 1887, the year before the first state antitrust law was enacted. To estimate the influence of small, non-agricultural businesses across states, we adopt the exclusion method used by Stigler (1985, 5) to generate *SBInt*. This approach attempts to measure small-business employment within a state to proxy for the political prowess of small-business interests in that state.[19] To do so, Stigler (1985) begins with total employment and then removes employment in sectors dominated by large enterprises during the period: railroads, public utilities, and manufacturing and mining industries with an average of more than 100 employees per establishment. Additionally, he excludes sectors unlikely to have been threatened by monopolies, such as agriculture and certain large handicraft occupations. Finally, we normalize this measure by dividing by total employment.[20]

To capture the intensity of monopolies—denoted by *MonInt*—we follow the approach described in footnote 25 of Stigler (1985) with two exceptions. First, to identify the set of monopolized (i.e., potentially monopolized) industries, Stigler relies on Nutter (1949), who uses industry classifications from 1929, while we identify monopolized industries at the turn of the century using industry classifications from the manufacturing census. Consequently, it was necessary for us to construct a crosswalk to match Nutter's industries with those reported in the 1900 census. We use reports on industry classification from the census of manu-

[19]This approach was also endorsed by Baxter (1980, 44) who opined that "[t]he portion of the work force employed in [small businesses] might serve adequately as a proxy for the political significance of these groups in any particular district."

[20]Following this method, we begin with the total number of occupied persons in each state, as reported in Table XXXIII of Volume I of the 1880 census (Bureau of the Census, 1883b). From this total, we subtract the number employed in agriculture, professional services, and large handicraft occupations. To identify manufacturing industries with large establishments, we use Table I from Volume II of the census to determine which industries had an average employment greater than 100. The corresponding employment figures by state are then drawn primarily from Table II, and in some cases from Table IV. For specific industries such as cotton, wool, and glass, employment data are taken from the industry-specific Special Reports. For mining, we identify industries with large establishments using Table LXIV of the Compendium of the Tenth Census. However, since employment by state is not reported for each mining industry, we allocate national employment across states based on each state's share of total production volume, as reported in Table LXVII. The full list of excluded industries and occupations, along with the corresponding data sources, is provided in Appendix 2.9.

factures, which were published every 5 years from 1904 to 1929, to track changes in industry classifications. Our mapping is detailed in Appendix 2.9 in Table 2.6. We then find the data for the "average number of wage-earners" for different manufacturing industries from Table 4 of the 1900 census of manufactures. For state-level non-agricultural labor force figures, we use Table 31 from "1900 Census Special Reports: Occupations at the Twelfth census."[21] Second, Stigler's (1985) Chi-squared specification uses a variable that simply measures whether the intensity of monopolies in a state is "below average" or "above average." We instead take full advantage of the more continuous measure by including the percentage of a state's non-agricultural employment in the identified monopolized industries.[22]

To capture the potential role of the agricultural sector in driving early adoption of antitrust legislation, we include a measure of the agricultural employment relative to total employment, $AgInt$. Finally, Equation 2.1 also controls for newspaper articles per capita—denoted by $ArtCap$—to account for state-level variations in the exposure of the state's population to newspaper articles. Finally, the regression includes an error term, ε.

Importantly, this specification allows for two expansions of the original Stigler exploration. First, Stigler's (1985) test fashioned itself as a test of whether special interest—*either* potential monopolists or small, non-agricultural business—acted to influence the passage of antitrust legislation. Equation 2.1, however, permits us to separately evaluate the independent influence of each of these special interest groups. Second, although Stigler's (1985) empirical test is designed to tease out the potential influence of special interest groups, Stigler (1985, 5) readily acknowledges the more general economic maxim that "[e]conomic theory tells us, as it told Adam Smith, that output is at a maximum under competition, so it would appear that most people would be beneficiaries of a rule commanding competition." This suggests that "most people," as reflected in newspaper accounts of the day, constituted a viable interest group, capable of accelerating the passage of the country's first antitrust statutes.[23] In particular, consumer-welfare-oriented legislation would afford "most people" protection from monopolistic practices while also allowing them to enjoy the efficiencies generated by economies of scale that increasingly emerged in the latter part of the nineteenth century.

Table 2.3 provides the results from estimating Equation 2.1. The results provide suggestive evidence that contemporaneous public concern with monopoly—captured through the frequency of newspaper mentions—was positively and significantly associated with adoption of antitrust legislation. Across all specifications, the coefficient on the variable measuring thousands of mentions of "monopoly" is positive and statistically significant at conventional levels, consistent with the view that public salience of monopoly concerns played

[21] Specifically, we define the non-agricultural labor force as the difference between "Persons Engaged in Gainful Occupations" and those in "Agricultural Pursuits."

[22] While Stigler (1985) identifies high collinearity between his measure of small, non-agricultural businesses and his measure of the intensity of monopoly within states, we find the correlation between our counterparts to be notably lower ($\rho = 0.21$), permitting estimation of the potentially independent influences that these measures may have on the passage of pre-Sherman Act antitrust legislation.

[23] In doing so, we acknowledge the standard collective action problems first identified by Olson (1971), while opening the door to the potential for the popular press to reduce those hurdles.

an important role in prompting legislative action.

By contrast, the estimated effects for other proposed explanations for the rise of antitrust—small-business intensity, monopoly intensity, and agricultural intensity—are statistically indistinguishable from zero. This suggests that variation in these factors alone may not explain which states acted first. However, these null effects should not be over-interpreted as evidence of absence. The sample necessarily consists of only 43 observations—the number of U.S. states and territories for which all requisite data could be assembled—and as such, statistical precision is limited. With such a small N, even moderately sized true effects may fail to reach conventional levels of significance. Accordingly, these results should be read as suggestive: they highlight the plausibility of public opinion as a proximate driver of early antitrust adoption, while leaving open the possibility that business or agrarian interests also played a meaningful—if statistically undetectable—role.

In sum, the regression evidence complements the qualitative and historical record by showing that states where discussion of monopoly was most intense were also those most likely to enact pre-Sherman antitrust laws. While the small number of observations constrains formal inference, the results suggest that popular pressure, as expressed through the press, was a distinctive and measurable force shaping the earliest state-level antitrust initiatives.

2.5 Conclusion

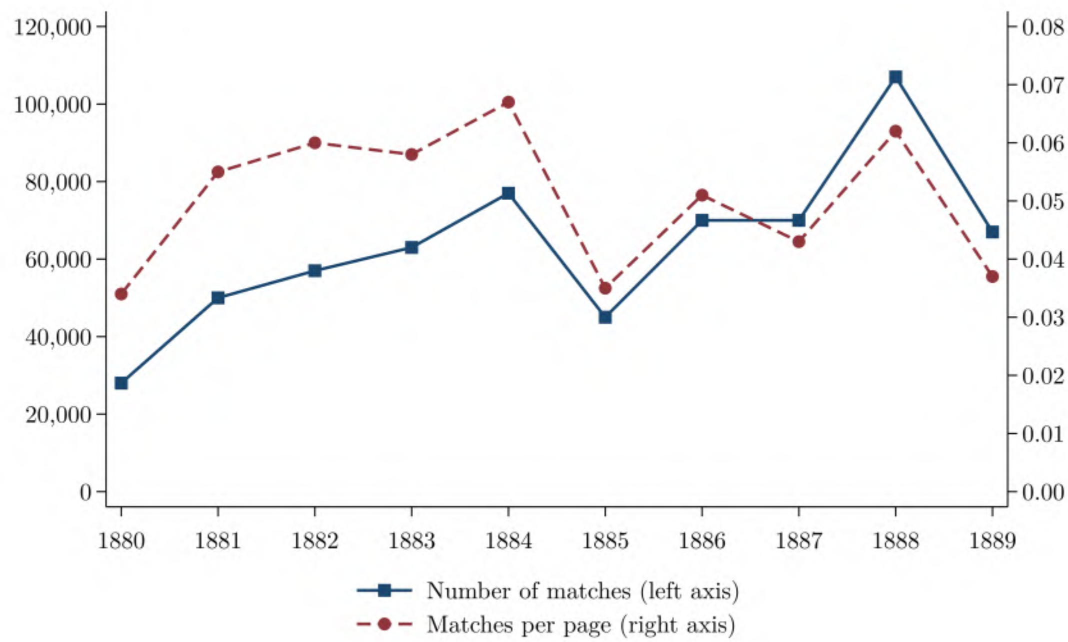
As has been described by numerous scholars, antitrust has enjoyed a nominal consensus for some time: that the principal, if not exclusive, motivating goal of antitrust was to protect consumer interests and competition. However, unresolved controversies have lurked beneath the surface. Prominent among them: Was the introduction of antitrust driven by the private interests of farmers and/or a portfolio of populist concerns centered on the preservation of small businesses?

In this paper, we have undertaken two approaches to providing additional evidence on this outstanding question. First, we focused on contemporaneous newspaper accounts from the period immediately prior to the passage of state-level antitrust legislation to shed light on the adoption of this legislation. This examination points largely (though not exclusively) toward the "consensus" perspective that the public's principal concerns were a combination of general antipathy toward monopoly and distress about the emergence of monopoly in particular industries. Contemporaneous newspaper accounts related to monopoly problems facing farmers in the period immediately prior to the passage of the pre-Sherman Act were sparse, suggesting that discontent among agrarian sectors may have been less important than has been previously suggested.

Second, building on Stigler's (1985) initial attempt to test the role of interest groups in the passage of state antitrust laws, we estimated a regression model examining how the relative strength of special-interest groups—namely, small, non-agricultural firms; would-be monopolists; and farmers—shaped adoption, compared to the broader "public interest," where we proxied the latter using newspaper mentions of "monopoly" and its derivatives. The regression results show that newspaper attention to monopoly is positively and significantly associated with pre-Sherman adoption of antitrust laws, while measures of small-business, monopoly, and agricultural intensity are statistically insignificant. These findings are consistent with the view that public salience, rather than organized special-interest pressure, best explains the pattern of early state antitrust enactments.

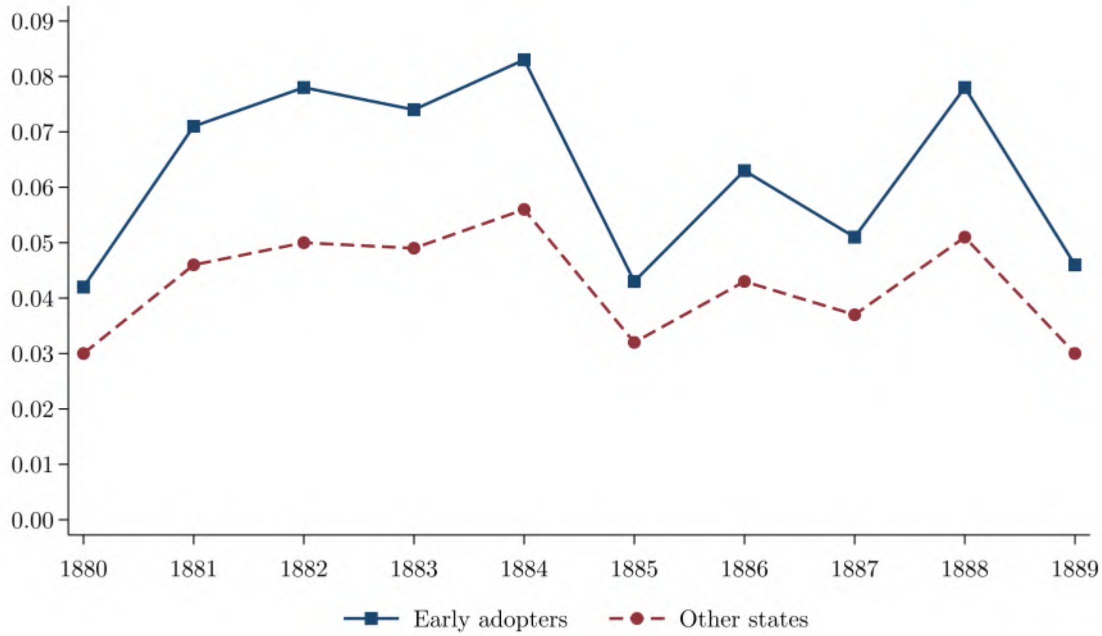
2.6 Figures

Figure 2.1: Matches for Keyword "Monopoly"



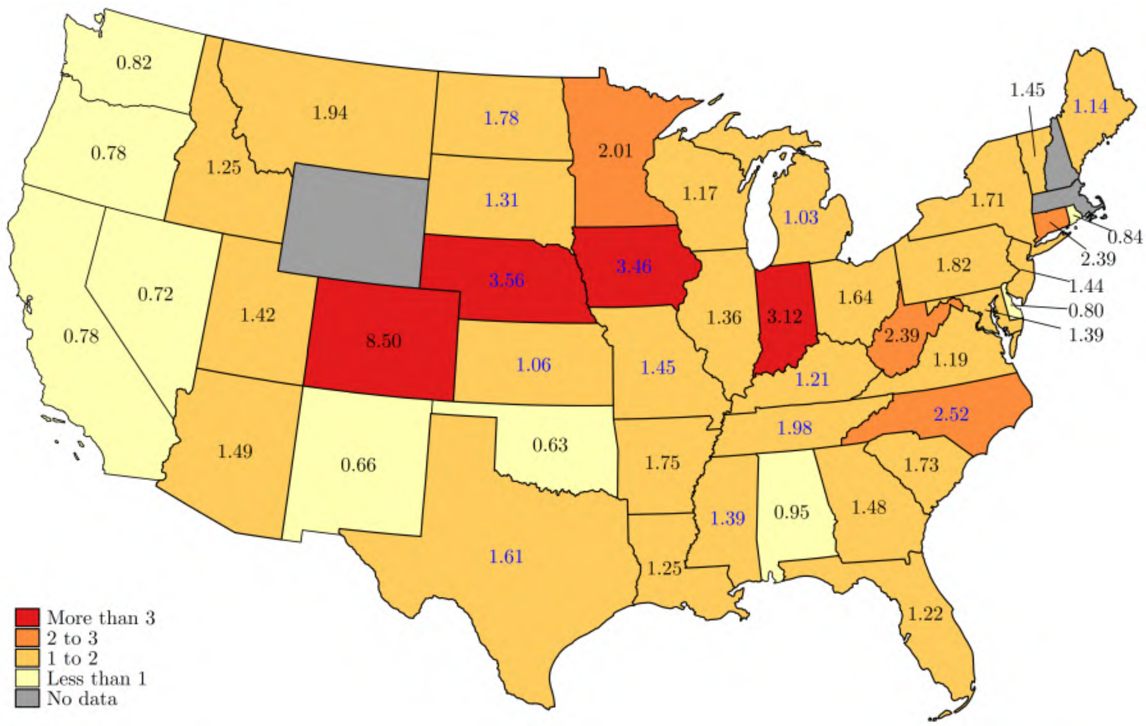
Notes: This figure plots the annual number of newspaper articles containing the keyword "monopoly" (left axis) and the number of matches per newspaper page (right axis) from 1880 through 1889. Data are from Newspapers.com. Each instance of the keyword is counted separately, and a single article may include multiple matches.

Figure 2.2: Matches for Keyword "Monopoly," by Enactment Timing



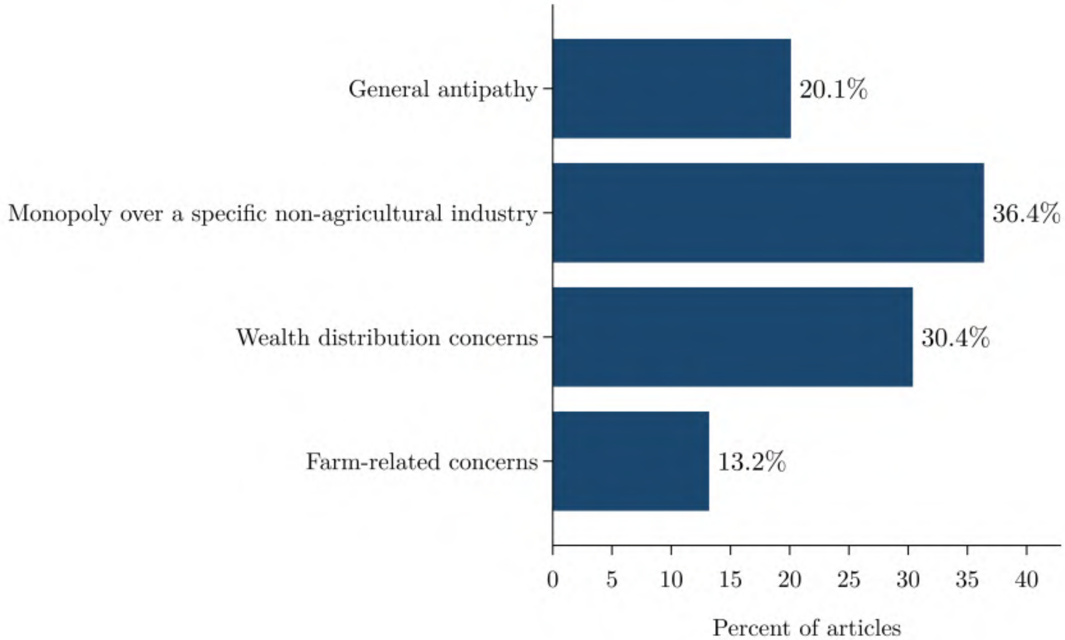
Notes: This figure plots the number of newspaper articles containing the keyword "monopoly" per page from 1880 through 1889, by enactment timing. "Early adopters" are states that enacted antitrust statutes prior to the Sherman Act of 1890. Data are from Newspapers.com. Each instance of the keyword is counted separately, and a single article may include multiple matches.

Figure 2.3: Map of Keyword "Monopoly" Matches per 1,000 Pages in 1887, by State



Notes: This figure maps the number of newspaper articles containing the keyword "monopoly" per 1,000 pages in 1887. Data are from Newspapers.com. Each instance of the keyword is counted separately, and a single article may include multiple matches. Blue labels mark states that would later adopt antitrust legislation prior to the Sherman Act of 1890.

Figure 2.4: Topic Distribution of Newspaper Articles Related to Monopoly



Notes: Among newspaper articles published in 1887 containing the keyword "monopoly" (or derivatives thereof), this chart shows the share pertaining to each topic, excluding the "not elsewhere classified" topic (category 5). Topics are defined in Table 2.2.

2.7 Tables

Table 2.1: Pre-Sherman State Antitrust Laws

State	Date of Enactment
Iowa	April 26, 1888
Maine	March 7, 1889
Kansas	March 9, 1889
North Carolina	March 11, 1889
Nebraska	March 29, 1889
Texas	March 30, 1889
Tennessee	April 4, 1889
Missouri	May 18, 1889
Michigan	July 1, 1889
Mississippi	February 22, 1890
North Dakota	March 3, 1890
South Dakota	March 7, 1890
Kentucky	May 20, 1890

Table 2.2: Classification of Articles Containing Keyword "Monopoly"

Category	Description
1	General, nonspecific antipathy for monopoly.
2	A specific concern about monopoly over a specific non-agricultural industry or trust(s), including but not limited to Bell Telephone, Standard Oil, natural gas, underground wire conduit, railroads/rail/railway, whiskey/beer/spirits/liquor, the Associated Press, salt, cable, tea, mining, and hotels.
3	Concerns about displacement of small businesses by larger ones, monopoly abuse of labor, small businesses, wealth distribution, or land monopoly.
4	Farm-related concerns, including farm-related monopolies and trusts (e.g., the Beef Trust, Barbed Wire Trust, Twine Trust, Cotton-Seed Oil Trust).
5	A concern about monopolies that is not most accurately captured by categories 1, 2, 3, or 4. This category includes articles that discuss the governmental grant of monopoly; the labeling of a politician or political party as "anti-monopoly" or "monopoly;" the use of the term "monopoly," "monopolized" or "monopolistic" in a non-commercial sense; the simple description or label of legislation; or foreign or indecipherable text.

Table 2.3: Main Regression Results

	(1)	(2)	(3)	(4)	(5)	(6)
Constant	-3.812	-0.213	-3.894	-0.165	-3.645	-0.065
	(2.825)	(0.606)	(3.038)	(0.658)	(3.150)	(0.656)
"Monopoly" mentions, 1000s	0.034^{**}	0.004^{**}	0.034^{**}	0.004^{*}	0.032^{*}	0.004^{*}
	(0.016)	(0.002)	(0.017)	(0.002)	(0.017)	(0.002)
Small business intensity	-1.880	-0.088	0.661	-0.226	0.324	-0.409
	(5.375)	(1.165)	(4.943)	(1.297)	(5.172)	(1.270)
Monopoly intensity	0.567	-0.008	-2.342	-0.063	-2.743	-0.191
	(4.766)	(1.061)	(5.809)	(1.116)	(5.643)	(1.112)
Agricultural intensity	5.312	0.998	5.649	0.993	5.548	0.932
	(3.565)	(0.777)	(3.794)	(0.809)	(3.921)	(0.805)
Newspapers per capita	-0.004	-0.001	-0.004	-0.001	-0.004	-0.001
	(0.012)	(0.002)	(0.012)	(0.002)	(0.012)	(0.002)
Granger law			0.244	0.033	0.233	0.029
			(0.868)	(0.260)	(0.882)	(0.257)
Constitutional provision			-0.235	-0.060	-0.485	-0.138
			(0.603)	(0.160)	(0.607)	(0.157)
Observations	43	43	43	43	43	43
R^2	0.319	0.182	0.323	0.139	0.333	0.154
Probit	✓		✓		✓	
OLS		✓		✓		✓
Dakota constitutional provision			✓	✓		

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable indicates whether a state had adopted an antitrust statute prior to the passage of the Sherman Act of 1890. Models (1), (3), and (5) are probit specifications; models (2), (4), and (6) are OLS. The "Granger law" variable equals 1 for states that enacted Granger-era railroad regulation laws in the 1870s. The "Constitutional provision" variable equals 1 if a state's constitution contained an explicit anti-monopoly clause prior to 1890. The "Dakota constitutional provision" row indicates whether the Dakota Territory is coded as having an anti-monopoly constitutional provision. (When statehood was granted to the Dakotas in 1889, North Dakota adopted an anti-monopoly clause in its first constitution, while South Dakota did not include such a provision until 1900—leaving it somewhat ambiguous as to whether the Dakotas should be counted as having an anti-monopoly provision in their state constitution prior to the Sherman Act.)

2.8 Reconciling State Antitrust Law Lists

A number of previous scholars have attempted to identify which states enacted antitrust laws before the passage of the Sherman Act in 1890, but they have reached different conclusions. For example, Stigler (1985, 6) identifies 17 states with antitrust laws before 1890 and 20 states by the end of 1890, while Collins (2013, 2336) lists only 13.[24] Chen (1988, 94) and Millon (1990, 146) each count 12 states with antitrust statutes enacted before the federal legislation. Seager and Gulick (1929, 341-343) assert that "at least thirteen" states had adopted antitrust statutes by the time the Sherman Act took effect. Others, such as Dameron (2016, 1077), distinguish between statutory and constitutional provisions, identifying 13 state antitrust statutes and five antitrust-like provisions in state constitutions that preceded the Sherman Act.

This appendix compares other scholars' lists of pre-Sherman antitrust laws to the list of state antitrust laws we count in this paper, explaining differences in classification, inclusion criteria, and treatment of constitutional provisions. Clarifying the precise enactment dates of state antitrust statutes helps to reconcile several of the discrepancies in different scholars' counts. Because Louisiana passed its antitrust statute in 1890 after the passage of the Sherman Act on June 2, 1890, by our count, 14 states passed an antitrust statute by the end of 1890, but 13 states passed an antitrust statute before the passage of the Sherman Act. Given this distinction, our list aligns with Collins (2013) and the list of statutory provisions in Dameron (2016). Our list of pre-Sherman state antitrust statutes also aligns with those given by Chen (1988) and Millon (1990) with the exception of North Carolina, which we include, but neither Chen nor Millon does.[25] Finally, our list aligns with the assertion of Seager and Gulick (1929, 341-343) that "at least thirteen" states had adopted antitrust statutes by the time the Sherman Act took effect; indeed, we find that *exactly* 13 such enactments took place.

Table 2.4 reconciles Stigler's influential list with the list used here, indicating whether each law Stigler (1985) counts is included in the present study, giving a rationale for the inclusion or lack thereof, and providing legal citations. Of the 20 states that Stigler (1985) identifies as having enacted antitrust laws by the end of 1890, we include 14 of these in our list—nine with the same years as Stigler and five with different years.[28]

[24] Boudreaux et al. (1995) also name 20 states as having enacted antitrust laws by the end of 1890. These scholars' list agrees with that of Stigler (1985) with one exception. Specifically, Boudreaux et al. (1995) date Arkansas's law to 1876, whereas Stigler (1985) places it in 1874. While Stigler (1985) presumably refers to the anti-monopoly provision in Arkansas's 1874 constitution, we could not identify any statute or amendment enacted in 1876 that plausibly aligns with the citation in Boudreaux et al. (1995).

[25] Chen (1988) does not mention North Carolina, while Millon (1990, 146) references an 1899 North Carolina statute. We instead include North Carolina's 1889 act "to prohibit trusts in the State of North Carolina, and to provide for the punishment of persons connected with them." Act of March 11, 1889, Chapter 374, 1889 North Carolina Acts 372.

[28] The states we count with the same years as Stigler (1985) are Kansas, Maine, Michigan, Missouri, Nebraska, North Carolina, Kentucky, Louisiana, and Mississippi. (Note that Louisiana passed its antitrust statute in 1890 *after* the Sherman Act was passed on June 2, 1890.) The states we count with different years as Stigler are Tennessee (Stigler counts an 1870 constitutional provision, while we count an 1889 statute), Texas (Stigler counts an 1876 constitutional provision, while we count an 1889 statute), Iowa (Stigler references an 1889 law that we could not locate, while we count an 1888 statute), North Dakota (Stigler counts an 1889 constitutional provision, while we count an 1890 statute), and South Dakota (Stigler counts an 1889 constitutional provision, while we count an 1890 statute). The remaining six states that Stigler identifies as having enacted antitrust laws by the end of 1890 are Maryland, Arkansas, Georgia, Indiana, Montana, and Washington. For each of these states, we count statutes enacted after 1890. Specifically, for Arkansas and Indiana, we count statutes enacted in 1897, while for Georgia and Montana, we count statutes enacted in 1896 and 1909, respectively. Maryland and Washington did not enact antitrust statutes until well into the twentieth century.

Table 2.4: State Antitrust Laws per "The Origin of the Sherman Act" (Stigler, 1985)

State	Year per Stigler (1985)	Year Agrees with Our Accounting	Presumed Citation
Maryland	1867	NO (Constitutional provision)	Md. Const. art. XLI
Tennessee	1870	NO (Constitutional provision)	Tenn. Const. art. I, §22
Arkansas	1874	NO (Constitutional provision)	Ark. Const. art. II, §19
Texas	1876	NO (Constitutional provision)	Tex. Const. art. I, §26
Georgia	1877	NO (Constitutional provision)	Ga. Const. art. IV, §5097
Indiana	1889	NO (No record found)	N/A
Iowa	1889	NO (No record found)	N/A
Kansas	1889	YES	Act of March 9, 1889, Chapter 257, 1889 Kansas Acts 389
Maine	1889	YES	Act of March 7, 1889, Chapter 266, 1889 Maine Acts 235
Michigan	1889	YES	Act of July 1, 1889, Chapter 225, 1889 Michigan Acts 331
Missouri	1889	YES	Act of May 18, 1889, 1889 Missouri Acts 96
Montana	1889	NO (Constitutional provision)	Mont. Const. art. XV, §20
Nebraska	1889	YES	Act of March 29, 1889, Chapter 69, 1889 Nebraska Acts 516
North Carolina	1889	YES	Act of March 11, 1889, Chapter 374, 1889 North Carolina Acts 372
North Dakota	1889	NO (Constitutional provision)	N.D. Const. art. VII, §146
South Dakota	1889	NO (Constitutional provision)	S.D. Const. art. XVII, §20
Washington	1889	NO (Constitutional provision)	Wash. Const. art. XII, §22
Kentucky	1890	YES	Act of May 20, 1890, Chapter 1621, 1890 Kentucky Acts 143
Louisiana	1890	YES	Act of July 5, 1890, Chapter 86, 1890 Louisiana Acts 90
Mississippi	1890	YES	Act of February 22, 1890, Chapter 36, 1890 Mississippi Acts 36
Alabama	1891	YES	Act of February 7, 1891, Chapter 202, 1891 Alabama Acts 438
Illinois	1891	YES	Act of June 11, 1891, 1891 Illinois Acts 206
Minnesota	1891	YES	Act of April 20, 1891, Chapter 10, 1891 Minnesota Acts 82
California	1893	NO (Does not meet inclusion criteria[26])	Act of February 27, 1893, Chapter 19, 1893 California Acts 30
New York	1897	NO (Earlier law exists)	Act of May 7, 1897, Chapter 383, 1897 New York Acts 310
Connecticut	†	YES	Act of August 15, 1911, Chapter 185, 1911 Connecticut Acts 1461
Florida	†	YES	Act of June 4, 1915, Chapter 6933, 1915 Florida Acts 281
Massachusetts	†	YES	Act of April 28, 1908, Chapter 454, 1908 Massachusetts Acts 409
New Hampshire	†	YES	Act of April 18, 1917, Chapter 177, 1917 New Hampshire Acts 698
Ohio	†	NO (Earlier law exists / No record found)	N/A
South Carolina	†	NO (Earlier law exists)	Act of February 26, 1902, Chapter 574, 1902 South Carolina Acts 1057
Vermont	†	NO (Does not meet inclusion criteria[27])	Act of April 1, 1915, Chapter 141, 1915 Vermont Acts 222
Virginia	†	YES	Act of September 9, 1919, Chapter 54, 1919 Virginia Acts 82
Wisconsin	†	NO (Earlier law exists / No record found)	N/A

Notes: Stigler (1985, 6) notes that the states marked with † enacted a law between 1900 and 1929, but "[e]xact dates...are not readily available."
Constitutional provision: The cited law appears in the state constitution, rather than in statutory law. This study limits its focus to statutory enactments.
No record found: No corresponding law could be located in state session laws, state constitutions, or published statutory compilations.
Does not meet inclusion criteria: The cited law was found but was excluded from this study for reasons explained in footnotes.
Earlier law exists: A different, earlier statute was identified and used in this study instead of the one cited by Stigler (1985).

Most of the discrepancies between our list and that of Stigler (1985) arise from the latter author counting state constitutional provisions, while we limit our focus to statutory enactments. Anti-monopoly provisions in state constitutions were typically brief and general in scope. As Figure 1.21 shows, they most commonly directed the state legislature to enact laws regulating monopolies, or declared monopolies, trusts, combinations, and restraints of trade to be unlawful in principle. However, these provisions did not, on their own, create criminal or civil liability, and designating a specific state official, such as the attorney general, to enforce the law was relatively uncommon. For these reasons, we exclude constitutional provisions from our count of state antitrust laws.

Table 2.5: Pre-Sherman Antitrust Statutes and Constitutional Provisions

	Anti-Monopoly Constitutional Provision		
Antitrust Statute	Yes	No	Total
Yes	4	9	13
No	9	26	35
Total	13	35	48

Notes: This table tallies the number of states that had antitrust statutes and/or anti-monopoly constitutional provisions before the Sherman Act was enacted on July 2, 1890. The 48 contiguous "states" are included in this tally, though some (Idaho, Wyoming, Utah, Oklahoma, New Mexico, and Arizona) were still territories at the time of the Sherman Act's passage.

Our decision to exclude constitutional provisions is also supported by the empirical pattern of adoption shown in Table 2.5: states did not treat constitutional provisions as substitutes for statutory enactments, but approached each independently. Only four states had both an antitrust statute and an anti-monopoly constitutional provision before the Sherman Act passed in 1890. Another nine had a statute but no constitutional provision, and nine had a provision but no statute. The remaining 26 had neither. These figures suggest states did not systematically treat statutes and constitutional provisions as complements or substitutes. Rather, adoption of one form of law was only weakly associated with adoption of the other. Some states used constitutional language to oppose monopolies without enacting legislation, while others passed statutes without embedding anti-monopoly language in their constitutions. Overall, the evidence in Table 2.5 indicates that states' enactment of antitrust statutes and anti-monopoly constitutional provisions were largely independent phenomena.

[26] California's 1893 statute applied only to the sale of livestock. We do not count statutes that applied to a single industry, rather than to all commerce, as antitrust statutes in this paper.

[27] Vermont's 1915 general incorporation law includes the following provision in §41: "The court of chancery within and for the county where the corporation has its principal office may, upon petition and after notice and hearing and upon such terms and conditions as it deems just, dissolve any corporation whenever it appears that its business transactions are repugnant to the law of this state. It shall so dissolve any corporation which has created a monopoly or unreasonably restrained competition in trade." We opted not to count this as an antitrust statute because it doesn't specifically outlaw anticompetitive conduct, such as price fixing.

The comparisons in this appendix underscore the ambiguity that can arise when sources differ in their treatment of constitutional provisions and narrow, industry-specific statutes. Without consistent standards, counts of early state antitrust laws can vary significantly, complicating research seeking to unpack these laws' origins. By applying clear inclusion criteria—focusing on statutory enactments of general applicability—and consulting original session laws, we offer a clarified accounting of state-level antitrust activity both preceding and following the enactment of federal legislation. This approach not only resolves discrepancies in the existing literature but also offers a firmer foundation for assessing the development of antitrust law in the United States.

2.9 Data Details

Table 2.6: Mapping Manufacturing Industries from 1929 to 1900

Industry from 1929 Used by Nutter (1949)	Corresponding Industries in 1900
Agricultural implements	Agricultural implements
Blast-furnaces products	Iron and steel
Cars, railroad, n.e.m.	Cars and general shop construction and repairs by steam railroad companies; Cars and general shop construction and repairs by street railroad companies; Cars, steam railroad, not including operations of railroads; Cars, street railroad, not including operations of railroads
Chewing gum; confectionery	Confectionery
Copper	Copper, smelting, and refining; Enameling and enameled goods; Stamped ware; Tin and terne plate; Tinsmithing, coppersmithing, and sheet-iron working
Cotton goods	Cotton goods; Cotton small wares; Cotton, compressing; Cotton, ginning
Electrical machinery	Electrical apparatus and supplies
Foundry and machine-shop products	Bells; Foundry and machine-shop products; Gas machines and meters; Hardware; Hardware, saddlery; Iron and steel, doors and shutters; Iron and steel, forgings; Iron and steel, nails and spikes; Iron and steel, pipes, wrought; Ironwork, architectural and ornamental; Plumber's supplies, n.e.s.; Plumbing, and gas and steam fitting; Pumps, not including steam pumps; Steam fittings and apparatus
Lead	Lead, bar, pipe, and sheet; Lead, smelting and refining
Leather	Leather board; Leather goods; Leather, tanned, curried, and finished; Pocketbooks; Saddlery and harness; Trunks and valises
Liquors, distilled	Liquors, distilled
Meat packing	Slaughtering and meat packing, wholesale
Paper; pulp	Paper and wood pulp; Pulp goods
Petroleum refining	Petroleum refining
Rubber goods, other	Rubber and elastic goods
Ships and boats	Ship and boat building, wooden; Ship building, iron and steel
Steel-mill products	Iron and steel
Tobacco products, other	Tobacco, chewing, smoking, and snuff; Tobacco, stemming, and rehandling
Woolen goods	Woolen goods
Worsted goods	Worsted goods

Table 2.7: Data Used to Measure Employment by Small, Non-Agricultural Businesses

Variable	Location in Bureau of the Census (1883b)
All occupations	Volume 1, Table XXXIII
Occupations excluded:	
Agriculture	
Professional and personal services	
Boatmen and Watermen	
Canalmen	
Clerks and bookkeepers in railroad offices	
Drayment, hackmen	
Employees of railroad companies	
Officials and empl. of street railroad firms	
Officials of telegraph companies	
Officials of telephone companies	
Officials of railroad companies	
Shippers and freighters	
Steamboat men and women	
Stewards and stewardesses	
Toll-gate and bridge keepers	
Blacksmiths	
Carpenters and joiners	
Plumbers and gasfitters	
Manufacturing industries excluded:	
Bagging, flax, hemp, jute; cars, railroad, street, and repairs; iron pipe, wrought; straw goods; sugar and molasses, refined; wire	Volume 2, Table II
Ammunition; belting and hose, rubber; boots and shoes, rubber; brass and copper, rolled; celluloid and celluloid goods; clocks; clothing, horse; envelopes; fire-arms; fruit-jar trimmings; glucose; jute and jute goods; rubber, vulcanized; sewing-machine cases; thread, linen; watches	Volume 2, Table IV
Cotton manufacturing	Report on Cotton Manufacturing of the United States by E. Atkinson, Table I
Wool hats; Worsted goods; Carpets, other than rags	Report on Wool Manufacture in All Its Branches by G. Bond
Iron and steel	Statistics of the Iron and Steel Production by J. Swank, Table VI
Glass	Report on the Manufacturing of Glass by J. Weeks, Table VI

CHAPTER 3

Federal Antitrust Enforcement and Industry Outcomes in the Early Twentieth Century

This project is joint work with Ezra Karger and Simcha Barkai.

3.1 Introduction

As large firms grow larger and concentration rises, some policymakers have suggested that antitrust policy could be a panacea for an economy lacking sufficient competition. The Sherman Act—the nation's first antitrust law, which remains the backbone of federal antitrust policy—has been in place for nearly a century and a half, yet the literature on the causal effects of antitrust enforcement lacks agreement. As a result, whether stronger antitrust enforcement is the cure-all some claim it will be remains to be seen. In the following chapter, we aim to shed light on this important and open question with historical evidence. Specifically, we study the effects of early federal antitrust enforcement on industry outcomes during the incipient years of U.S. competition policy.

We study the period from 1890—the year the Sherman Act was signed into law—through 1951, a time that several scholars have characterized as including the "golden age" of U.S. antitrust law.[1] As Figure 3.1 shows, enforcement of the federal antitrust laws ebbed and flowed over the course of the early twentieth century. In the first decade of the Sherman Act's existence, federal prosecutors pursued fewer than 20 antitrust cases. Enforcement picked up during the presidential administrations of Theodore Roosevelt and William Taft, the former of whom established the Bureau of Corporations to study industrial combinations and release reports on its findings to the general public. After lulls during World War I and the Great Depression, federal antitrust enforcement skyrocketed during the tenure of Thurman Arnold, who Franklin Roosevelt appointed in 1938 to serve as Assistant Attorney General for the Antitrust Division at the U.S. Department of Justice. Arnold aggressively enforced the nation's antitrust laws during his tenure.[2] However, by the early 1940s, World War II had taken center stage and political support for Arnold's enforcement campaign had run dry. Roosevelt appointed Arnold to the U.S. Court of Appeals for the District of Columbia Circuit in 1943, after which time federal antitrust enforcement declined sharply. Enforcement picked up again during the administration of Harry Truman, who vowed to "vigorously enforce the antitrust laws" in his State of the Union address in 1947 (Truman, 1947).

[1] For example, Scott and Abreu (2006) argue that 1897 to 1911, a period in which the U.S. Supreme Court began enforcing the Sherman Act and broke up the Standard Oil Trust, was antitrust's "golden age." Taking an alternate view, Tim Wu (2024) defined "peak antitrust" as the period from 1935 to 1981 in a recent lecture. Somewhat similarly, Stucke and Ezrachi (2017) situate antitrust's "golden era" in the period spanning the 1940s through the late 1970s.

[2] As Weber Waller (2005, 87) notes, the Antitrust Division's budget more then quadrupled and its staff grew from 18 employees to almost 500 employees between 1933 and 1943, the year Arnold left DOJ.

Our work contributes to a long-standing and unsettled literature on the causal effects of antitrust enforcement in the United States. Despite more than a century of policy experimentation and empirical research, scholars continue to disagree about whether enforcement efforts enhance competition and economic welfare. A substantial body of work finds evidence of benefits: antitrust enforcement can generate sustained increases in business formation and employment (Babina et al., 2023), stimulate long-term innovation by breaking up dominant research monopolies (Watzinger et al., 2020), and yield social welfare and consumer surplus gains that far exceed government enforcement costs and firms' compliance expenditures (Baker, 2003). Early studies also suggested that antitrust policy modestly reduced concentration and curbed monopoly profits (Stigler, 1966; Shepherd, 1982).

Yet a parallel literature has raised doubts about these optimistic conclusions. Several studies find that enforcement actions can raise rather than lower prices (Sproul, 1993; González and Moral, 2019), negatively shock productivity (Young and Shughart, 2010), or impose collateral losses on competitors of targeted firms (Bittlingmayer and Hazlett, 2000). Others argue that the consumer benefits of enforcement are difficult to detect empirically (Crandall and Winston, 2003) and that post-war trends in industrial concentration were largely unaffected by merger policy (Peltzman, 2014).

Against this backdrop, we revisit the question of how antitrust enforcement affects market structure and performance by turning to an earlier era in which federal enforcement took shape. By examining enforcement episodes between 1890 and 1951—when the Department of Justice and the federal courts were still defining the meaning and scope of the Sherman Act—we provide new historical evidence on whether and how early antitrust policy shaped industrial outcomes. Our analysis thus speaks to a broader debate that remains central to modern antitrust: under what conditions does enforcement meaningfully promote competition?

3.2 Data Sources and Descriptive Statistics

3.2.1 DOJ-Initiated Antitrust Cases

We assembled a dataset on all federal antitrust actions initiated by the Department of Justice (DOJ) from the Commerce Clearing House (CCH) *Trade Regulation Reporter*—also known as the antitrust "Blue Book"—a publication that contains brief summaries of all DOJ antitrust cases in order of filing date (Commerce Clearing House, Inc., 1952).[3] Our dataset spans July 2, 1890—the date the Sherman Act was signed into law—to January 31, 1952. Legal scholars and economists have long used CCH case data to characterize federal

[3]For each case, we collected a number of variables, including the section of federal antitrust law alleged to have been violated; the filing date; the court in which the case was originally filed; the names of any firms mentioned in the case description; the names of any individuals mentioned in the case description; the number of firms mentioned in the case description; the number of individuals mentioned in the case description; the type of violation; the outcome of the case; the court that oversaw appeals, if any; an indicator for whether the case was appealed to the U.S. Supreme Court; and the date the case was resolved. The violation types we coded are horizontal *per se* violations, exclusionary practices, price fixing, monopoly violations, vertical restraints, horizontal merger violations, and non-horizontal merger violations. See Table 3.5 for definitions of each of these.

antitrust enforcement.[4] We used the narrative information to match each case to the Standard Industrial Classification (SIC) codes corresponding to the industry or industries affected by the alleged anticompetitive conduct.[5] We also matched each case to the state or states affected by the alleged anticompetitive conduct.[6]

As Figure 3.2 shows, nearly half of the antitrust cases that DOJ initiated between 1890 and 1951 were in manufacturing, which motivates our use of data from the census of manufactures on the left-hand side of our models. Moreover, as can be seen in Figure 3.3, manufacturing industries represented a sizeable portion of all DOJ-initiated antitrust cases throughout the study period. Of the 58 years between 1890 and 1951 in which the DOJ initiated at least one antitrust case, all but seven of these 58 years included at least one enforcement action in manufacturing. Wholesale trade was the second-most affected industry category between 1890 and 1951.[7] About 16 percent of the antitrust cases that DOJ pursued between 1890 and 1951 were in wholesale trade-related industries.

Early federal antitrust enforcement was characterized by a tendency to resolve cases relatively swiftly and an emphasis on horizontal forms of anticompetitive conduct. Figure 3.4 shows that most DOJ-initiated antitrust cases were resolved quickly after filing. Over one-quarter (26.7 percent) concluded within the same year, and nearly two-thirds were resolved within three years. Only a small minority—less than 10 percent—remained open for more than six years. In keeping with this pattern, only a small subset of cases (148 out of 1,124 cases or about 13 percent) were appealed to the U.S. Supreme Court. Figure 3.5 shows that most DOJ-initiated antitrust actions between 1890 and 1951 targeted horizontal conduct. Over half of all cases involved alleged horizontal *per se* violations such as collusive agreements among competitors, followed by exclusionary practices and price fixing. Monopoly cases were also common but less frequent, while challenges to vertical restraints and mergers were comparatively rare.

3.2.2 State-by-Industry Manufacturing Outcomes

To analyze the effects of federal antitrust enforcement on the manufacturing sector, we digitized state-by-industry tabulations from the census of manufactures. We digitized data covering all 30 manufacturing censuses from 1850 to 1992, and we use data covering 1850 through 1947 in this paper.[8] The source documents

[4] Posner (1970), who examines federal antitrust enforcement from 1890 through 1969, was one of the first do do this. Another prominent paper using CCH case data is by Gallo et al. (2000), who examine federal antitrust enforcement through 1997.

[5] We matched about 52 percent of cases to a single SIC code, about 31 percent of cases to two SIC codes, about 13 percent of cases to three SIC codes, and about 4 percent of cases to four or more SIC codes. We used the industry descriptions in Technical Committee on Industrial Classification (1949) to identify the SIC codes corresponding to non-manufacturing industries and the industry descriptions in Technical Committee on Industrial Classification (1945) to identify the SIC codes corresponding to manufacturing industries.

[6] The narrative information in Commerce Clearing House, Inc. (1952) does not always clarify the state or states affected by the alleged anticompetitive conduct. In cases where the location cannot be ascertained, we use the state in which the case was filed as a proxy.

[7] The Technical Committee on Industrial Classification (1949, 57) defines wholesale trade as "[including] establishments or places of business primarily engaged in selling merchandise to retailers; to industrial, commercial, institutional, or professional users; or to other wholesalers; or acting as agents in buying merchandise for or selling merchandise to such persons or companies."

[8] There were 21 manufacturing censuses conducted between 1850 and 1947. These 21 censuses were conducted in 1850, 1860, 1870, 1880, 1890, 1900, 1905, 1909, 1914, 1919, 1921, 1923, 1925, 1927, 1929, 1931, 1933, 1935, 1937, 1939, and 1947.

for this dataset are listed in Table 3.2 and a dictionary of the variables included in this dataset is available in Table 3.3. Because establishment-level schedules are lost for most of the period we study, these data provide the most comprehensive historical record of U.S. manufacturing activity disaggregated by both state and industry. The census of manufactures did not introduce standardized industry codes until 1947, so we map historical industry descriptions to four-digit SIC codes using the contemporaneous Standard Industrial Classification Manual (Technical Committee on Industrial Classification, 1945).

These data have several advantages over the county-by-industry census of manufactures data that other scholars previously digitized (Lafortune et al., 2019). First, the state-by-industry data encompass all manufacturing activity in the United States, while the county-by-industry data are limited in geographic scope and include only the largest urban areas for most years.[9] For example, in 1880, the 100 largest cities in the United States are included, which produced about half of the nation's manufacturing output that year. The state-by-industry tabulations allow us to analyze all manufacturing activity, including activity outside of large urban centers. Given likely differences in industry composition in rural areas relative to urban areas, the county-by-industry data provide a limited scope for external validity. Second, the state-by-industry data are available at more regular time intervals. The county-by-industry data include only one census per decade from 1860 through 1940—a total of nine censuses—but the census of manufactures was conducted 19 times between 1860 and 1940. Digitizing the state-by-industry tabulations provides a more complete time series. Third, the state-by-industry data extend beyond 1940, the last year in the county-by-industry data. We add manufacturing censuses from 1947 through 1992 to the panel, allowing us to examine longer-term effects of the antitrust cases in our dataset. Fourth, the state-by-industry tabulations are an important dataset with a wide range of applications in historical economic research. We hope that these records will generate positive spillovers for other scholars in the field.

3.2.3 Labor Market Outcomes from Decennial Population Censuses

We also use decennial population censuses, which document the industries in which individuals were employed, to assess how federal antitrust enforcement affected labor market outcomes (Ruggles et al., 2021). Decennial population censuses have the benefit of being comprehensive and covering all industries in the economy, not just manufacturing industries. Thus, these data allow us to examine employment changes in

[9] The data digitized by Lafortune et al. (2019) cover each decade from 1860 through 1940. The published census volumes that were the original source for these data contain county-by-industry tabulations in 1860 and 1870 and city-by-industry tabulations in each decade from 1880 through 1940. County-by-industry tabulations provide data for all U.S. counties. City-by-industry tabulations exist only for the most populous cities in the United States, and the Census Bureau changed threshold for inclusion in these tabulations from census to census. Specifically, published census volumes provide city-by-industry data for the 100 largest cities in 1880, the 165 largest cities in 1890, cities with more than 20,000 inhabitants in 1900, and cities with more than 50,000 inhabitants in 1910 and 1920. In 1930 and 1940, statistics are provided for the "industrially important" cities in each state. Lafortune et al. (2019) provide a county-level dataset that matches cities to their corresponding counties and merges counties over time to ensure consistency in the geographic unit of analysis.

industries that experienced antitrust enforcement, before and after the enforcement took place. The 1940 and 1950 population censuses also include information on respondents' wage and salary income, which we use to examine the effects of later antitrust cases on wages. To merge our case dataset with the decennial census we mapped the SIC codes in our dataset to their corresponding North American Industry Classification System (NAICS) codes.[10] We then use the crosswalk provided by Ruggles et al. (2021), which maps NAICS codes to the `ind1950` industry codes used in the decennial population censuses.

3.3 Identification Strategy

We employ difference-in-differences and event study models to understand the effects of early federal antitrust cases.[11] The difference-in-differences model we estimate is given by

$$y_{ist} = \beta \times enforcement_{ist} + BX_{st} + \delta_{is} + \delta_{it} + \delta_{st} + \varepsilon_{ist} \tag{3.1}$$

and the event study model we estimate is given by

$$y_{ist} = \sum_{j \neq 0} \beta_j 1[t = j] \times enforcement_{ist} + BX_{st} + \delta_{is} + \delta_{it} + \delta_{st} + \varepsilon_{ist} \tag{3.2}$$

where i denotes an industry, s denotes a state, and t denotes a year. The variable $enforcement_{ist}$ is an indicator that switches from zero to one in the year the DOJ files an antitrust case involving industry i and state(s) s, remaining one in all subsequent years. The model also includes a vector of time-varying controls, X_{st}; industry-by-state fixed effects, δ_{is}; industry-by-year fixed effects, δ_{it}; and state-by-year fixed effects, δ_{st}. In all specifications, we cluster standard errors at the state-by-industry level.

Causal identification in equations (3.1) and (3.2) relies on the assumption that, absent federal antitrust enforcement, treated and untreated state-industry pairs would have followed parallel trends in the outcomes of interest. The inclusion of state-by-year, industry-by-year, and industry-by-state fixed effects helps mitigate concerns about differential shocks that vary along any two of these dimensions, such as state-specific business cycle conditions or industry-wide technological changes. Remaining variation comes from within-state-industry deviations from these broader trends, which are assumed to be unrelated to unobserved factors influencing both enforcement timing and outcomes. The event-study specification further allows visual inspection of pre-trends to assess the plausibility of the parallel trends assumption.

[10] Mapping SIC codes to NAICS codes can be difficult, as Yuskavage (2007) describes in detail. One reason for this difficulty is that the NAICS system replaced the SIC system in the 1990s, and industry codes generally become more specific over time, so a single SIC code sometimes maps to multiple NAICS codes. To overcome this issue, we reviewed the mappings carefully, referring to the original cases to select the most appropriate NAICS codes.

[11] Because DOJ initiated antitrust cases at different times across industries and states, treatment timing in this setting is staggered. In future versions of this paper, we will account explicitly for this staggered timing using recently developed econometric approaches for difference-in-differences and event-study designs.

3.4 Preliminary Results

Preliminary results provide little evidence that early federal antitrust actions meaningfully altered the competitive landscape in the manufacturing sector in the short to medium run. Panel A in Figure 3.6 shows that the number of establishments exhibited little systematic change in the years surrounding a DOJ filing. Coefficients are small and statistically indistinguishable from zero in most years before and after enforcement, suggesting limited effects on firm entry or exit. Panels B and C similarly provide limited evidence of declines in profitability following case filings. Figure 3.7 shows similar null effects on most labor market outcomes. Panel A provides an exception, showing a negative effect on average employment per establishment. However, the estimate does not attain statistical significance until 30 years after filing.

Turning to estimates of economy-wide impacts beyond the manufacturing sector, Figure 3.8 plots the effect of antitrust case filings on industry employment rates in the decades surrounding enforcement using data from decennial population censuses. Pre-filing coefficients are small and statistically indistinguishable from zero, indicating no evidence of differential pre-trends between industries later targeted by enforcement and those that were not. Following filing, the coefficients remain statistically indistinguishable from zero through forty years after enforcement. While employment appears to rise slightly ten and twenty years following enforcement, the estimated effects are not statistically significant. These results suggest that, on average, federal antitrust enforcement did not prompt a local reallocation of labor toward the targeted industries or an expansion in their employment share.

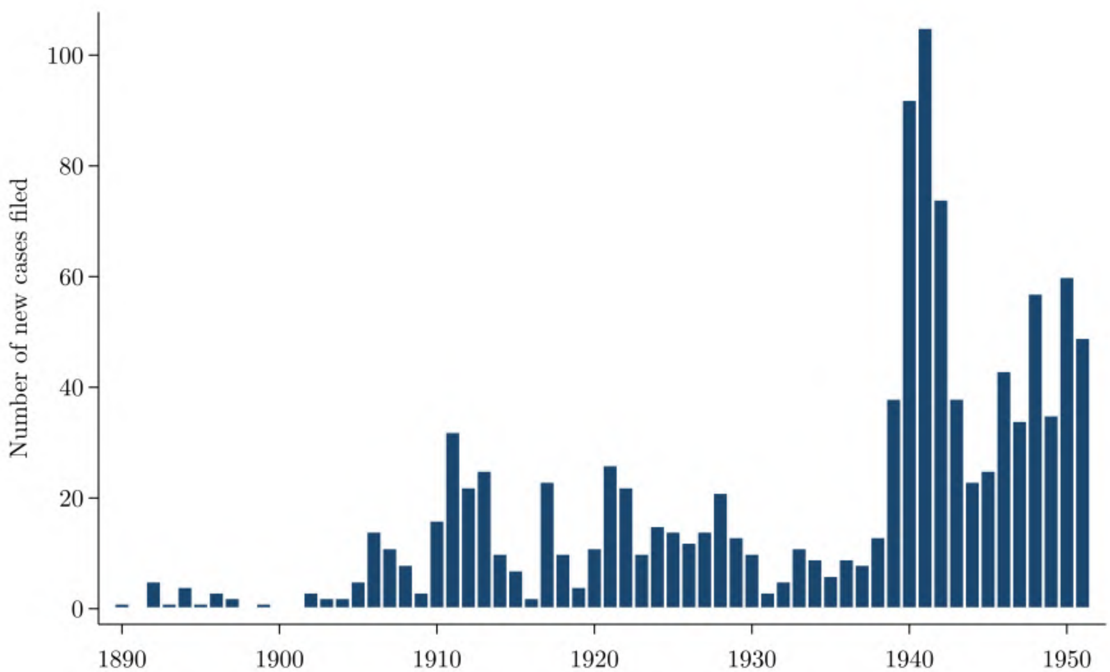
Figure 3.9 disaggregates these results by sector, reporting difference-in-differences estimates of the effects of federal antitrust enforcement on county-level employment rates. The estimated effects vary across industries, with five sectors exhibiting statistically significant changes. Of these, four—Manufacturing; Mining; Personal Services; and Transportation, Communication, and Other Utilities—show positive and significant effects, while only one sector—Wholesale and Retail Trade—displays a negative and significant coefficient. The remaining sectors have small, imprecisely estimated effects centered near zero. Taken together, these results suggest that antitrust enforcement was not associated with widespread changes in employment, but enforcement corresponded to increases in local employment within several key sectors.

3.5 Next Steps

This paper provides new evidence on the economic effects of early federal antitrust enforcement, yet several aspects of our empirical design warrant refinement in future iterations of this work. First, our coding of the geographic scope of each antitrust case likely misrepresents the true reach of enforcement. A large number of case descriptions provide few, if any, clues about the geographies affected, and in these instances we have defaulted to coding the relevant state as the one in which the case was filed. This approach almost certainly biases our estimates toward zero by misclassifying cases with regional or national reach as local in scope. Future work should explore alternative sources of information about the geographic reach of cases. Second, our current framework treats all antitrust cases as equally important, despite substantial heterogeneity in their scope and economic impact. Some cases—such as the breakups of major trusts—were landmark decisions that restructured entire industries, while others involved relatively narrow disputes or attempts to apply antitrust statutes to labor unions. Developing a measure of significance and weighting cases by this measure could yield more precise estimates of enforcement intensity. Addressing these issues will allow us to more accurately capture the intensity and reach of early federal antitrust enforcement and, in turn, better understand its role in shaping industrial development.

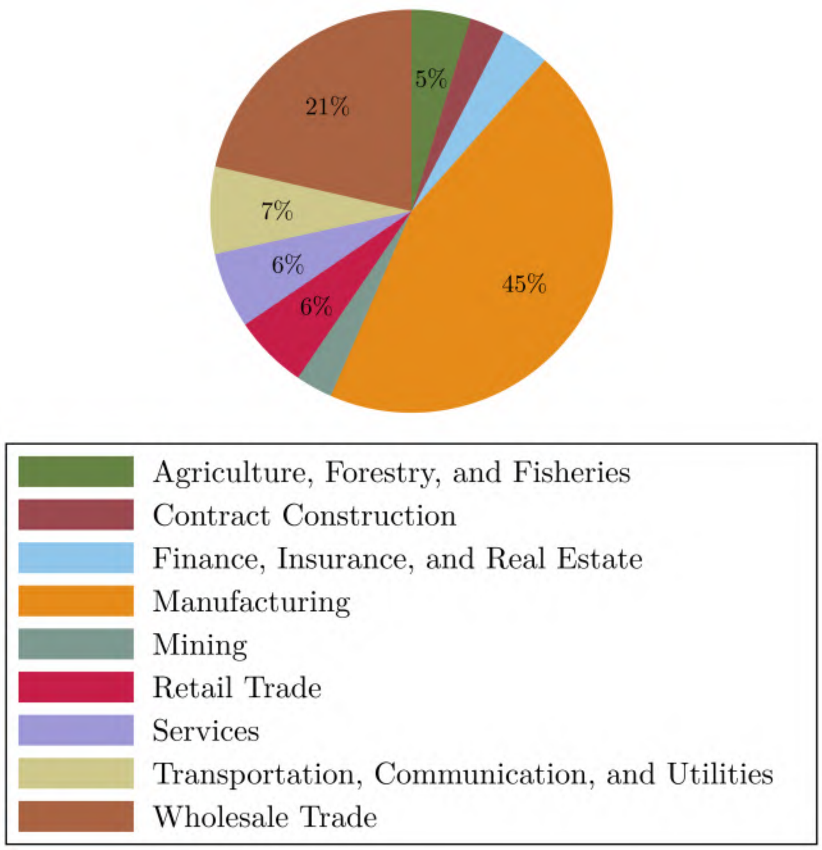
3.6 Figures

Figure 3.1: DOJ-Initiated Antitrust Actions, 1890–1951



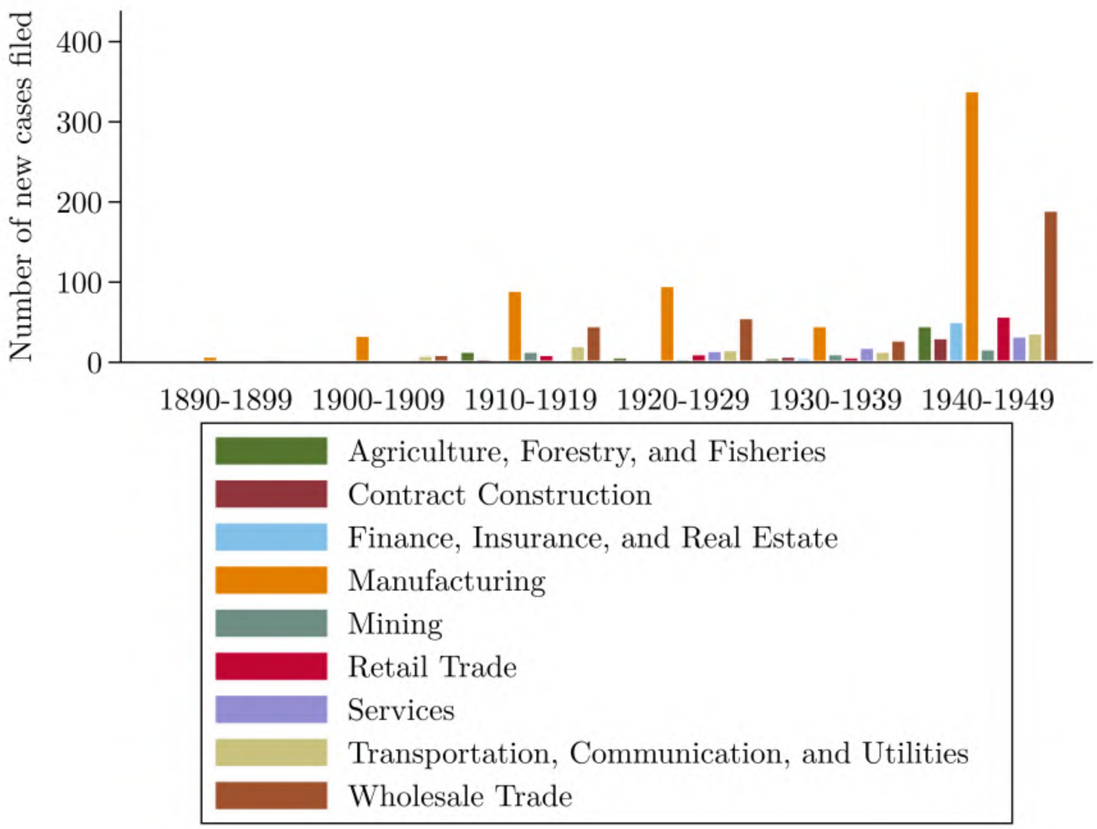
Notes: This figure shows the number of new antitrust cases the DOJ filed each year from 1890 through 1951. Cases filed in 1952 are not included in this graphic since our dataset only partially covers 1952.

Figure 3.2: Primary Industry Categories among Antitrust Cases, 1890–1951



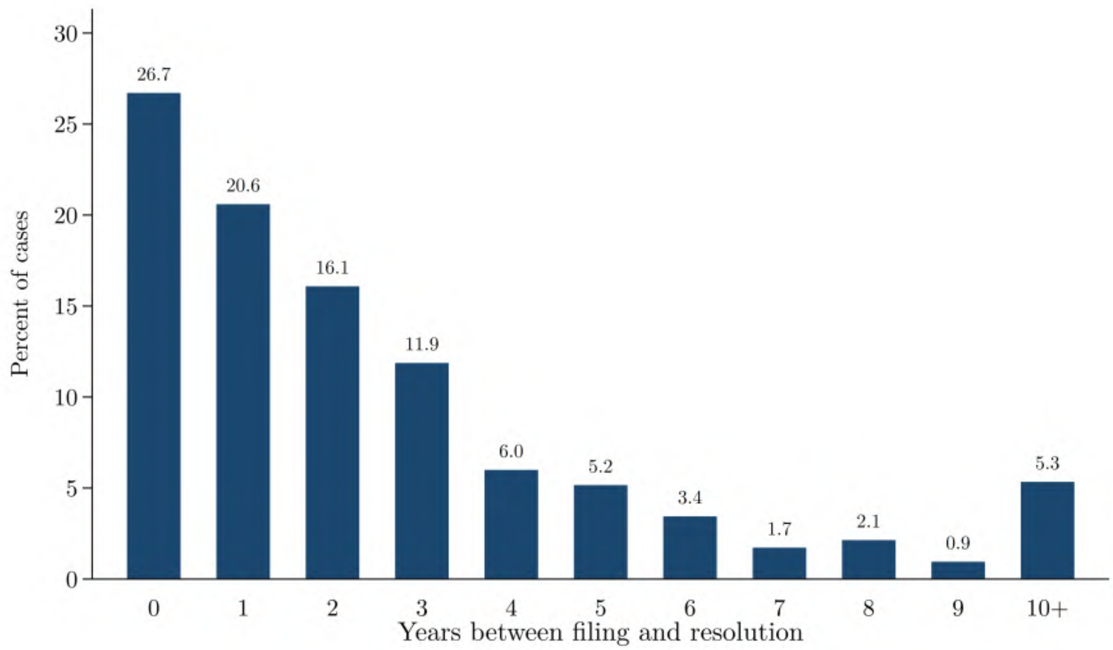
Notes: This figure shows the overall industry mix of DOJ-initiated antitrust cases from 1890—the year the Sherman Act was signed into law—through 1951. The industry categories shown in this figure are the SIC code "divisions" listed in Technical Committee on Industrial Classification (1949) and Technical Committee on Industrial Classification (1945). This figure represents about 75 percent of the antitrust cases in our dataset. In about 25 percent of cases, we were unable to identify a "primary" industry category because cases were coded as affecting multiple industry categories, and no industry category comprised the majority of industry category codes.

Figure 3.3: DOJ-Initiated Antitrust Actions by Primary Industry Category, 1890–1951



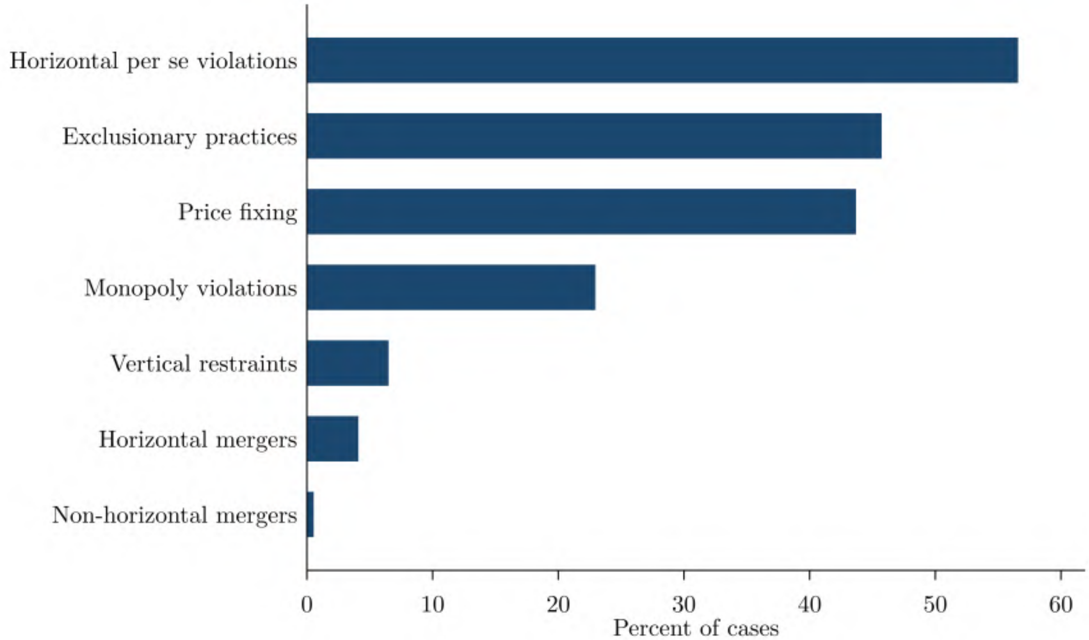
Notes: This figure shows how the overall industry mix of DOJ-initiated antitrust cases changed from 1890—the year the Sherman Act was signed into law—through 1951. The industry categories shown in this figure are the SIC code "divisions" listed in Technical Committee on Industrial Classification (1949) and Technical Committee on Industrial Classification (1945). "Missing" industry categories correspond to instances in which we were unable to identify a "primary" industry category because cases were coded as affecting multiple industry categories, and no industry category comprised the majority of industry category codes. In future iterations of this project, we hope to refine our industry coding to address these "missing" industry categories.

Figure 3.4: DOJ-Initiated Antitrust Actions by Case Duration, 1890–1951



Notes: The figure shows the share of cases resolved within each number of years after filing. Cases still pending at the end of the sample period are excluded from this distribution.

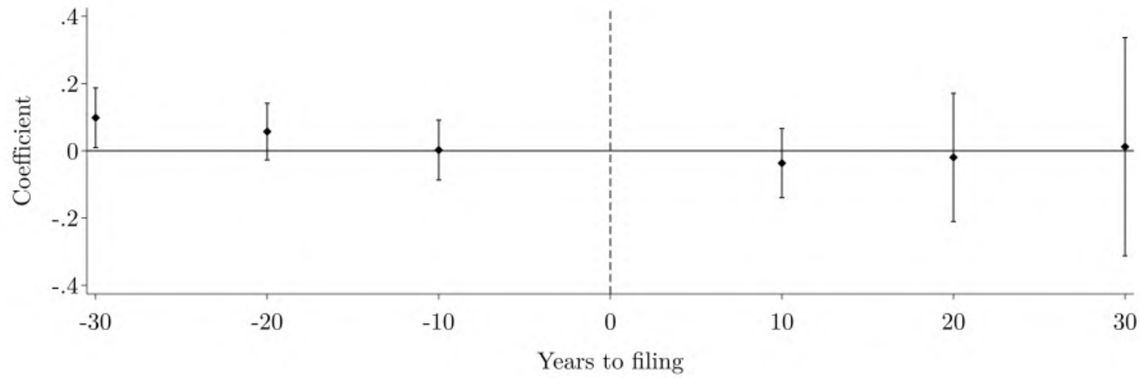
Figure 3.5: DOJ-Initiated Antitrust Actions by Violation Type, 1890–1951



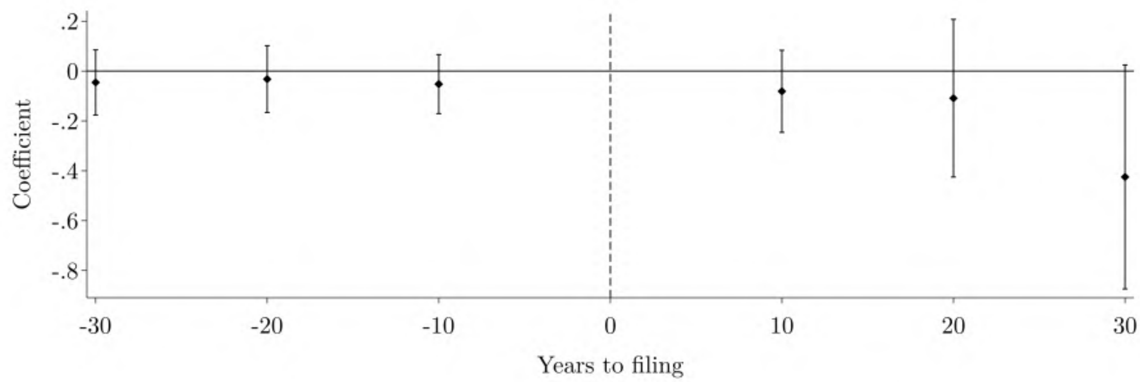
Notes: Cases may involve multiple types of alleged violations; categories are not mutually exclusive. Violations are defined in Table 3.1.

Figure 3.6: Event Study Results for Establishment Outcomes in the Manufacturing Sector

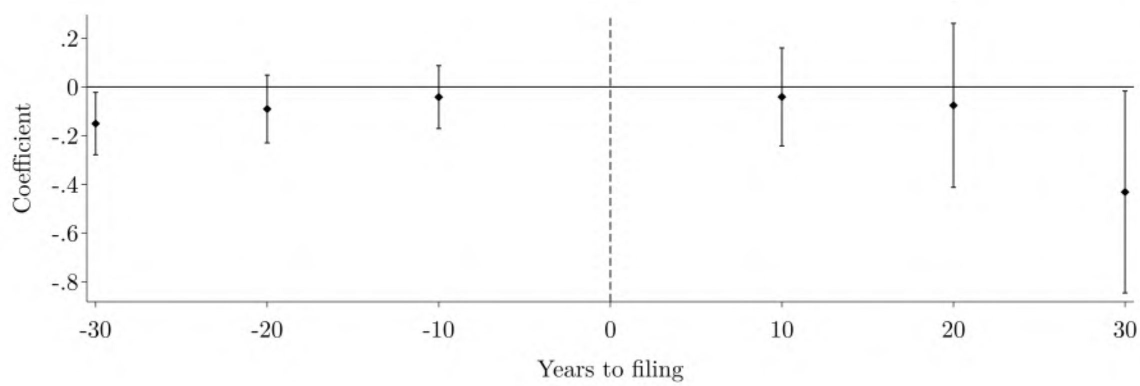
Panel A: Log Number of Establishments



Panel B: Log Total Profit



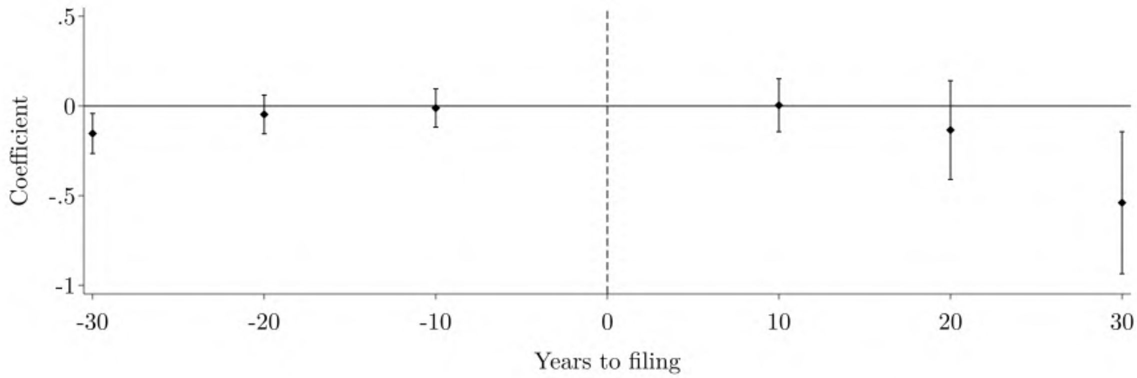
Panel C: Log Average Profit per Establishment



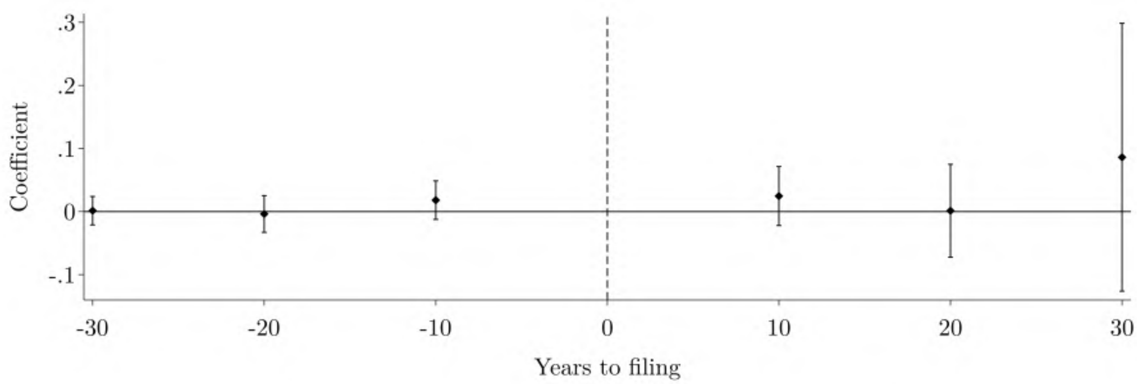
Notes: Data are from state-by-industry tabulations from the census of manufactures. Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state-by-industry level. These results do not yet include controls. Industry-by-state, industry-by-year, state-by-year fixed effects are included. Observations are weighted by both the inverse frequency of the census of manufactures and revenue. Specifically, observations from decennial censuses receive a weight of 1, those from quinquennial censuses receive a weight of 0.5, and those from biennial censuses receive a weight of 0.2. Observations are then additionally weighted by their share of manufacturing revenue in the observed census year.

Figure 3.7: Event Study Results for Labor Market Outcomes in the Manufacturing Sector

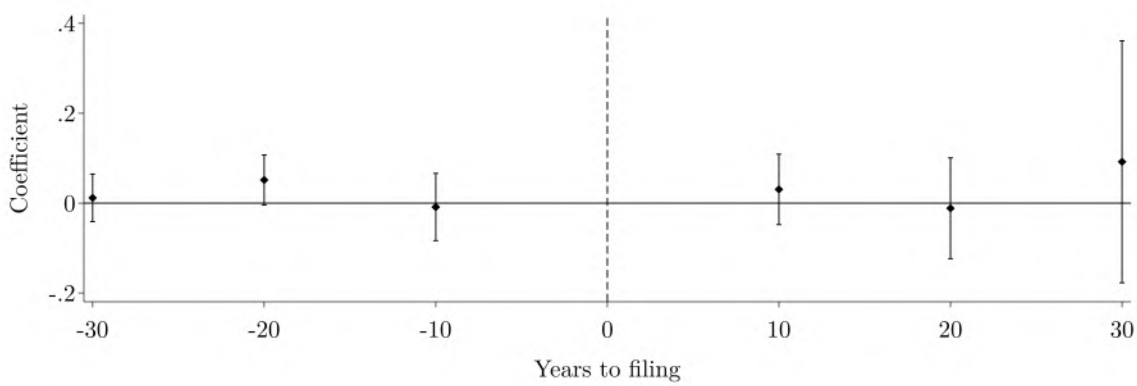
Panel A: Log Average Employment per Establishment



Panel B: Log Average Wage

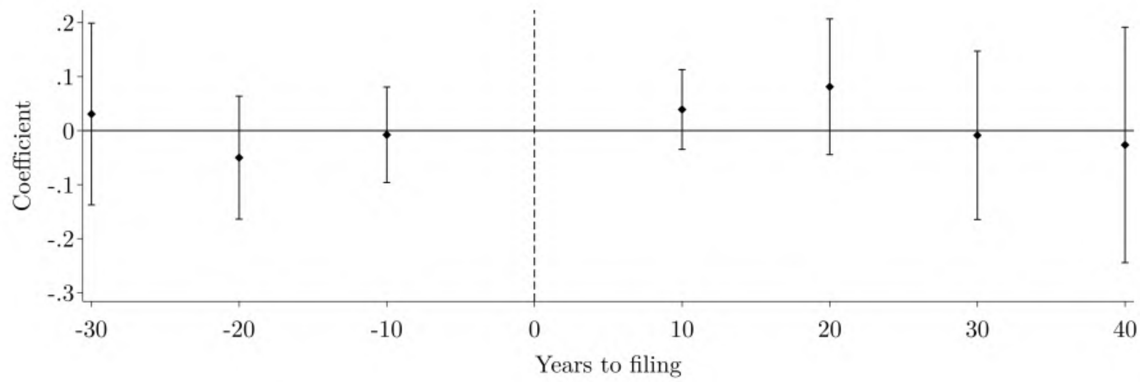


Panel C: Log Labor Share



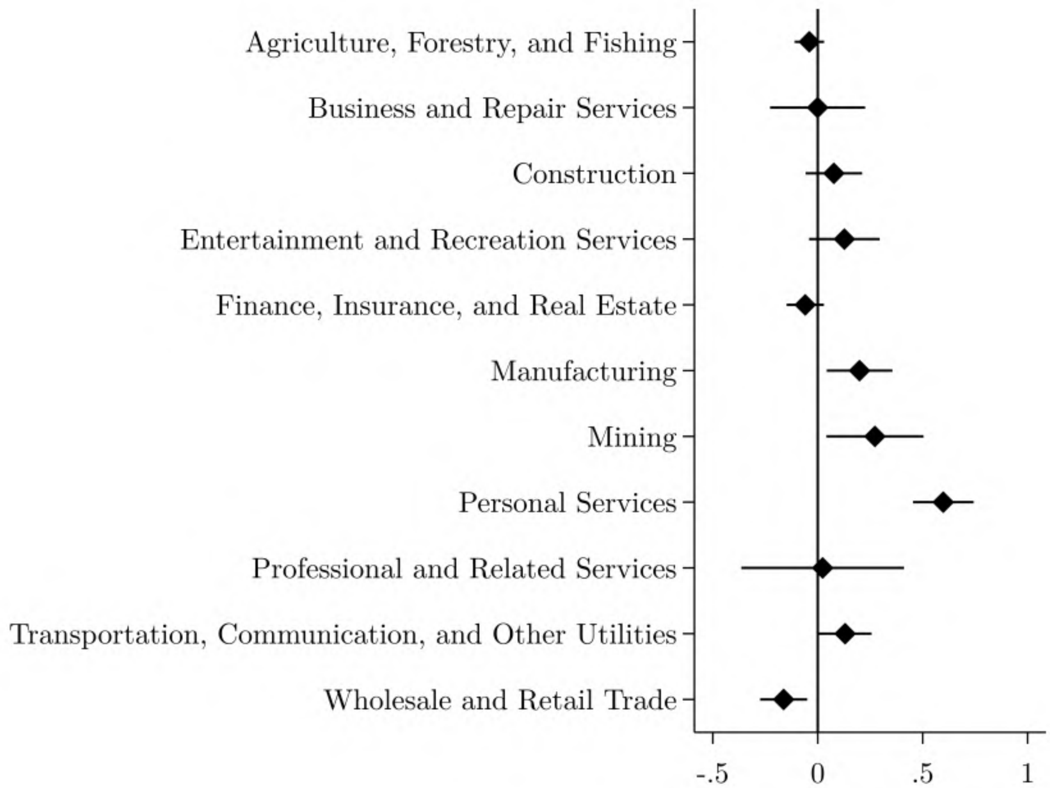
Notes: Data are from state-by-industry tabulations from the census of manufactures. Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state-by-industry level. These results do not yet include controls. Industry-by-state, industry-by-year, state-by-year fixed effects are included. Observations are weighted by both the inverse frequency of the census of manufactures and revenue. Specifically, observations from decennial censuses receive a weight of 1, those from quinquennial censuses receive a weight of 0.5, and those from biennial censuses receive a weight of 0.2. Observations are then additionally weighted by their share of manufacturing revenue in the observed census year.

Figure 3.8: Event Study Results for Industry Employment Rate



Notes: The dependent variable is the percent of the population of county c employed in industry i. Data are from decennial population censuses (Ruggles et al., 2021). Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state-by-industry level. These results do not yet include controls. Industry fixed effects are included. Observations are weighted by their share of the workforce in the observed census year.

Figure 3.9: Difference-in-Differences Results for Industry Employment Rate, by Sector



Notes: The dependent variable is the percent of the population of county c employed in industry i. Data are from decennial population censuses (Ruggles et al., 2021). Bars indicate 95 percent confidence intervals. Robust standard errors are clustered at the state-by-industry level. These results do not yet include controls. Industry fixed effects are included. Observations are weighted by their share of the workforce in the observed census year.

3.7 Tables

Table 3.1: Definitions of Alleged Violation Types

Violation type	Definition
Horizontal *per se* violations	Involves competitors conspiring to eliminate competition among themselves by coordinating prices, bids, or markets.
Exclusionary practices	Includes predatory pricing, price discrimination, boycott, reciprocity, tying arrangements, misuse of patents, threatening patent actions, and exclusive dealings.
Price fixing	Refers to competitors colluding with each other to set prices.
Monopoly violations	Includes monopolization and the act of acquiring a monopoly.
Vertical restraints	Includes resale price maintenance and other restrictions imposed by the manufacturer, a franchisee, or a dealer.
Horizontal mergers	Involves the merger of direct competitors.
Non-horizontal mergers	Involves the merger of companies that are not direct competitors but may be linked vertically (supply chain) or complementarily (related products or markets).

Notes: Cases may involve multiple types of alleged violations; categories are not mutually exclusive.

Table 3.2: State-by-Industry Census of Manufactures Data Sources by Year

Year	Source
1850	Bureau of the Census (1859, pp. 5–129)
1860	Bureau of the Census (1865, pp. 1–673)
1870	Bureau of the Census (1872, pp. 493–584)
1880	Bureau of the Census (1883a, pp. 88–192)
1890	Bureau of the Census (1895, pp. 333–639)
1900	Bureau of the Census (1902, pp. 1–979)
1904	Bureau of the Census (1907, pp. 5–1219)
1909	Bureau of the Census (1912, pp. 13–1401)
1914	Bureau of the Census (1919, pp. 15–1677)
1919	Bureau of the Census (1923, pp. 19–1694)
1921	Bureau of the Census (1924, pp. 1372–1522)
1923	Bureau of the Census (1926, pp. 1244–1391)
1925	Bureau of the Census (1928, pp. 1302–1380)
1927	Bureau of the Census (1930, pp. 24–109)
1929	Bureau of the Census (1933, pp. 41–576)
1931	Bureau of the Census (1935a, pp. 1188–1281)
1933	Bureau of the Census (1935b)[a]
1935	Bureau of the Census (1937)[a]
1937	Bureau of the Census (1939, pp. 1329–1427)
1939	Bureau of the Census (1942, pp. 47–1152)
1947	Bureau of the Census (1950, pp. 66–658)
1954	Bureau of the Census (1957, pp. 101-1–150-20)
1958	Bureau of the Census (1961, pp. 1-2–51-12)
1963	Bureau of the Census (1966, pp. 1-1–51-8)
1967	Bureau of the Census (1971a, pp. 1-1–27-10); Bureau of the Census (1971b, pp. 28-1–51-7)
1972	Bureau of the Census (1975a, pp. 1-1–27-11); Bureau of the Census (1975b, pp. 28-1–51-8)
1977	Bureau of the Census (1981a, pp. 1-1–27-13); Bureau of the Census (1981b, pp. 28-1–51-10)
1982	Bureau of the Census (1985)[b]
1987	Bureau of the Census (1990)[b]
1992	Bureau of the Census (1996)[b]

a The Bureau of the Census did not publish state-by-industry tables in the main volumes for 1933 and 1935. Instead, these data appeared in separately issued, unpublished addenda ((Bureau of the Census, 1935b, 4–5); (Bureau of the Census, 1937, 4)). Page numbers are omitted here because each state's table was separately paginated in these addenda.

b In 1982, 1987, and 1992, the Bureau of the Census published separate manufacturing reports for each state. This citation is representative; identical reports are available for all other states.

Table 3.3: State-by-Industry Census of Manufactures Variables and Availability, by Census Year

Variable	**Description**	1850	1860	1870	1880	1890	1900	1904	1909	1914	1919	1921	1923	1925	1927	1929	1931	1933	1935	1937	1939	1947	1954	1958	1963	1967	1972	1977	1982	1987	1992
`year`	Census year.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`census_day`	Benchmark date for census.[a]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`obs_number`	Observation number, by census year. Created during data digitization.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`page_number`	Page(s) on which observation was recorded in original census volume.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`line_number`	Line number from original census table. Created during data digitization.					✓	✓	✓	✓	✓	✓									✓											
`state_name`	Name of U.S. state or territory.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`statefip`	Federal Information Processing Standards (FIPS) codes to identify U.S. states and territories.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`ind_sic`	Standard Industrial Classification (SIC) code matching industry name in original census volume. Codes for 1939 and before were manually added using the industry descriptions in the 1946 SIC Manual. Codes for 1947 and later were provided in the original census volumes.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`ind_sic_desc`	Description corresponding to SIC code. In 1939 and before, these correspond to the descriptions in the 1946 SIC Manual. In 1947 and later, these were provided in the original census volumes.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`ind_orig`	Industry name, as given in original census volume.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`ind_raw`	Industry name as in original census volume but with standardized capitalization and punctuation.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`ind_sic_group`	Group corresponding to SIC code. Groups for 1937 and before were manually added using the industry descriptions in the 1946 SIC Manual. Groups for 1939 and later were provided in the original census volumes.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`ind_sic_len`	Length of assigned SIC code.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`national_total`	Indicator for nationwide industry sum.[b, c]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`aggregate_obs`	Indicator for statewide sum across industries.[c]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`overflow_obs`	Indicator for the catch-all "all other industries" category that aggregates across state-by-industry cells that correspond to too few establishments to be reported.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`estimated_data`	Percent of data estimated using administrative records.																				✓	✓	✓	✓	✓						
`establishments`	Number of establishments.[d]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`establishments_1_19`	Number of establishments with 1 to 19 employees.																			✓											
`establishments_20_99`	Number of establishments with 20 to 99 employees.																			✓											
`establishments_100_plus`	Number of establishments with 100 or more employees.																			✓											

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Variable	Description	1850	1860	1870	1880	1890	1900	1904	1909	1914	1919	1921	1923	1925	1927	1929	1931	1933	1935	1937	1939	1947	1954	1958	1963	1967	1972	1977	1982	1987	1992
establishments_large	Number of establishments with 20 or more employees.														✓	✓	✓	✓	✓	✓	✓	✓	✓								
cap	Total value of capital.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																				
cap_expend	Spending on new capital and equipment.																														
cap_plant_land	Value of plant capital in land.					✓	✓	✓																							
cap_plant_bldg	Value of plant capital in buildings.					✓	✓	✓																							
cap_plant_mach	Value of plant capital in machinery, tools, and implements.					✓	✓	✓																							
cap_live	Value of livestock.					✓																									
cap_misc	Value of miscellaneous capital.						✓	✓																							
hp_total	Total horsepower.			✓					✓	✓	✓		✓			✓															
prime_hp	Horsepower of "prime movers," which includes engines, turbines, and water wheels.															✓															
se_hp	Horsepower of steam engines.			✓						✓	✓																				
se_number	Number of steam engines.			✓																											
st_hp	Horsepower of steam turbines										✓																				
ww_hp	Horsepower of water wheels.			✓						✓	✓																				
ww_number	Number of water wheels.			✓																											
comb_hp	Horsepower of internal combustion engines.									✓	✓																				
elec_hp	Horsepower from purchased electric energy.									✓	✓					✓															
own_elec_hp	Horsepower generated in house.									✓	✓																				
coal	Coal consumed in short tons.												✓																		
mat	Total cost of materials.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓					✓	✓	✓	✓	✓	✓
mat_princ	Cost of principal materials.						✓	✓	✓	✓	✓					✓					✓										
mat_energy	Cost of energy, which is the sum of the cost of fuel and the cost of purchased electric energy in 1939.						✓	✓	✓	✓	✓					✓					✓										
mat_fuel	Cost of fuel.																				✓										
mat_elec	Cost of purchased electric energy.																				✓										
misc_exp	Total miscellaneous expenses.					✓	✓	✓	✓	✓	✓																				
misc_exp_works	Miscellaneous expenses for rent of works (i.e., factory space).						✓	✓	✓	✓	✓																				
misc_exp_taxes	Miscellaneous expenses for taxes. This amount includes internal revenue (i.e., federal taxes) in 1909, 1914, and 1919 but not in 1900 and 1904.						✓	✓	✓	✓	✓																				
misc_exp_sundry	Miscellaneous expenses for rent of offices, interest, and other items.						✓	✓	✓																						
misc_exp_contract	Miscellaneous expenses for contract labor.						✓	✓	✓	✓	✓	✓									✓										
expenses_total	Total expenses, including wages, materials and miscellaneous expenses.								✓																						
costs_total	Total costs, including materials and miscellaneous expenses.																				✓										

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Variable	Description	1850	1860	1870	1880	1890	1900	1904	1909	1914	1919	1921	1923	1925	1927	1929	1931	1933	1935	1937	1939	1947	1954	1958	1963	1967	1972	1977	1982	1987	1992
`persons`	Total number of persons engaged in industry.								✓	✓	✓	✓	✓																		
`proprietors`	Number of proprietors and firm members.						✓	✓	✓	✓	✓	✓	✓								✓										
`labor`	Total number of people employed.[e]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`labor_orig`	Range specifying the total number of people employed. (In later years, the tables do not disclose the exact number of employees in cases where the number of establishments is small.)																									✓	✓	✓	✓	✓	✓
`labor_ofcs`	Number of officers employed.[f]					✓	✓	✓	✓	✓	✓	✓	✓			✓					✓										
`labor_mgrs`	Number of managers employed.								✓	✓	✓																				
`labor_clerks`	Number of clerks employed.								✓	✓	✓										✓										
`labor_clerks_male`	Number of male clerks employed.								✓	✓	✓																				
`labor_clerks_female`	Number of female clerks employed.								✓	✓	✓																				
`labor_ops`	Number of operatives employed.					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`labor_ops_hours`	Total number of hours worked by operatives.																						✓	✓	✓	✓	✓	✓	✓	✓	✓
`labor_male`	Number of men employed.	✓	✓	✓	✓	✓																									
`labor_ops_max_month`	Month in which the number of operatives was at its maximum.								✓	✓	✓	✓																			
`labor_ops_max`	Number of operatives during the month when the number of operatives was at its maximum.								✓	✓	✓	✓																			
`labor_ops_min_month`	Month in which the number of operatives was at its minimum.								✓	✓	✓	✓																			
`labor_ops_min`	Number of operatives during the month when the number of operatives was at its minimum.								✓	✓	✓	✓																			
`labor_ops_stock`	Number of operatives on December 15, or the nearest representative day.								✓	✓	✓																				
`labor_ofcs_male`	Number of male officers employed.					✓																									
`labor_ops_male`	Number of male operatives employed.					✓	✓	✓	✓	✓	✓																				
`labor_piece_male`	Number of male pieceworkers employed.					✓																									
`labor_female`	Number of women employed.	✓	✓	✓	✓	✓																									
`labor_ofcs_female`	Number of female officers employed.					✓																									
`labor_ops_female`	Number of female operatives employed.					✓	✓	✓	✓	✓	✓																				
`labor_piece_female`	Number of female pieceworkers employed.					✓																									
`labor_youth`	Number of youths employed.[g]			✓	✓	✓	✓	✓	✓	✓	✓																				
`labor_ops_youth`	Number of youth operatives employed.					✓	✓	✓	✓	✓	✓																				
`labor_ops_male_youth`	Number of male youth operatives employed.								✓	✓	✓																				
`labor_ops_female_youth`	Number of female youth operatives employed.								✓	✓	✓																				
`labor_piece_youth`	Number of youth pieceworkers employed.					✓																									
`labor_dist`	Number of distribution workers employed.																				✓										
`labor_const`	Number of construction workers employed.																				✓										
`labor_other`	Number of other workers employed.																				✓										

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Variable	Description	1850	1860	1870	1880	1890	1900	1904	1909	1914	1919	1921	1923	1925	1927	1929	1931	1933	1935	1937	1939	1947	1954	1958	1963	1967	1972	1977	1982	1987	1992
`wage_bill`	Total cost of labor.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`wages_ofcs`	Wages paid to officers.					✓	✓	✓	✓	✓	✓	✓	✓			✓					✓										
`wages_mgrs`	Wages paid to managers.								✓	✓	✓										✓										
`wages_clerks`	Wages paid to clerks.								✓	✓	✓										✓										
`wages_ops`	Wages paid to operatives.					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`wages_male`	Wages paid to men.					✓																									
`wages_ofcs_male`	Wages paid to male officers.					✓																									
`wages_ops_male`	Wages paid to male operatives.					✓	✓	✓																							
`wages_piece_male`	Wages paid to male pieceworkers.					✓																									
`wages_female`	Wages paid to women.					✓																									
`wages_ofcs_female`	Wages paid to female officers.					✓																									
`wages_ops_female`	Wages paid to female operatives.					✓	✓	✓																							
`wages_piece_female`	Wages paid to female pieceworkers.					✓																									
`wages_youth`	Wages paid to youths.					✓	✓	✓																							
`wages_ops_youth`	Wages paid to youth operatives.					✓	✓	✓																							
`wages_piece_youth`	Wages paid to youth pieceworkers.					✓																									
`wages_dist`	Wages paid to distribution workers.																				✓										
`wages_const`	Wages paid to construction workers.																				✓										
`wages_other`	Wages paid to other workers.																				✓										
`product`	Total value of products manufactured (in unadjusted dollars).	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓
`value_added`	Total value of value added (in unadjusted dollars). Value added is the total value of products minus the total value of materials.	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`value_added_new`	Total value of value added (in unadjusted dollars). Value added is the total value of products minus the total value of materials, but for industries where contract work is important, the calculation also subtracts the cost of contract work.																		✓												

a For example, in 1850, the Census Day was June 1, 1850. This means that in the 1850 census of manufactures, operators of manufacturing establishments were asked to report on their activity for the twelve months preceding June 1, 1850. (However, the census volumes note that some reported figures for the previous calendar year instead.) Starting with the 1905 census of manufactures (which covered calendar year 1904), respondents were asked to report information corresponding to the preceding calendar year.

b The 1850 census of manufactures reported state data by industry, with national totals for each industry; later censuses reversed this structure, reporting state-level data disaggregated by industry. This variable is available in all years but only equal to one in 1850.

c Researchers should exclude these observations when analyzing disaggregated data to prevent duplication of underlying counts.

d In 1929 and earlier, the Census of Manufactures excluded establishments with products valued at less than $500. In the later years of this period, data on establishments producing between $500 and $5,000 in goods were sometimes limited. For example, in 1921 the Census reported only the number of wage earners and the value of products for these smaller establishments, while in 1923 and 1925 it reported only the value of products. From 1931 to 1939, the Census excluded all establishments with less than $5,000 in product value. Beginning in 1947, the Census Bureau eliminated the product-value threshold and instead included all establishments that employed at least one person at any time during the census year.

e In cases where the exact number of employees is suppressed, as noted in the `labor_orig` description, we use the midpoint of each range as an estimate.

f Most censuses specify that these workers are salaried.

g "Youth" were typically defined as individuals younger than 16. However, there are some irregularities. For example, the 1880 census of manufactures defines adult male workers as "males above 16 years" and adult female workers as "females above 15 years."

References

Aghion, P., A. Bergeaud, T. Boppart, P. J. Klenow, and H. Li (2023). A Theory of Falling Growth and Rising Rents. *The Review of Economic Studies 90*(6), 2675–2702.

Aghion, P., N. Bloom, R. Blundell, R. Griffith, and P. Howitt (2005). Competition and Innovation: an Inverted-U Relationship. *The Quarterly Journal of Economics 120*(2), 701–728.

Aguzzoni, L., G. Langus, and M. Motta (2013). The Effect of EU Antitrust Investigations and Fines on a Firm's Valuation. *The Journal of Industrial Economics 61*(2), 290–338.

Aldrich, M. (1980). A Note on Railroad Rates and the Populist Uprising. *Agricultural History 54*(3), 424–432. Publisher: Agricultural History Society.

Alexander, B. (1994). The Impact of the National Industrial Recovery Act on Cartel Formation and Maintenance Costs. *The Review of Economics and Statistics 76*(2), 245–254.

Asker, J. and V. Nocke (2021). Collusion, Mergers, and Related Antitrust Issues. In K. Ho, A. Hortaçsu, and A. Lizzeri (Eds.), *Handbook of Industrial Organization*, pp. 177–279. Amsterdam, Netherlands: Elsevier.

Atack, J. (1985). Industrial Structure and the Emergence of the Modern Industrial Corporation. *Explorations in Economic History 22*(1), 29–52.

Atack, J. and F. Bateman (1999). Nineteenth-Century U.S. Industrial Development through the Eyes of the Census of Manufactures: A New Resource for Historical Research. *Historical Methods: A Journal of Quantitative and Interdisciplinary History 32*(4), 177–188.

Autor, D., D. Dorn, L. F. Katz, C. Patterson, and J. Van Reenen (2020). The Fall of the Labor Share and the Rise of Superstar Firms. *The Quarterly Journal of Economics 135*(2), 645–709.

Babina, T., S. Barkai, J. Jeffers, E. Karger, and E. Volkova (2023). Antitrust Enforcement Increases Economic Activity.

Baily, M. N. and B. P. Bosworth (2014). U.S. Manufacturing: Understanding Its Past and Its Potential Future. *Journal of Economic Perspectives 28*(1), 3–26.

Bajgar, M., G. Berlingieri, S. Calligaris, C. Criscuolo, and J. Timmis (2019). Industry Concentration in Europe and North America. *OECD Productivity Working Papers*.

Baker, A. C., D. F. Larcker, and C. C. Y. Wang (2022). How Much Should We Trust Staggered Difference-in-Differences Estimates? *Journal of Financial Economics 144*(2), 370–395.

Baker, J. B. (2003). The Case for Antitrust Enforcement. *Journal of Economic Perspectives 17*(4), 27–50.

Baker, J. B., J. Farrell, A. I. Gavil, M. Gaynor, M. Kades, M. L. Katz, G. Kimmelman, A. D. Melamed, N. L. Rose, S. C. Salop, F. M. Scott Morton, and C. Shapiro (2020). Joint Response to the House Judiciary Committee on the State of Antitrust Law and Implications for Protecting Competition in Digital Markets.

Baker, R. B., C. Frydman, and E. Hilt (2023). Political Discretion and Antitrust Policy: Evidence from the Assassination of President McKinley. *The Journal of Law and Economics 66*(4), 837–873.

Barkai, S. (2020). Declining Labor and Capital Shares. *The Journal of Finance 75*(5), 2421–2463.

Barkai, S., E. Karger, and A. Schaller (2025). Federal Antitrust Enforcement and Industry Outcomes in the Early Twentieth Century.

Barrie, C., A. Palmer, and A. Spirling (2024). Replication for Language Models.

Baxter, W. F. (1980). The Political Economy of Antitrust. In R. D. Tollison (Ed.), *The Political Economy of Antitrust: Principal Paper by William Baxter*. D.C. Heath and Company.

Beach, B. and W. W. Hanlon (2022, June). Historical Newspaper Data: A Researcher's Guide and Toolkit. Technical Report w30135, National Bureau of Economic Research.

Becker, R. A., W. B. Gray, and J. Marvakov (2021). NBER-CES Manufacturing Industry Database (1958-2018, version 2021a).

Berger, D., K. Herkenhoff, and S. Mongey (2022). Labor Market Power. *American Economic Review 112*(4), 1147–1193.

Berkes, E. (2018). Comprehensive Universe of U.S. Patents (CUSP): Data and Facts.

Besley, T., N. Fontana, and N. Limodio (2021). Antitrust Policies and Profitability in Nontradable Sectors. *American Economic Review: Insights 3*(2), 251–265.

Binder, J. J. (1988). The Sherman Antitrust Act and the Railroad Cartels. *The Journal of Law and Economics 31*(2), 443–468.

Bittlingmayer, G. (1985). Did Antitrust Policy Cause the Great Merger Wave? *The Journal of Law and Economics 28*(1), 77–118.

Bittlingmayer, G. (1992). Stock Returns, Real Activity, and the Trust Question. *The Journal of Finance 47*(5), 1701–1730.

Bittlingmayer, G. (1993). The Stock Market and Early Antitrust Enforcement. *The Journal of Law and Economics 36*(1).

Bittlingmayer, G. and T. W. Hazlett (2000). DOS *Kapital*: Has antitrust action against Microsoft created value in the computer industry? *Journal of Financial Economics 55*(3), 329–359.

Bork, R. H. (1966). Legislative Intent and the Policy of the Sherman Act. *The Journal of Law and Economics 9*, 7–48.

Borusyak, K., X. Jaravel, and J. Spiess (2024). Revisiting Event-Study Designs: Robust and Efficient Estimation. *The Review of Economic Studies 91*(6), 3253–3285.

Bosch, J.-C. and E. W. Eckard (1991). The Profitability of Price Fixing: Evidence From Stock Market Reaction to Federal Indictments. *The Review of Economics and Statistics 73*(2), 309–317. Publisher: The MIT Press.

Boudreaux, D. J. and T. J. DiLorenzo (1993). The Protectionist Roots of Antitrust. *The Review of Austrian Economics 6*(2), 81–96.

Boudreaux, D. J., T. J. DiLorenzo, and S. Parker (1995). Antitrust Before the Sherman Act. In F. S. McChesney and W. F. Shughart II (Eds.), *The Causes and Consequences of Antitrust: The Public Choice Perspective*. Chicago: The University of Chicago Press.

Bradford, A., A. Chilton, and K. Linos (2025). The Limits and Promise of Global Antitrust Law. *Harvard International Law Journal 66*(2), 435–521.

Bresnahan, T. F. and P. C. Reiss (1991). Entry and Competition in Concentrated Markets. *Journal of Political Economy 99*(5), 977–1009.

Bringhurst, B. (1979). *Antitrust and the Oil Monopoly: The Standard Oil Cases, 1890-1911*. Westport, Connecticut: Greenwood Press.

Brinkerhoff, Jr., J. C. (2017). Ropes of Sand: State Antitrust Statutes Bound by Their Original Scope. *Yale Journal on Regulation 34*, 353–390.

Buccirossi, P., L. Ciari, T. Duso, G. Spagnolo, and C. Vitale (2013). Competition Policy and Productivity Growth: An Empirical Assessment. *The Review of Economics and Statistics 95*(4), 1324–1336.

Bunting, D. (1971). The Truth About "The Truth About the Trusts". *The Journal of Economic History 31*(3), 664–671.

Bureau of Corporations (1915). *Trust Laws and Unfair Competition*. Government Printing Office.

Bureau of Economic Analysis (2025). BEA Interactive Data Application.

Bureau of the Census (1859). *Abstract of the Statistics of Manufactures, According to the Returns of the Seventh Census*. Washington, DC: Government Printing Office.

Bureau of the Census (1865). *Manufactures of the United States in 1860; Compiled from the Original Returns of the Eighth Census, Under the Direction of the Secretary of the Interior*. Washington, DC: Government Printing Office.

Bureau of the Census (1872). *The Statistics of the Wealth and Industry of the United States*, Volume III. Washington, DC: Government Printing Office.

Bureau of the Census (1883a). *Report on the Manufactures of the United States at the Tenth Census*, Volume II of *Tenth Census of the United States*. Washington, DC: Government Printing Office.

Bureau of the Census (1883b). *Tenth Census of the United States, 1880*. Washington, D.C.: Government Printing Office.

Bureau of the Census (1895). *Report on Manufacturing Industries in the United States at the Eleventh Census: 1890. Part I. Totals for States and Industries*, Volume VI of *Eleventh Census of the United States*. Washington, DC: Government Printing Office.

Bureau of the Census (1902). *Manufactures. Part II: States and Territories*, Volume VIII of *Census Reports, Twelfth Census of the United States, Taken in the Year 1900*. Washington, DC: Government Printing Office.

Bureau of the Census (1907). *Manufactures: 1905. Part II: States and Territories*. Washington, DC: Government Printing Office.

Bureau of the Census (1912). *Manufactures: 1909. Reports by States, with Statistics for Principal Cities*, Volume IX of *Thirteenth Census of the United States, Taken in the Year 1910*. Washington, DC: Government Printing Office.

Bureau of the Census (1919). *Census of Manufactures: 1914. Reports for Selected Industries and Detail Statistics for Industries, by States*, Volume II. Washington, DC: Government Printing Office.

Bureau of the Census (1923). *Manufactures: 1919. Reports for States, with Statistics for Principal Cities*, Volume IX of *Fourteenth Census of the United States, Taken in the Year 1920*. Washington, DC: Government Printing Office.

Bureau of the Census (1924). *Biennial Census of Manufactures: 1921*. Washington, DC: Government Printing Office.

Bureau of the Census (1926). *Biennial Census of Manufactures: 1923*. Washington, DC: Government Printing Office.

Bureau of the Census (1928). *Biennial Census of Manufactures: 1925*. Washington, DC: Government Printing Office.

Bureau of the Census (1930). *Biennial Census of Manufactures: 1927*. Washington, DC: Government Printing Office.

Bureau of the Census (1933). *Manufactures: 1929. Reports by States: Statistics for Industrial Areas, Counties, and Cities*, Volume III of *Fifteenth Census of the United States*. Washington, DC: Government Printing Office.

Bureau of the Census (1935a). *Biennial Census of Manufactures: 1931*. Washington, DC: Government Printing Office.

Bureau of the Census (1935b). *Biennial Census of Manufactures: 1933. State Summaries by Industries and by Cities and Counties*. Washington, DC: Government Printing Office.

Bureau of the Census (1937). *Biennial Census of Manufactures: 1935. Miscellaneous Reports. State Reports: Summary by Industries*, Volume IV. Washington, DC: Government Printing Office.

Bureau of the Census (1939). *Biennial Census of Manufactures: 1937*, Volume I. Washington, DC: Government Printing Office.

Bureau of the Census (1942). *Manufactures: 1939. Reports for States and Outlying Areas*, Volume III of *Sixteenth Census of the United States: 1940*. Washington, DC: Government Printing Office.

Bureau of the Census (1950). *Census of Manufactures: 1947. Statistics by States: Selected General Statistics for States, Metropolitan Areas, Counties, and Urban Places*, Volume III of *Census of Manufactures: 1947*. Washington, DC: Government Printing Office.

Bureau of the Census (1957). *Census of Manufactures: 1954. Area Statistics: Industry Statistics for Geographic Divisions, States, Standard Metropolitan Areas, Counties and Cities*, Volume III. Washington, DC: Government Printing Office.

Bureau of the Census (1961). *Census of Manufactures: 1958. Area Statistics: Industry Statistics for Geographic Divisions, States, Standard Metropolitan Statistical Areas, Counties, and Cities*, Volume III. Washington, DC: Government Printing Office.

Bureau of the Census (1966). *1963 Census of Manufactures. Area Statistics*, Volume III. Washington, DC: Government Printing Office.

Bureau of the Census (1971a). *1967 Census of Manufactures. Area Statistics. Part 1: Alabama–Montana*, Volume III. Washington, DC: Government Printing Office.

Bureau of the Census (1971b). *1967 Census of Manufactures. Area Statistics. Part 2: Nebraska–Wyoming*, Volume III. Washington, DC: Government Printing Office.

Bureau of the Census (1975a). *1972 Census of Manufactures. Area Statistics. Part 1: Alabama–Montana*, Volume III. Washington, DC: Government Printing Office.

Bureau of the Census (1975b). *1972 Census of Manufactures. Area Statistics. Part 2: Nebraska–Wyoming*, Volume III. Washington, DC: Government Printing Office.

Bureau of the Census (1981a). *1977 Census of Manufactures. Geographic Area Statistics. Part 1: General Summary, Alabama–Montana*, Volume III. Washington, DC: Government Printing Office.

Bureau of the Census (1981b). *1977 Census of Manufactures. Geographic Area Statistics. Part 2: General Summary, Nebraska–Wyoming*, Volume III. Washington, DC: Government Printing Office.

Bureau of the Census (1985). *1982 Census of Manufactures: Geographic Area Series: Alabama*. Washington, DC: Government Printing Office.

Bureau of the Census (1990). *1987 Census of Manufactures: Geographic Area Series: Alabama*. Washington, DC: Government Printing Office.

Bureau of the Census (1996). *1992 Census of Manufactures: Geographic Area Series: Alabama*. Washington, DC: Government Printing Office.

Burns, M. R. (1977). The Competitive Effects of Trust-Busting: A Portfolio Analysis. *Journal of Political Economy 85*(4), 717–739.

Calabresi, S. G. and L. C. Leibowitz (2013). Monopolies and the Constitution: A History of Crony Capitalism. *Harvard Journal of Law & Public Policy 36*, 983.

Callaway, B. and P. H. C. Sant'Anna (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics 225*(2), 200–230.

Cengiz, D., A. Dube, A. Lindner, and B. Zipperer (2019). The Effect of Minimum Wages on Low-Wage Jobs. *The Quarterly Journal of Economics 134*(3), 1405–1454.

Chen, Y. (1988). Sherman's Predecessors: Pioneers in State Antitrust Legislation. *Journal of Reprints for Antitrust Law and Economics 18*(2), 93–96.

Chicu, M., C. Vickers, and N. L. Ziebarth (2013). Cementing the Case for Collusion Under the National Recovery Administration. *Explorations in Economic History 50*(4), 487–507.

Clark, J. D. (1931). *The Federal Trust Policy*. Baltimore, Maryland: The Johns Hopkins Press.

Clay, K. and W. Troesken (2002). Strategic Behavior in Whiskey Distilling, 1887-1895. *The Journal of Economic History 62*(4), 999–1023.

Collins, W. D. (2013). Trusts and the Origins of Antitrust Legislation. *Fordham Law Review 81*, 2279–2348.

Columbia Law Review (1961). The Commerce Clause and State Antitrust Regulation. *Columbia Law Review 61*(8), 1469–1496.

Commerce Clearing House, Inc. (1952). *The Federal Antitrust Laws: With Summary of Cases Instituted by the United States, 1890-1951*. Commerce Clearing House.

Cook, W. W. (1888). *Trusts: The Recent Combinations in Trade, Their Character, Legality and Mode of Organization, and the Rights, Duties and Liabilities of Their Managers and Certificate Holders*. New York, New York: L.K. Strouse & Company.

Crandall, R. W. and C. Winston (2003). Does Antitrust Policy Improve Consumer Welfare? Assessing the Evidence. *Journal of Economic Perspectives 17*(4), 3–26.

Crane, D. A. (2014). The Tempting of Antitrust: Robert Bork and the Goals of Antitrust Policy. *Antitrust Law Journal 79*(3), 835–853.

Crane, D. A. (2023). De-Nazifying by De-Cartelizing: The Legacy of the American Decartelization Project in Germany. In D. A. Crane and W. J. Novak (Eds.), *Antimonopoly and American Democracy*. Oxford University Press.

Dameron, C. S. (2016). Present at Antitrust's Creation: Consumer Welfare in the Sherman Act's State Statutory Forerunners. *Yale Law Journal 125*(4), 796–1149.

Dasgupta, S. and A. Žaldokas (2019). Anticollusion Enforcement: Justice for Consumers and Equity for Firms. *The Review of Financial Studies 32*(7), 2587–2624.

de Chaisemartin, C. and X. D'Haultfœuille (2020). Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *American Economic Review 110*(9), 2964–2996.

De Loecker, J., J. Eeckhout, and G. Unger (2020). The Rise of Market Power and the Macroeconomic Implications. *The Quarterly Journal of Economics 135*(2), 561–644.

Dell, M., J. Carlson, T. Bryan, E. Silcock, A. Arora, Z. Shen, L. D'Amico-Wong, Q. Le, P. Querubin, and L. Heldring (2023). American Stories: A Large-Scale Structured Text Dataset of Historical U.S. Newspapers.

Delle Donne, C. R. (1973). *Federal Census Schedules, 1850-1880: Primary Sources for Historical Research*. National Archives and Recorods Service Reference Information Paper No. 67. General Services Administration.

Department of Commerce and Labor (1904). *Eighteenth Annual Report of the Commissioner of Labor*. Washington: Government Printing Office.

Department of Commerce and Labor (1906). *Bulletin of the Bureau of Labor No. 65*. Washington: Government Printing Office.

Department of Commerce and Labor (1912). *Retail Prices: 1890 to 1911. Bulletin of the United States Bureau of Labor. Whole Number 105: Part II.* Washington: Government Printing Office.

Deshpande, M. and Y. Li (2019). Who Is Screened Out? Application Costs and the Targeting of Disability Programs. *American Economic Journal: Economic Policy 11*(4), 213–248.

Dilorenzo, T. J. (1985). The Origins of Antitrust: An Interest-Group Perspective. *International Review of Law and Economics 5*(1), 73–90.

Dilorenzo, T. J. and J. C. High (1988). Antitrust and Competition, Historically Considered. *Economic Inquiry 26*(3), 423–435.

Donaldson, D. and R. Hornbeck (2016). Railroads and American Economic Growth: A 'Market Access' Approach. *The Quarterly Journal of Economics 131*(2), 799–858.

Dong, A., M. Massa, and A. Žaldokas (2019). The Effects of Global Leniency Programs on Margins and Mergers. *The RAND Journal of Economics 50*(4), 883–915.

Easterlin, R. (1960). Interregional Differences in Per Capita Income, Population, and Total Income, 1840-1950. In *Trends in the American Economy in the Nineteenth Century*, pp. 73–140. Princeton University Press.

Easterlin, R. A. (1957). State Income Estimates. In S. Kuznets and D. S. Thomas (Eds.), *Population Redistribution and Economic Growth: United States, 1870-1950*, pp. 701–759. Philadelphia, Pennsylvania: The American Philosophical Society.

Federal Reserve Bank of Minneapolis. Consumer Price Index, 1800-.

Federal Reserve Bank of St. Louis (2024). Gross Domestic Product.

Feldman, R. and M. Lemley (2022). Atomistic Antitrust. *William & Mary Law Review 63*(6), 1869.

Fisher, F. M. and J. J. McGowan (1983). On the Misuse of Accounting Rates of Return to Infer Monopoly Profits. *American Economic Review 73*(1), 82–97.

Flynn, J. J. (1988). The Reagan Administration's Antitrust Policy, 'Original Intent' and the Legislative History of the Sherman Act. *The Antitrust Bulletin 33*(2), 259–307.

Forrest, J. D. (1896). Anti-Monopoly Legislation in the United States. *American Journal of Sociology 1*(4), 411–425.

Gallo, J. C., K. Dau-Schmidt, J. L. Craycraft, and C. J. Parker (2000). Department of Justice Antitrust Enforcement, 1955-1997: An Empirical Study. *Review of Industrial Organization 17*(1), 75–133.

Geloso, V. (2020). Collusion and Combines in Canada, 1880–1890. *Scandinavian Economic History Review 68*(1), 66–84.

Gentzkow, M., J. M. Shapiro, and M. Sinkinson (2011). The Effect of Newspaper Entry and Exit on Electoral Politics. *American Economic Review 101*(7), 2980–3018.

Gil, R. (2015). Does Vertical Integration Decrease Prices? Evidence from the Paramount Antitrust Case of 1948. *American Economic Journal: Economic Policy 7*(2), 162–191.

Gilligan, T., W. Marshall, and B. Weingast (1989). Regulation and the Theory of Legislative Choice: The Interstate Commerce Act of 1887. *Journal of Law and Economics 32*(1), 35–61.

González, X. and M. J. Moral (2019). Effects of Antitrust Prosecution on Retail Fuel Prices. *International Journal of Industrial Organization 67*, 102537.

Goodman-Bacon, A. (2021). Difference-in-Differences with Variation in Treatment Timing. *Journal of Econometrics 225*(2), 254–277.

Gordon, S. (1963). Attitudes Towards Trusts Prior to the Sherman Act. *Southern Economic Journal 30*, 156–167.

Gormley, T. A. and D. A. Matsa (2011). Growing Out of Trouble? Corporate Responses to Liability Risk. *The Review of Financial Studies 24*(8), 2781–2821.

Grandy, C. (1993). Original Intent and the Sherman Antitrust Act: A Re-Examination of the Consumer-Welfare Hypothesis. *The Journal of Economic History 53*(2), 359–376.

Gross, D. P. (2020). Collusive Investments in Technological Compatibility: Lessons from U.S. Railroads in the Late 19th Century. *Management Science 66*(12), 5683–5700.

Grullon, G., Y. Larkin, and R. Michaely (2019). Are US Industries Becoming More Concentrated? *Review of Finance 23*(4), 697–743.

Haines, M. R. (1989). A State and Local Consumer Price Index for the United States in 1890.

Hashmi, A. R. (2013). Competition and Innovation: The Inverted-U Relationship Revisited. *The Review of Economics and Statistics 95*(5), 1653–1668.

Hawley, J. D. (2025). Sen. Hawley: Break Up Big Tech to Reclaim American Freedom. *New York Post*.

Hazlett, T. W. (1992). The Legislative History of the Sherman Act Re-Examined. *Economic Inquiry 30*(2), 263–276.

Higgs, R. (1970). Railroad Rates and the Populist Uprising. *Agricultural History 44*(3), 291–298.

Higgs, R. (1971). *The Transformation of the American Economy, 1865-1914: An Essay in Interpretation*. Auburn, Alabama: John Wiley & Sons, Inc.

Hofstadter, R. (1979). *The Paranoid Style in American Politics, and Other Essays*. Chicago, Illinois: University of Chicago Press.

Hopkins, D. J., E. Kim, and S. Kim (2017). Does Newspaper Coverage Influence or Reflect Public Perceptions of the Economy? *Research & Politics 4*(4).

Hornbeck, R., S. H.-M. Hsu, A. Humlum, and M. Rotemberg (2024). Gaining Steam: Incumbent Lock-in and Entrant Leapfrogging.

Hornbeck, R. and M. Rotemberg (2019). Railroads, Reallocation, and the Rise of American Manufacturing.

Hornbeck, R. and M. Rotemberg (2024). Growth Off the Rails: Aggregate Productivity Growth in Distorted Economies. *Journal of Political Economy 132*(11), 3547–3602.

Hovenkamp, H. (1983). State Antitrust in the Federal Scheme. *Indiana Law Journal 58*(3), 375–432.

Hovenkamp, H. and P. Areeda (2022). *Antitrust Law: An Analysis of Antitrust Principles and Their Application* (5th ed.). New York, New York: Aspen Publishers, Inc.

Huang, J. (2025). Market Manipulation by a Monopsony Cartel: Evidence from the U.S. Meatpacking Industry.

Hylton, K. N. and F. Deng (2007). Antitrust Around the World: An Empirical Analysis of the Scope of Competition Laws and Their Effects. *Antitrust Law Journal 74*(2), 271–341.

Industrial Commission (1900). *Preliminary Report on Trusts and Industrial Combinations. Volume 1 of the Commission's Reports.* Washington: Government Printing Office.

Inter-University Consortium for Political and Social Research (1999). United States Historical Election Returns, 1824-1968.

Jacobson, J. (2018). The 'Patent Monopoly'. *Antitrust 32*(3), 3–4.

Jacquemin, A. and M. E. Slade (1989). Cartels, Collusion, and Horizontal Merger. In *Handbook of Industrial Organization*, Volume 1, pp. 415–473. Elsevier.

James, J. A. (1983). Structural Change in American Manufacturing, 1850-1890. *The Journal of Economic History 43*(2), 433–459.

John, R. R. (2017). Proprietary Interest: Merchants, Journalists, and Antimonopoly in the 1880s. In B. J. Schulman and J. E. Zelizer (Eds.), *Media Nation: The Political History of News in Modern America*, pp. 10–35. University of Pennsylvania Press.

Johnston, L. and S. H. Williamson (2024). Annual Real and Nominal GDP for the United States, 1790-Present.

Kang, H. (2025). Price-Fixing Cartels and Firm Innovation. *Management Science*.

Kent A. Clark Center for Global Markets (2020). Antitrust in the Digital Econmy.

Khan, L. M. (2017). Amazon's Antitrust Paradox. *The Yale Law Journal*, 710–805.

Klein, A. (2009). Personal Income of U.S. States: Estimates for the Period 1880–1910. *Warwick Economic Research Papers*.

Kolko, G. (1963). *The Triumph of Conservatism: A Reinterpretation of American History, 1900-1916*. New York, New York: The Free Press of Glencoe.

Kwon, S. and A. C. Marco (2021). Can Antitrust Law Enforcement Spur Innovation? Antitrust Regulation of Patent Consolidation and its Impact on Follow-On Innovataions. *Research Policy 50*(9), 104295.

Lafortune, J., E. Lewis, and J. Tessada (2019). People and Machines: A Look at the Evolving Relationship between Capital and Skill in Manufacturing, 1860–1930, Using Immigration Shocks. *The Review of Economics and Statistics 101*(1), 30–43.

Lamb, D. (2001). Avoiding Impotence: Rethinking the Standards for Applying State Antitrust Laws to Interstate Commerce. *Vanderbilt Law Review 54*(4), 1705.

Lamoreaux, N. R. (1985). *The Great Merger Movement in American Business, 1895-1904*. Cambridge: Cambridge University Press.

Lamoreaux, N. R. (2023). Antimonopoly and State Regulation of Corporations in the Gilded Age and Progressive Era. In D. A. Crane and W. J. Novak (Eds.), *Antimonopoly and American Democracy*. Oxford University Press.

Lamoreaux, N. R. and L. Phillips-Sawyer (2021). Voting Trusts and Antitrust: Rethinking the Role of Shareholder Litigation in Public Regulation, from the 1880s to the 1930s. *Law and History Review 39*(3), 569–600.

Lande, R. (1982). Wealth Transfers as the Original and Primary Concern of Antitrust: The Efficiency Interpretation Challenged. *The Hastings Law Journal 34*, 65–151.

Lande, R. H. (1988). The Rise and (Coming) Fall of Efficiency as the Ruler of Antitrust. *The Antitrust Bulletin 33*, 429–465.

Letwin, W. (1965). *Law and Economic Policy in America: The Evolution of the Sherman Antitrust Act*. New York, New York: Random House.

Letwin, W. L. (1956). Congress and the Sherman Antitrust Law: 1887-1890. *The University of Chicago Law Review 23*(2), 221–258.

Levinsohn, J. and A. Petrin (2003). Estimating Production Functions Using Inputs to Control for Unobservables. *The Review of Economic Studies 70*(2), 317–341.

Libecap, G. D. (1992). The Rise of the Chicago Packers and the Origins of Meat Inspection and Antitrust. *Economic Inquiry 30*(2), 242–262.

Library of Congress (2024). Chronicling America.

Manson, S., J. Schroeder, D. Van Riper, T. Kugler, and S. Ruggles (2022). *IPUMS National Historical Geographic Information System: Version 17.0 [dataset].* Minneapolis, MN: IPUMS.

May, J. (1987). Antitrust Practice and Procedure in the Formative Era: The Constitutional and Conceptual Reach of State Antitrust Law. *University of Pennsylvania Law Review 135*(3), 495–594.

May, J. (1990). The Role of the States in the First Century of the Sherman Act and the Larger Picture of Antitrust History. *Antitrust Law Journal 59*, 93–107.

Mayo, J. W. and M. Schiffer (2006). Antitrust Economics Meets Antitrust Psychology: A View from the Firms. *International Journal of the Economics of Business 13*(2), 281–306.

McCraw, T. K. (1981). Rethinking the Trust Question. In T. K. McCraw (Ed.), *Regulation in Perspective: Historical Essays*, pp. 1–55. Cambridge, Massachusetts: Harvard University Press.

Melitz, M. J. and G. I. P. Ottaviano (2008). Market Size, Trade, and Productivity. *Review of Economic Studies 75*(1), 295–316.

Michel, J.-B., Y. K. Shen, A. P. Aiden, A. Veres, M. K. Gray, J. P. Pickett, D. Hoiberg, D. Clancy, P. Norvig, J. Orwant, S. Pinker, M. A. Nowak, and E. L. Aiden (2011). Quantitative Analysis of Culture Using Millions of Digitized Books. *Science 331*(6014), 176–182.

Miller, N. H. (2025). Industrial Organization and *The Rise of Market Power*. *International Journal of Industrial Organization 98*, 1–16.

Millon, D. (1988). The Sherman Act and The Balance of Power. *Southern California Law Review 61*, 1219–1292.

Millon, D. (1990). The First Antitrust Statute. *Washburn Law Journal 29*, 141.

Moody, J. (1904). *The Truth About The Trusts: A Description and Analysis of the American Trust Movement*. New York: Moody Publishing Company.

Mullin, G. L., J. C. Mullin, and W. P. Mullin (1995). The Competitive Effects of Mergers: Stock Market Evidence from the U.S. Steel Dissolution Suit. *The RAND Journal of Economics 26*(2), 314–330.

Mullin, W. P. and C. M. Snyder (2021). A Simple Method for Bounding the Elasticity of Growing Demand with Applications to the Analysis of Historic Antitrust Cases. *American Economic Journal: Microeconomics 13*(4), 172–217.

Nolette, P. (2012). Litigating the 'Public Interest' in the Gilded Age: Common Law Business Regulation by Nineteenth-Century State Attorneys General. *Polity 44*(3), 373–399.

Noll, R. G. and M. T. Kanazawa (1994). The Origins of State Railroad Regulation: The Illinois Constitution of 1870. In C. Goldin and G. D. Libecap (Eds.), *The Regulated Economy: A Historical Approach to Political Economy*, pp. 13–54. University of Chicago Press.

Nutter, G. W. (1949). *A Quantitative Study of the Extent of Enterprise Monopoly in the United States: 1899-1939.* Ph. D. thesis, The University of Chicago.

Nutter, G. W. and H. A. Einhorn (1969). *Enterprise Monopoly in the United States: 1899-1958*. New York: Columbia University Press.

Ochieng, M., V. Gumma, S. Sitaram, J. Wang, V. Chaudhary, K. Ronen, K. Bali, and J. O'Neill (2024). Beyond Metrics: Evaluating LLMs' Effectiveness in Culturally Nuanced, Low-Resource Real-World Scenarios.

Olson, M. (1971). *The Logic of Collective Action: Public Goods and the Theory of Groups, Second Printing with a New Preface and Appendix*. Harvard University Press.

Peltzman, S. (2014). Industrial Concentration under the Rule of Reason. *The Journal of Law and Economics 57*, S101–S120.

Perlman, E. R. and S. S. Schuster (2016). Delivering the Vote: The Political Effect of Free Mail Delivery in Early Twentieth Century America. *The Journal of Economic History 76*(3), 769–802.

Petralia, S., P.-A. Balland, and D. L. Rigby (2016). Unveiling the Geography of Historical Patents in the United States from 1836 to 1975. *Scientific Data 3*(1), 160074.

Philippon, T. (2019). *The Great Reversal: How America Gave Up on Free Markets*. Cambridge, Massachusetts: Harvard University Press.

Phillips-Sawyer, L. (2018). *American Fair Trade: Proprietary Capitalism, Corporatism, and the 'New Competition,' 1890–1940*. Cambridge: Cambridge University Press.

Phillips-Sawyer, L. (2019). U.S. Antitrust Law and Policy in Historical Perspective. In *Oxford Research Encyclopedia of American History*.

Pittman, R. W. and G. J. Werden (1990). The Divergence of SIC Industries from Antitrust Markets: Indications from Justice Department Merger Cases. *Economics Letters 33*(3), 283–286.

Porter, R. H. (1983). A Study of Cartel Stability: The Joint Executive Committee, 1880-1886. *The Bell Journal of Economics 14*(2), 301–314.

Posner, R. A. (1970). A Statistical Study of Antitrust Enforcement. *The Journal of Law and Economics 13*(2), 365–419.

Prager, R. A. (1992). The Effects of Horizontal Mergers on Competition: The Case of the Northern Securities Company. *The RAND Journal of Economics 23*(1), 123–133.

Pratt, J. A. (1980). The Petroleum Industry in Transition: Antitrust and the Decline of Monopoly Control in Oil. *The Journal of Economic History 40*(4), 815–837.

ProQuest (2024). ProQuest Historical Newspapers™.

Reed, T., M. P. López, A. U. Arrieta, and L. Iacovone (2022). Cartels, Antitrust Enforcement, and Industry Performance: Evidence from Mexico. *World Bank Policy Research Working Paper 20169*.

Rogers, R. P. (2009). *An Economic History of the American Steel Industry*. London: Routledge.

Ruggles, S., C. A. Fitch, R. Goeken, J. D. Hacker, M. A. Nelson, E. Roberts, M. Schouweiler, and M. Sobek (2021). *IPUMS Ancestry Full Count Data: Version 3.0 [dataset]*. Minneapolis, MN: IPUMS.

Scherer, F. M. and D. Ross (1990). *Industrial Market Structure and Economic Performance* (3rd edition ed.). Boston, Massachusetts: Houghton Mifflin Company.

Schumpeter, J. A. (1942). *Capitalism, Socialism and Democracy*. New York: Harper and Brothers.

Schwert, G. W. (1990). Indexes of U.S. Stock Prices from 1802 to 1987. *The Journal of Business 63*(3), 399–426.

Scott, T. D. and V. C. Abreu (2006). Is a Second Golden Age of Antitrust Here? *The National Law Journal*.

Seager, H. R. H. R. and C. A. Gulick (1929). *Trust and Corporation Problems*. New York, New York: Harper and Brothers.

Shapiro, C. and A. Yurukoglu (2024). Trends in Competition in the United States: What Does the Evidence Show? *Journal of Political Economy Microeconomics*.

Shepherd, W. G. (1982). Causes of Increased Competition in the U.S. Economy, 1939-1980. *The Review of Economics and Statistics 64*(4), 613.

Smith, D. A. and S. Ocampo (2025). The Evolution of U.S. Retail Concentration. *American Economic Journal: Macroeconomics 17*(1), 71–101.

Sproul, M. F. (1993). Antitrust and Prices. *Journal of Political Economy 101*(4), 741–754.

Stigler, G. J. (1966). The Economic Effects of the Antitrust Laws. *The Journal of Law and Economics 9*, 225–258.

Stigler, G. J. (1985). The Origin of the Sherman Act. *The Journal of Legal Studies 14*(1), 1–12.

Stucke, M. E. and A. Ezrachi (2017). The Rise, Fall, and Rebirth of the U.S. Antitrust Movement.

Sun, L. and S. Abraham (2021). Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects. *Journal of Econometrics 225*(2), 175–199.

Symeonidis, G. (2000). Price Competition and Market Structure: The Impact of Cartel Policy on Concentration in the UK. *The Journal of Industrial Economics 48*(1), 1–26.

Symeonidis, G. (2008). The Effect of Competition on Wages and Productivity: Evidence from the United Kingdom. *The Review of Economics and Statistics 90*(1), 134–146.

Symeonidis, G. (2019). Competition, Innovation and the Use of Innovations. *The Journal of Industrial Economics 67*, 565–592.

Symeonidis, G. (2024). Unprofitable Cartels: Evidence from a Natural Experiment in the UK. *Review of Industrial Organization 64*(3), 421–447.

Syverson, C. (2004). Market Structure and Productivity: A Concrete Example. *Journal of Political Economy 112*(6), 1181–1222.

Syverson, C. (2024). Spatial Competition, Strategic Entry Responses, and the North Dakota Railroad War of 1905.

Technical Committee on Industrial Classification (1945). *Standard Industrial Classification Manual, Volume I: Manufacturing Industries*. Washington, DC: U.S. Government Printing Office.

Technical Committee on Industrial Classification (1949). *Standard Industrial Classification Manual, Volume II: Nonmanufacturing Industries*. Washington, DC: U.S. Government Printing Office.

Thatchenkery, S. and R. Katila (2023). Innovation and Profitability Following Antitrust Intervention Against a Dominant Platform: The Wild, Wild West? *Strategic Management Journal 44*(4), 943–976.

Thorelli, H. B. (1955). *Federal Antitrust Policy: Origination of an American Tradition*. Baltimore: Johns Hopkins Press.

Tirole, J. (1988). *The Theory of Industrial Organization*. Cambridge, Massachusetts: The MIT Press.

Troesken, W. (1995). Antitrust Regulation before the Sherman Act: The Break-Up of the Chicago Gas Trust Company. *Explorations in Economic History 32*, 109–136.

Troesken, W. (1998). Exclusive Dealing and the Whiskey Trust, 1890-1895. *The Journal of Economic History 58*(3), 755–778.

Troesken, W. (2000). Did the Trusts Want a Federal Antitrust Law? An Event Study of State Antitrust Enforcement and Passage of the Sherman Act. In J. C. Heckelman, J. C. Moorhouse, and R. M. Whaples (Eds.), *Public Choice Interpretations of American Economic History*, pp. 77–104. Boston: Kluwer Academic Publishers.

Truman, H. S. (1947). Annual Message to the Congress on the State of the Union.

Vickers, C. and N. L. Ziebarth (2014). Did the National Industrial Recovery Act Foster Collusion? Evidence from the Macaroni Industry. *The Journal of Economic History 74*(3), 831–862.

Vickers, C. and N. L. Ziebarth (2019). The Census of Manufactures: An Overview. In C. Diebolt and M. J. Haupert (Eds.), *Handbook of Cliometrics*, Volume 2.

Viscusi, W. K., J. E. Harrington Jr., and D. E. M. Sappington (2018). *Economics of Regulation and Antitrust* (5th edition ed.). Cambridge, Massachusetts: The MIT Press.

von Halle, E. (1899). *Trusts or Industrial Combinations and Coalitions in the United States.* New York: The Macmillan Company.

Wang, T. (2025). The Electric Telegraph, News Coverage, and Political Participation. *The Journal of Economic History 85*(1), 1–32.

Warren, E. A. (2024). What's Next for Big Tech.

Watzinger, M., T. A. Fackler, M. Nagler, and M. Schnitzer (2020). How Antitrust Enforcement Can Spur Innovation: Bell Labs and the 1956 Consent Decree. *American Economic Journal: Economic Policy 12*(4), 328–359.

Watzinger, M. and M. Schnitzer (2022). The Breakup of the Bell System and its Impact on U.S. Innovation.

Weber Waller, S. (2005). *Thurman Arnold: A Biography*. New York: New York University Press.

Weeks, J. D. (1886). *Report on the Average Retail Prices of Necessaries of Life in the United States*. Washington: Government Printing Office.

Werden, G. J. (2018). Establishment of the Antitrust Division of the U.S. Department of Justice. *St. John's Law Review 92*(3), 419–430.

Werden, G. J. (2020). *The Foundations of Antitrust*. Durham, North Carolina: Carolina Academic Press.

Werden, G. J. and L. M. Froeb (2018). Don't Panic: A Guide to Claims of Increasing Concentration. *Antitrust*, 74–79.

Whinston, M. D. (2006). *Lectures on Antitrust Economics*. Cambridge, Massachusetts: The MIT Press.

Willis, J. F. (2021). Antitrust in Arkansas Politics during the Progressive Era. *The Arkansas Historical Quarterly 80*(4), 436–467.

Works Progress Administration (1940). *State Antitrust Laws*. Government Printing Office.

Wright, C. D. (1900). *The History and Growth of the United States Census*. Washington: U.S. Government Printing Office.

Wu, T. (2018). *The Curse of Bigness: Antitrust in the New Gilded Age*. New York: Columbia Global Reports.

Wu, T. (2024). The American Schools of Antitrust.

Yates, J. (1986). The Telegraph's Effect on Nineteenth Century Markets and Firms. *Business and Economic History 15*, 149–163.

Yeager, M. (1981). *Competition and Regulation: The Development of Oligopoly in the Meat Packing Industry*. Greenwich, Connecticut: Jai Press, Inc.

YouGov (2023). YouGov Survey: Antitrust Policy.

Young, A. T. and W. F. Shughart (2010). The Consequences of the U.S. DOJ's Antitrust Activities: A Macroeconomic Perspective. *Public Choice 142*, 409–422.

Yuskavage, R. E. (2007). Converting Historical Industry Time Series Data from SIC to NAICS.

Zeballos, E., X. Dong, and E. Islamaj (2023). A Disaggregated View of Market Concentration in the Food Retail Industry. *U.S. Department of Agriculture Economic Research Service*.